

# Learning a Worker-Selection Policy for Dynamo’s KV Router in Offline Replay

A simulation study on Qwen3-32B (vLLM, TP2, H100), aggregated serving

Dynamo routing team

**PENDING** [author list]

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**Draft: simulation results with a live GPU check of the ranking.**

The headline and every result outside Section 11 come from AISim offline replay, not from GPUs; Section 11 checks the ranking on GPUs. Boxed PENDING items mark the open items.

## Abstract

Dynamo’s KV router sends each request to one of  $N$  workers by a fixed cost that weighs prefix-cache reuse against load. Published routers weigh the two differently, and which rule serves a workload best shows only under sustained load, which is costly to reproduce on GPUs for every rule and setting. We describe the router’s worker-selection API, in which every routing rule is a composition of filters, scorers and a picker over declared inputs, and express Dynamo’s default cost function and nine other catalog policy types, most of them ports of published routers, in it. Inside that API we learn a policy, **learned-choice**: a conditional-logit choice model whose utility is linear in signals the live router already computes and which contains the default router as one coefficient setting. We tune it with CMA-ES in AISim offline replay, where Dynamo’s router drives mock vLLM engines timed by AISimulate’s performance model, on windowed goodput defined by inter-token latency and end-to-end slowdown, with common-random-number workload replicates. Every heuristic receives the same tuning budget, restarts and validation-based selection. The headline test, registered before training, is a one-sided Wilcoxon test over the 12 independent segments of a frozen 60-cell test set that holds out time windows, agent plays, session seeds, a trace family, transform ranges and worker counts. In AIS-timed simulation, the learned router M1-v2 beats the branch’s ported heuristics, each tuned with an equal budget: against the validation-selected best of them, tuned Ramjet, it is ahead on 10 of 12 segments with a segment-mean clipped log-ratio of goodput of +0.0316 (one-sided exact Wilcoxon  $p = 0.0017$ ), and it stays first, ahead of Ramjet, under router-state lag of up to 200 ms and under engine-timing perturbations. The claim is scoped: the effect is below the pre-registered minimum detectable effect, so only its sign is claimed; the default cost plus a faithful LMetric term, which the learned model’s class contains and the baseline pool lacks, reproduces most of the margin on validation; the gain is a request-count property, with long prompts slower than under Ramjet; and black-box search first found a degenerate policy that a sign constraint on load coefficients removed. The headline is simulated. On six test cells served live on H100 GPUs, the policy ranking agrees with simulation and M1-v2 leads Ramjet in every cell where simulation predicts a lead; absolute goodput and the size of the gaps between policies are not validated.

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## 1 Introduction

**The decision.** A KV-aware router places each request on the worker whose cache already holds the longest prefix of its prompt, unless that worker is busy enough that recomputing the prefix elsewhere is cheaper. On the deployment studied here, Qwen3-32B on vLLM with two H100 GPUs per worker, both sides of this trade are large in AIS’s timing model. Reuse saves prefill: a 120,000-token prompt has a time to first token of 17.8s on an idle worker that lacks its prefix and 56 ms on an idle worker that already holds the first 119,808 tokens. Load costs decode speed: one decode step at 8,192 tokens of context takes 15.3ms at batch size 1 and 109.9ms at batch size 256, so each request placed on a worker slows the token stream of every other request there. Dynamo’s default router makes the trade with a fixed cost, the prefill blocks a worker would still have to compute plus its decode load (Equation (1)). Published routers draw the line elsewhere. LMetric multiplies a worker’s prefill work by its number of running requests instead of adding the two [63]. DualMap hashes a prefix of each prompt to two candidate workers and picks between them by their current state [61]. Lodestar narrows its candidates to two consistent-hash choices in the same way, but only while cluster memory utilization is above a threshold, 80% in its evaluation [23]. SMetric keeps a session’s follow-up turns on the worker that holds its context while that worker can still meet a latency deadline [54].

These rules disagree, and which one wins depends on the traffic. LMetric’s authors report that the product beats linear scores tuned for each workload, Dynamo’s among them, on their traces [63], whose ideal prefix reuse SMetric’s authors put at about 50–70%. On an agentic workload with 82% ideal reuse and no global KV tier, which is the setting of our replay, SMetric’s authors measured LMetric 36% below a tuned linear score in throughput within SLO, because its reuse ratio fell to 45% against 74% [54]. Even the best linear weight moves: LMetric’s own sweep put it at 0.7 on one workload and 0.55 on another with a similar access pattern [63]. An operator has to choose one rule and tune it for a deployment whose traffic may change.

**Why the choice is hard to settle.** Three properties of the problem make the comparison expensive.

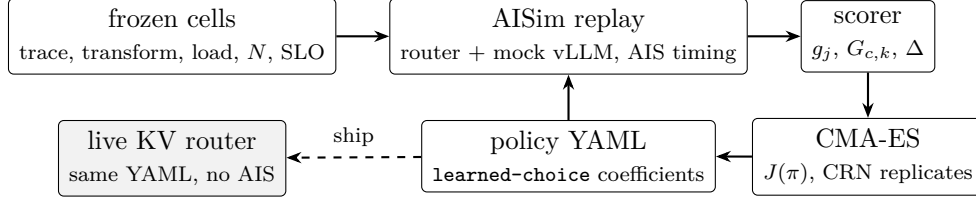
1. *No per-decision label.* A routing decision changes the cache contents and queues that later requests meet, so its quality shows only in aggregate, over minutes of traffic at a realistic load. A policy can be scored only on whole episodes.
2. *Small differences next to workload variation.* On this deployment, the two strongest tuned heuristics finished within 0.002 of each other in mean validation score, and their order flipped between the mean over cells (+0.0018) and the mean over independent segments (−0.0087) (Section 9.1). Detecting effects of a few percent needs many cells, replicates and independent workload segments.
3. *Regime dependence.* In classical load balancing, how much waiting a dispatch policy can avoid depends on the number of servers and on the load [53]. In a measured study of learned routing for disaggregated serving, the learned router tied join-the-shortest-queue at decode width 3, led at width 4 and tied round-robin again at width 6 [52]. A policy tuned at one worker count can therefore lose at another.

Settling the choice by measurement means many long runs. LMetric’s authors note that they could not sweep all configurations because replaying each trace on their testbed consumes substantial GPU time [63]. The tuning in this study is larger: 66 CMA-ES restarts of 400 evaluations each, every evaluation replaying all 34 train cells on 2 workload replicates, plus validation. Each train cell simulates 6 to 29 minutes of serving on 4 or 8 workers, that is, 8 to 16 GPUs.

**Approach.** We first describe the router’s worker-selection API (Section 2). A routing policy is a YAML instance of a catalog type that composes filters, scorers and a picker over inputs it declares: cache overlap, load, capacity and request facts. Dynamo’s default cost function and every ported heuristic in this study are such instances, and so is the learned policy, so all of them are configured, tuned and replayed through the same interface. We then use AISim offline replay as a development environment (Section 4). It runs Dynamo’s router, including the builtin policy catalog the live router loads, against mock vLLM engines in simulated time, and times every forward pass with AISimulate’s performance model for the configured model, GPU and parallelism. A 2,000-request trace replays in about 1.5s of one CPU core, and identical inputs replay identically, so competing policies can be compared on common random numbers. In that environment we tune a worker-selection function, **learned-choice** (Section 5). It is a conditional-logit choice model: it scores every candidate worker with one shared utility, linear in features the live router already computes, and picks the highest. CMA-ES tunes the coefficients on windowed SLO goodput (Section 6). The simulator only trains and scores: the headline policy reads no performance-model signal, so the YAML file that training writes runs in the live router unchanged. A separately reported league of variants may read AISimulate estimates at runtime (Section 9.10).

The choice-model form gives three properties the study relies on. It is defined for any candidate set, so a policy tuned at  $N \in \{4, 8\}$  can run at  $N = 2$  or 32 and be tested there. It contains the default router: the coefficient vector  $\theta = -\mathbf{e}_0$  reproduces the default router’s choice (Section 5.6), so training starts from the status quo and every learned coefficient reads as a change to it. And it is cheap: a decision costs  $O(Nd)$  operations for  $d$  features (8 in the base set, 23 in the extended set that won).

**How “better” is decided.** The comparison is on equal footing. Every policy, the learned arms and each of 11 tuned heuristic baselines, is one configuration tuned on the pooled train cells with



**Figure 1:** The training loop. A policy is a YAML file read by the router’s builtin policy catalog, so each CMA-ES candidate costs a new YAML and a set of replays, with no rebuild. AIS times the simulated engines and so shapes the score; the headline policy never reads it.

the same CMA-ES budget and restarts and selected on validation; the best baseline, and by the same rule the headline learned arm, are chosen on fresh validation replicates (Sections 7 and 8). The comparison is made once, on a test set frozen with SHA-256 hashes before any tuning. Its 60 cells come from 12 independent workload segments and hold out time windows, agent plays, session seeds, a trace family, transform ranges and worker counts. The headline is a one-sided exact Wilcoxon signed-rank test over the segments, registered before training. Reported but not tested are robustness to the SLO scale and definition, to stale router state (replay gives the router perfectly fresh cache and load information, a live router does not) and to engine timing, and a check of the ranking on GPUs.

**What we found.** Stated as the final audit allows: in AIS-timed simulation, the learned router M1-v2 beats the branch’s ported heuristics, each tuned with an equal budget. Against the validation-selected best baseline, tuned **ramjet**, it is ahead on 10 of the 12 test segments and on all 8 informative ones (the 4 session segments tie at ceiling), with a segment-mean clipped log-ratio of +0.0316 and one-sided exact Wilcoxon  $p = 0.0017$  (Section 9.2). It keeps first place among the robustness finalists under router-state lag of 10 to 200 ms and under four engine-timing perturbations, and it ranks first under all four alternative definitions of a good request. Four qualifications travel with the result.

- Only the sign is a claim: the effect is below the pre-registered minimum detectable effect (0.038) and below the spread the timing perturbations induce (0.060).
- It is not “learning beats every heuristic”. The baseline pool holds the branch’s ports as implemented, and its **lmetric** port lacks LMetric’s queued-prefill term. Faithful LMetric lies inside the learned model’s class; untuned, it beats every tuned baseline on validation, and with the default cost it reproduces 53–66% of the learned model’s validation margin (Section 9.11).
- It is a request-count property. A token-weighted goodput is mixed, and prompts of 32K to 64K tokens see higher time to first token than under **ramjet** (Section 10).
- Black-box search games a request-count objective. The first tuned model kept a worker whose requests were all bad; constraining load coefficients so that load cannot attract removed it, and every later learned arm runs under that constraint (Section 10).

A pre-registered live check then served six of the  $N = 4$  test cells on H100 GPUs with Dynamo’s router running the same policy files (Section 11). The live ranking of the six policies checked agrees with simulation (pooled Kendall  $\tau_K$ -b 1.00), and M1-v2 leads **ramjet** in each of the five cells where simulation predicts a lead, each time by more than the registered noise bar. The check validates the ranking and that sign only: with one run per policy on six cells chosen after the test pass, it says nothing about the headline’s magnitude, and live goodput departs from simulation by policy-dependent factors of 0.72 to 2.18, enough that on one cell the simulated gain over the default

router is not reproduced beyond live noise.

### Contributions.

- A description of Dynamo’s worker-selection API as routing primitives, with the default cost function and nine other catalog policy types written in that vocabulary, and a list of the major caveats the campaign uncovered in the API, the replay host and the workloads, each with its evidence and status (Section 2).
- **learned-choice**, a deployable worker-selection policy for Dynamo’s router plugin catalog: a discrete-choice model over router-observable features that contains the default router as one coefficient setting and reads no simulator signal (Section 5).
- A tuning method for routing policies in replay: CMA-ES on an episodic, windowed goodput objective with common-random-number workload replicates, applied with the same budget, restarts and selection rule to the learned model and to every heuristic baseline (Sections 6 and 7).
- An objective and evaluation protocol for routing studies: goodput defined by inter-token latency and end-to-end slowdown over a request’s own uncontended latency, measured in a window that excludes warm-up and drain; held-out worker counts; a pre-registered test over independent workload segments; and staged adversarial audits (Sections 3 and 8).
- Results with their scope: the headline and its strata, worker-count extrapolation, the model ladder, the learned coefficients, robustness, the objective-gaming finding and its remedy, the secondary metrics that bound the claim, and a pre-registered live GPU check of the ranking (Sections 9 to 11).

**Reading guide.** Section 2 introduces the routing primitives, the policies and the caveats. Section 3 defines the decision and the objective, Section 4 the simulator and workloads, Section 5 the policy and Section 6 its training. Section 7 describes how the baselines are tuned, Section 8 the evaluation protocol, Section 9 the results and Section 10 the concentration findings. Section 11 reports the live GPU validation. Section 12 discusses limitations and threats to validity, and Section 13 relates the work to prior research. Labels of the form  $An$  refer to the campaign contract’s numbered amendments and  $LR-n$  to the numbered lessons of its literature review; both are part of the campaign record (Section C). Every number traces to a campaign record named in a comment in the L<sup>A</sup>T<sub>E</sub>X source.

## 2 Background: routing primitives and Dynamo’s worker-selection API

Every routing rule in this study, Dynamo’s default, the ported heuristics and the learned policy, runs through one interface: Dynamo’s worker-selection API, a contract between the router *host*, which owns the request path and the shared state, and a *policy*, which turns that state into a choice. This section describes the interface as a small set of routing primitives, writes the default cost function and every other catalog policy in its vocabulary, shows how the learned policy generalizes them, and lists the major caveats the campaign uncovered along the way. The vocabulary is the one the code uses; type and method names are those of the router’s plugin API on the campaign branch.

## 2.1 The decision and what the host owns

For each request  $j$  the host builds a *candidate table*: one row per eligible worker (or worker and data-parallel rank). Eligibility is the host’s: a worker must be in the request’s allowed set, available, not marked overloaded by the host’s busy thresholds, routable, and must satisfy the request’s routing constraints. If the request is pinned to a worker, that worker is the only candidate, and a policy may also ask the host to treat a session-affinity target as exclusive, in which case an eligible target is pinned the same way. Pins and eligibility always win over a policy’s preference.

The host also owns the state a policy reads. It keeps the KV index that the engines’ cache events feed and computes each candidate’s prefix overlap with the request, per cache tier; it books the load of every request it dispatches (prefill tokens in flight, decode blocks, active requests); and it knows each worker’s configured KV capacity. A policy owns only its decision and whatever private state it chooses to keep, such as a random generator, a session map or a sliding window. This split is what makes a policy portable: the same policy file runs wherever a host can fill the inputs it declares, including the offline replay used here (Section 4.1).

## 2.2 Filters, scorers and pickers

A worker-selection policy is an ordered composition of three kinds of component (Figure 2).

**WorkerFilter** keeps or drops one eligible candidate. Filters run in declaration order and can only shrink the set.

**WorkerScorer** writes one finite, lower-is-better cost for every surviving candidate in a single call.

The host validates each contribution and adds the contributions of all scorers, so scorers compose additively.

**WorkerPicker** receives the surviving candidates with their total costs and returns one row. It may ignore the costs entirely.

None of the builtin policies in this study uses a filter; candidate restriction, where a policy needs it, happens inside its picker. Every builtin policy is therefore “scorers plus a picker”, and the interesting differences are in which inputs each reads and whether its choice for one worker depends on the other candidates.

**Declared inputs.** A component declares the optional worker signals it needs, as a set of **WorkerInputs**, and the host materializes only the union of what the policy’s components declared. Each component then sees only its own declarations, even when another component of the same policy asked for more.

**CACHE** the accounting cache estimate (effective overlap blocks and estimated cached tokens), the per-tier overlaps (device, host, disk) and shared-cache hits.

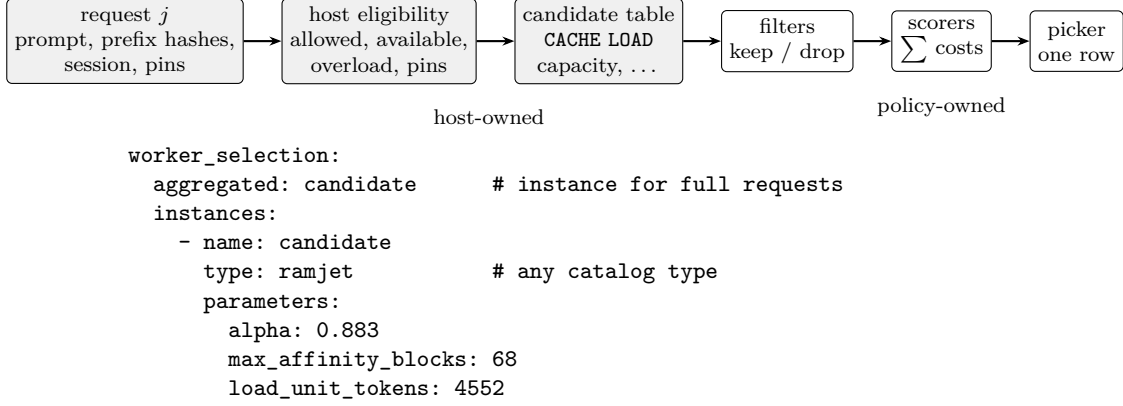
**LOAD** active prefill tokens, decode cost blocks and active requests, with a flag that separates an unobserved worker from an idle one.

**PREFILL\_TIME** the host’s modeled prefill backlog in milliseconds, available only when the router runs a prefill-load model (`router_prefill_load_model: ais`).

**OCCUPANCY** host-owned active-request counts.

**PREFERRED\_TAINT** a cost multiplier from the request’s preferred routing constraints.

No policy in this study declares **OCCUPANCY**, and only the AIS-informed variants read **PREFILL\_TIME** (Section 7.3).



**Figure 2:** Top: one worker selection. The host fixes the eligible candidates and fills the inputs the policy’s components declared; the policy’s filters, scorers and picker turn them into one row. Bottom: the policy file that selects a catalog type and its parameters; the parameters shown are the tuned **ramjet** configuration that serves as the best baseline in this study.

**Request facts.** Request-level values arrive through **WorkerSelectionContext**: the prompt length in tokens and in KV blocks and the block size; the *prefix hashes*, chained hashes of the prompt’s complete blocks, so that two requests share their first  $i + 1$  blocks exactly when their entry  $i$  is equal; whether the request is tracked for prefill load; the session context; the session-affinity target the host resolved; the expected output length, if the request carries one; each worker’s advertised KV capacity; the policy class; the request’s priorities; and a per-request router temperature override. These are the only request facts a policy can use.

**The policy catalog.** A deployment selects policies with a YAML file. Its **worker\_selection** block names one instance per worker role (aggregated, prefill, decode, encode) and lists named instances, each a catalog **type** with a **parameters** mapping (Figure 2). A registered provider for each type validates the parameters once, at startup, and returns a factory that builds one policy instance per routing partition and worker role, so every instance owns its state. Unknown types and malformed parameters fail at startup with a message naming the problem. The name **default** is reserved for the builtin selector. On this branch the offline replay host runs the same catalog; on Dynamo’s main branch at the time of the study, replay rejected catalog policies selected by YAML.

### 2.3 Dynamo’s default cost function

In the API’s terms, the default router (**dynamo-default-cost-fn** in the catalog) is one scorer and one picker. The scorer declares **LOAD**, **PREFERRED\_TAINT** and, unless its overlap credit is zero, **CACHE**, and assigns each candidate a cost in KV blocks:

$$c_i^{\text{def}} = \beta_{\text{pf}} \max\left(0, \frac{L_i^{\text{pf}} + \text{ISL}_j}{16} - \beta_{\text{ov}} \gamma_i o_i\right) + L_i^{\text{dec}} + \beta_{\text{req}} L_i^{\text{req}}, \quad (1)$$

where  $o_i$  is the candidate’s device-tier overlap in blocks,  $L_i^{\text{pf}}$  its active prefill tokens,  $L_i^{\text{dec}}$  its decode cost blocks (including this request’s new blocks),  $L_i^{\text{req}}$  its active requests, 16 the KV block size, and  $\gamma_i = (1 + \beta_{\text{decay}} (L_i^{\text{pf}} - \min_{i' \in S} L_{i'}^{\text{pf}}) / (16 b_j))^{-1}$  discounts the overlap credit of workers with more queued prefill than the least-loaded one. The shipped defaults are an overlap credit  $\beta_{\text{ov}} = 1$ , a decay  $\beta_{\text{decay}} = 0$ , a prefill load scale  $\beta_{\text{pf}} = 1$  and a decode request weight  $\beta_{\text{req}} = 0$ , so the default router minimizes the prefill blocks the worker would still have to compute plus its decode load.

The full scorer also credits host- and disk-tier hits and shared-cache hits, has a separate branch for decode-only workers, and multiplies the cost by a preferred-taint factor; none of these enter the aggregated, device-tier-only setting studied here, in which replay requests carry no taint preferences.

The picker takes the minimum cost. With a `router_temperature` above zero it instead samples from a softmax over costs normalized to their range across the candidates, so at positive temperature the default is set-dependent. Ties are broken at random; on Dynamo’s main branch the generator is unseeded, so identical replays of one cell differ, and this campaign added a `seed` parameter (Section 2.6). The picker deliberately ignores the per-request temperature override, and the default policy asks the host to treat an eligible session-affinity target as exclusive.

## 2.4 The ported cost functions as instances of the API

Table 1 writes every catalog policy type of the study in the API’s vocabulary: its components, the inputs and request facts it reads, whether its choice for one worker depends on the other candidates (set dependence), the state it keeps, and the parameters this study tunes. The pedagogical value of the table is that the policies differ along a few axes only. Most are a single picker over `CACHE` and `LOAD`; they differ in how they combine overlap with load (additive, multiplicative, thresholded or lexicographic), in whether they restrict candidates by a hash of the prompt prefix, and in whether they compare a worker with the rest of the set.

**Table 1:** Catalog policy types as compositions of API components. S: scorer; P: picker (every policy has exactly one). Inputs: declared `WorkerInputs` (C = `CACHE`, L = `LOAD`, PT = `PREFILL_TIME`, T = `PREFERRED_TAINT`). Request facts beyond the prompt length: hashes = prefix hashes, session = session context, capacity = advertised KV capacity. Set-dependent: whether a candidate’s score or eligibility depends on the other candidates at the shipped parameters.

| Type (variants)                                             | Components                                          | Inputs, request facts        | Set dependence                                                     | State            | Tuned here   |
|-------------------------------------------------------------|-----------------------------------------------------|------------------------------|--------------------------------------------------------------------|------------------|--------------|
| <code>dynamo-default-cost-fn</code>                         | S default cost; P min cost (softmax if $\tau > 0$ ) | C, L, T                      | through $\gamma_i$ if $\beta_{\text{decay}} > 0$ , and the softmax | random generator | M0: 5 knobs  |
| <code>dynamo-two-tier-cost-fn</code>                        | P lexicographic (load tier, cache tier)             | C, L                         | yes: request spread and best overlap across the set                | none             | 3 thresholds |
| <code>lmetric</code>                                        | P min of a product                                  | C, L, hashes                 | hot-spot classes (holders against non-holders)                     | class window     | 2            |
| <code>ramjet</code>                                         | P max of affinity minus load                        | C, L                         | yes: affinity relative to the warmest candidate                    | tie rotation     | 3            |
| <code>dualmap</code>                                        | P two hashed candidates, warmer unless over budget  | C, L, hashes                 | yes: candidates fixed by hash                                      | prefix window    | 3            |
| <code>chwb1</code>                                          | P hashed owner within a load bound                  | L, hashes                    | yes: bound relative to the set mean                                | none             | 2            |
| <code>llm-d-precise-prefix</code>                           | S prefix, S queue, S KV utilization; P max score    | C, L, capacity, policy class | yes: queue score scaled by the set’s range                         | tie rotation     | 4            |
| <code>llm-d-optimized-baseline</code> (throughput, modeled) | S token load; P sticky filter and TTFT gate         | C, L (PT if modeled)         | yes: sticky set and TTFT gate                                      | tie rotation     | 3            |

| Type (variants)                          | Components                                                   | Inputs, request facts                   | Set dependence                              | State                     | Tuned here           |
|------------------------------------------|--------------------------------------------------------------|-----------------------------------------|---------------------------------------------|---------------------------|----------------------|
| <b>sticky-session</b><br>(hard, bounded) | S default cost; P session’s worker or min cost               | C, L, T, session                        | bounded: request count against the set mean | session map, generator    | 5 or 6               |
| <b>thunderagent</b>                      | S prefill and decode load; P classifier’s target or min cost | L, affinity target                      | no                                          | in its request classifier | not evaluated        |
| <b>learned-choice</b><br>(this work)     | S default cost (feature 0); P utility argmax                 | C, L, T, capacity, session, hashes (v2) | v1 at rank 0: no; v2 and context terms: yes | session map, generator    | 7 to 22 coefficients |

We describe each rule in the router’s terms. For request  $j$  with candidates  $S$ , worker  $i$  has overlap  $o_i$  blocks,  $L_i^{\text{pf}}$  active prefill tokens,  $L_i^{\text{dec}}$  decode cost blocks,  $L_i^{\text{req}}$  active requests and a configured capacity of  $\text{cap}_i$  blocks; a KV block holds 16 tokens. The ports read the device-tier overlap, which in replay equals  $o_i$ , so worker  $i$ ’s uncached prompt is  $\max(\text{ISL}_j - 16 o_i, 0)$  tokens.

**Round-robin.** The router’s round-robin mode is not a catalog policy: it sends requests to the workers in turn and reads neither cache nor load. It has no parameters and serves as a floor.

**Two-tier.** `dynamo-two-tier-cost-fn` ports the thresholds and selection order of the experimental SGLang router’s `cache_aware_zmq` policy. Instead of one additive cost it applies the first of three rules that holds. (1) If the spread of active requests across candidates exceeds `balance_abs_threshold` (default 32) and the largest count exceeds `balance_rel_threshold` (1.1) times the smallest, it takes the worker with the fewest active requests. (2) Otherwise, if the largest overlap  $\max_i o_i$  exceeds `cache_threshold` (0.5) times the request’s  $b_j$  blocks, strictly, it takes the least-loaded worker among those holding that overlap. (3) Otherwise it takes the least-loaded worker. Load means  $L_i^{\text{req}}$  alone, and ties go to the lowest worker ID. The port reads Dynamo’s KV-event overlap where the SGLang router keeps its own approximate cache history.

**LMetric.** Following Zhang et al. [63], the `lmetric` port routes to

$$\arg \min_{i \in S} \max(\text{ISL}_j - 16 o_i, 1) (L_i^{\text{req}} + 1), \quad (2)$$

the product of the worker’s uncached prompt tokens and its batch size counting this request, with ties to the lowest worker ID. A product needs no weight between its factors: a weight on either one scales every candidate’s score alike. A hot-spot detector handles a popular prefix that few workers hold. Requests sharing their first `class_prefix_blocks` (4) blocks form a class, and the workers whose overlap covers those blocks hold it. A class is alarmed when the odds of its share of the last `window_requests` (1000) decisions exceed the ratio of holders to non-holders. While it is alarmed, once twice as many consecutive class requests as there are holders would have gone to a holder, later ones go to the best non-holder. The paper measures the window in time; the port counts decisions so that replay stays deterministic. The paper’s prefill factor also counts the prefill already queued on the worker,  $L_i^{\text{pf}}$ ; the port omits that term, a gap that matters for the scope of the headline (Sections 7.2 and 9.11).

**Ramjet.** The `ramjet` port follows Helix’s Ramjet load balancer [14]. Each worker scores its credited prefix affinity minus `alpha` (default 4) times its load units, and the highest score wins,

then the deeper raw overlap, then a rotating tie-break. Affinity counts the cached prefix in 512-token blocks (`affinity_block_tokens`). With the default `relative` basis, which the port’s documentation says Ramjet used in its published comparison with Dynamo, a worker is credited the warmest candidate’s affinity capped at `max_affinity_blocks` (32), minus how far it trails that candidate, also capped; differences near the leader therefore survive the cap. A worker’s load units are  $L_i^{\text{req}} + L_i^{\text{pf}}/8192$  (`load_unit_tokens`). Ramjet charges each in-flight request between 1 and 8 units according to its uncached prompt, but Dynamo exposes load per worker rather than per request, so the port approximates Ramjet’s phase-aware mode.

**DualMap.** In the `dualmap` port, following Yuan et al. [61], two rendezvous hashes of a prompt-prefix key pick two distinct candidates, so requests that share a prefix share candidates. A candidate’s backlog is  $L_i^{\text{pf}} + \max(\text{ISL}_j - 16 o_i, 0)$ , the prefill tokens it would carry after taking the request. The request goes to the candidate with more overlap unless that candidate’s backlog would exceed `pending_prefill_token_budget` (65,536 tokens); then, and whenever the two overlaps are equal, it goes to the candidate with the smaller backlog. The key hashes the first `hash_prefix_blocks` (4) blocks. A prefix is hot when its share of the last `window_requests` (1000) requests exceeds  $2/|S|$ , and a hot prefix’s key doubles in depth, up to seven times, which spreads its requests over more candidate pairs.

**CHWBL.** `chwb1` is consistent hashing with bounded loads [32], ported from KubeAI’s balancer with the defaults of SGLang’s `prefix_hash` gateway policy. The key is the prefix hash of the prompt’s first 16 blocks (`prefix_tokens` = 256), or of all its full blocks if the prompt is shorter. The worker with the highest rendezvous weight owns the key and takes the request while  $L_i^{\text{req}} \leq 1.25 (1 + \sum_{i' \in S} L_{i'}^{\text{req}})/|S|$  (`load_factor` = 1.25); otherwise the request walks the other workers in hash order to the first one within that bound, and returns to the owner if none is. The policy reads no cache state, so it routes the same with or without KV events.

**llm-d precise prefix.** `llm-d-precise-prefix` ports the precise prefix-cache profile of llm-d’s inference scheduler [25]. It is the one policy here built from several scorers: a prefix scorer, a queue scorer and a KV-utilization scorer, weighted 2:1:1, with a picker that takes the highest total. At the shipped prefix-length weight of 0 it routes to

$$\arg \max_{i \in S} 2 \min \left( 1, \frac{o_i}{\lfloor \text{ISL}_j / 16 \rfloor} \right) + \frac{\max_{i' \in S} L_{i'}^{\text{req}} - L_i^{\text{req}}}{\max_{i' \in S} L_{i'}^{\text{req}} - \min_{i' \in S} L_{i'}^{\text{req}}} + 1 - \min \left( 1, \frac{L_i^{\text{dec}}}{\text{cap}_i} \right), \quad (3)$$

where the queue term is 1 when every candidate has the same  $L_i^{\text{req}}$ , and ties rotate. A nonzero `prefix_match_length_weight` blends the first term with the squared matched length relative to `prefix_match_length_scale_tokens` (8192), and per-class weights can follow the request’s policy class, which replay never sets.

**llm-d optimized baseline.** `llm-d-optimized-baseline` ports llm-d’s optimized-baseline profile, a prefix-cache affinity filter followed by a token-load scorer [24]. Workers whose overlap covers at least `affinity_threshold` (0.8) of the prompt’s complete blocks are sticky. Each worker’s TTFT is estimated from its prefill backlog: with `ttft_source: throughput` as  $1000 L_i^{\text{pf}}/15,928$  ms, where `peak_prefill_tokens_per_second` = 15,928 is llm-d’s calibration for Qwen3-32B at TP2 on H100, and with `ttft_source: modeled` from the host’s `PREFILL_TIME` backlog. If the best sticky estimate trails the best non-sticky one by more than `max_ttft_penalty_ms` (18,000), stickiness breaks and every worker stays in play; otherwise only the sticky workers do, or all workers if none is sticky.

Among those, the request goes to the worker with the lowest token load  $L_i^{\text{pf}} + \max(\text{ISL}_j - 16 o_i, 0)$  (scored as one minus that load over `queue_threshold_tokens`, 4,194,304, capped at 1), and ties rotate. The port implements llm-d’s filter inside its picker, since the API’s filters cannot see the other candidates. The headline league uses the throughput source; the modeled source reads a performance-model estimate and belongs to the secondary AIS league (Section 7.3).

**Sticky sessions.** Dynamo’s live router can pin a session to a worker, but offline replay never sets the router’s affinity target. The campaign therefore added `sticky-session`, which keeps a bounded least-recently-used map of up to 65,536 sessions, each mapped to the worker that served its last request. A request without a session, or the first of its session, takes the default cost function’s choice, built from the router’s configured default-cost knobs. In `hard` mode a later request goes to its session’s worker whenever that worker is eligible; otherwise it takes the default choice and the session rebinds. In `bounded` mode it also leaves the session’s worker when that worker’s active requests exceed `load_factor` (1.25) times the candidates’ mean. The policy mirrors the intent of the live router’s session affinity; it is not a port of a published method. On the flat traces (Mooncake and FAST25) open-loop requests carry no session and closed-loop requests carry a single-use one, so both modes make the default choice there, apart from tie-breaking.

**ThunderAgent.** `thunderagent`’s worker selection is a scorer of each worker’s prefill tokens plus its decode blocks in tokens and a picker that follows the affinity target its paired request classifier sets, falling back to the lowest cost [19]. Its program state lives in the classifier, a separate kind of plugin that runs before worker selection. Replay runs worker-selection policies but not request classifiers, so this policy was not evaluated (Section 7.3).

## 2.5 The learned policy as a generalization

Each rule above hand-picks a functional form over a few of the declared inputs. `learned-choice` (Section 5) instead scores every candidate with one utility that is linear in a fixed vector of features computed from the same inputs, and lets the coefficients be tuned. It is built from the same parts as the rules it generalizes: the default cost function’s own scorer supplies its first feature, and a picker computes the rest and takes the argmax. Several of the heuristics are points, or limits, of its parameter space.

- $\theta = -\mathbf{e}_0$  is the default router (Section 5.6).
- A large positive session coefficient is hard stickiness (Section 5.8).
- With the extended feature set `v2`,  $\theta = -(\mathbf{e}_8 + \mathbf{e}_{10})$  takes the argmin of LMetric’s product with the queued-prefill term, and  $-(\mathbf{e}_9 + \mathbf{e}_{10})$  reproduces the branch’s `lmetric` port without its hot-spot filter (Section 5.2).
- `v2`’s `hash_home` indicator marks the two rendezvous-hashed candidates of a prompt prefix, the candidate set of DualMap and Lodestar [23, 61].
- Set-relative features and context terms let the comparison between two workers depend on the others (Section 5.4), as the queue score of `llm-d-precise-prefix` or the relative basis of `ramjet` do by construction.

What it cannot express are hard thresholds and lexicographic rules (two-tier’s tiers, DualMap’s budget, CHWBL’s bound, the optimized baseline’s TTFT gate) and history-dependent windows, which a smooth utility can only approximate. The study’s question is whether tuning a generalization of this kind, with the same budget as each hand-picked rule, finds a better trade on one deployment.

## 2.6 Major caveats uncovered by the campaign

Building, calibrating and auditing the study surfaced a number of properties of the API, of the replay host and of the workloads that can change which policy wins. Table 2 lists the major ones with their evidence and status: *fixed* (a fix exists on the campaign branch or upstream), *follow-up* (recorded for an upstream change, worked around here), or *inherent* (a property of the setting that this study can only scope). Several of them shape the results in later sections, and we refer back to them there.

**Table 2:** Major caveats the campaign uncovered. Evidence names the campaign record; “status” says what was done.

| # | Caveat                                                                       | Evidence and effect                                                                                                                                                                                                                                                                                                                                | Status                                                                                                                                                                                                                                     |
|---|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Replay’s router state is perfectly fresh.                                    | Replay applies KV events, prefill completions and request completions to the router synchronously, so policies always see exact overlap and load; a live router sees them late. Policies that lean on exact state could lose their advantage live.                                                                                                 | Follow-up: a lag knob on a separate branch; M1-v2 stays first and ahead of <b>ramjet</b> at 10–200 ms on test and ahead at 1,000 ms on validation (Section 9.9), and the live ranking agrees with replay’s on six test cells (Section 11). |
| 2 | <b>expected_output_tokens</b> equals the true output length in replay.       | The replay host fills the field from the trace’s output length, a value no live router has. A policy reading it would see the future.                                                                                                                                                                                                              | Inherent to replay; no learned feature reads it, and two leakage audits found no decision that depends on it (Section 5.1).                                                                                                                |
| 3 | Prefix hashes are unique per request when the host does not assume KV reuse. | With <b>router_assume_kv_reuse=false</b> the tracking hashes are random, so prefix-keyed policies ( <b>chwbl</b> , <b>dualmap</b> , <b>lmetric</b> ’s classes, <b>learned-choice</b> ’s <b>hash_home</b> ) lose their keys and decode-block accounting changes; turning off <b>router_track_prefill_tokens</b> zeroes the active-prefill features. | Inherent to the API contract, documented in the policy stack; the learned policy must run with the training-time host knobs (Section 5.1).                                                                                                 |
| 4 | Synthesized closed-loop session IDs reach the router.                        | For a flat trace in closed loop, replay gives every request a single-use session ID and passes it to the router; the same trace in open loop, or a live client, sends none. A session-keyed feature then behaved differently by load mode (2 against 8 workers per prefix key).                                                                    | Follow-up: a fix with a regression test is in AISimulate pull request #378, and Dynamo needs a pin bump after its release; the learned feature was re-keyed on the prompt prefix.                                                          |
| 5 | One public trace is a relabeled copy of another.                             | The FAST’25 tool-agent trace matches Mooncake on all 23,608 output lengths, maps one-to-one onto its hash blocks and scales its timestamps by 0.9825. It would have put one workload in two families and possibly two splits.                                                                                                                      | Fixed: excluded from every split (Section 4.4).                                                                                                                                                                                            |
| 6 | The default router’s tie-breaks are unseeded.                                | Identical replays of one cell gave goodput 2.8934 and 2.8290 under the catalog default, so re-runs could not be paired.                                                                                                                                                                                                                            | Fixed on the campaign branch (a <b>seed</b> parameter); follow-up for the policy stack.                                                                                                                                                    |

| #  | Caveat                                                                                            | Evidence and effect                                                                                                                                                                                                                                                                                                                    | Status                                                                                                                              |
|----|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 7  | Open-loop agent traffic has a load cliff.                                                         | With Poisson play arrivals, between 0.002 and 0.0025 plays per second per worker the number of plays in flight grows without bound at every $N$ , so no stationary open-loop load reaches the heaviest calibration band.                                                                                                               | Inherent: AgentX runs only in closed-loop lanes (Section 4.4.1).                                                                    |
| 8  | The slowdown clause has a knife edge at slowdown 1, and drain-window scoring dilutes differences. | A request that ran alone has $E2E = E_0$ exactly; float noise and an off-by-one decode context made it fail a strict comparison in a policy-dependent way (round-robin’s ratio 0.57 instead of 0.88 on one cell). Scoring closed-loop windows through the end-of-run drain diluted policy differences by 23–41%, once flipping a sign. | Fixed: $E_0$ aligned with replay’s decode context plus a $10^{-6}$ relative tolerance; windows end at full occupancy (Section 3.3). |
| 9  | Logged arrival timestamps are quantized.                                                          | Mooncake’s arrivals sit on a 3,053 ms grid, with up to 47 requests per instant, and replay delivers each burst at once at any speedup. At $N = 2$ the default’s 85th-percentile slowdown was 8.32 with bursts and 1.46 with arrivals spread inside their slot.                                                                         | Follow-up: the harness spreads arrivals per replicate (Section 4.3).                                                                |
| 10 | No AgentX play fits a 32K context.                                                                | Every one of the 82 plays that fits the 131,072-token context peaks at 44,048 tokens or more, so the deployment runs at 128K with YaRN, where AIS’s accuracy is least validated.                                                                                                                                                       | Inherent; disclosed (Section 12.1).                                                                                                 |
| 11 | The per-request temperature override is ignored by the default policy.                            | The plugin default picker reads only its own temperature; a documentation example still shows the override.                                                                                                                                                                                                                            | Intentional upstream; documentation follow-up.                                                                                      |
| 12 | The policy file could change between hashing and use.                                             | A policy file edited during a replay benchmark run could be reported under a hash that does not match the policy that ran.                                                                                                                                                                                                             | Fixed in the policy stack (#15450): the benchmark hashes the file once and rejects a change before reporting.                       |
| 13 | AISim prices a mixed prefill pass at the batch mean.                                              | Adding a short fresh prompt to a long-context chunk makes the pass cheaper than the chunk alone (0.73–0.82× in four probes), a simulator artifact a search could exploit.                                                                                                                                                              | Follow-up (not patched, it would change every cached result); the exploit check is negative for the headline policy (Section 10).   |
| 14 | Catalog policies saw no device-tier overlap in replay on Dynamo’s main branch.                    | Replay left the device-tier overlap at zero, which reduces every cache-aware port to load-only routing.                                                                                                                                                                                                                                | Fixed in the policy stack the campaign builds on, with a regression test.                                                           |
| 15 | A request-count objective can be gamed.                                                           | The first tuned learned model kept one worker whose requests were all bad (Section 10).                                                                                                                                                                                                                                                | Remedied by a sign constraint on load coefficients; the remaining concentration is reported on every learned row.                   |

### 3 Problem and objective

This section fixes what a routing policy decides, what it may observe, how load is applied, and the score by which every policy, learned or heuristic, is compared.

### 3.1 The routing decision

**Deployment.** A deployment runs  $N$  identical aggregated workers, each a vLLM engine serving Qwen3-32B with tensor parallelism 2 on H100 SXM GPUs, so prefill and decode share a worker. Each worker holds 18,863 KV blocks of 16 tokens, 301,808 tokens in all. The study covers  $N \in \{2, 4, 6, 8, 16, 32\}$ ; heterogeneous worker sets are out of scope (Amendment A5.2). The router never holds a request back: `router_queue_threshold` is left unset, which disables router-side queueing; an ablation that tuned it jointly found no measurable gain (Section 9.7).

**A sequential discrete choice.** When request  $j$  arrives at time  $a_j$ , the router’s worker-selection policy  $\pi$  picks one worker from the eligible candidate set  $S \subseteq \{1, \dots, N\}$ . Three properties shape the problem.

- *The choice is sequential.* The chosen worker computes the uncached part of  $j$ ’s prompt and keeps the resulting KV blocks, which later requests that share the prefix can reuse and which may evict other prefixes’ blocks. Request  $j$  also adds to that worker’s prefill and decode load until it completes. A decision’s cost or benefit therefore lands on other requests, often much later in simulated time, and no per-decision label exists. Policies are scored on whole episodes (Section 3.3).
- *The worker set varies.*  $N$  differs across deployments, and  $S$  can be smaller than  $N$  when the host marks workers ineligible. A policy defined on a fixed-length vector of all workers’ states would need retraining for every  $N$ . A policy that scores each candidate with one shared function and picks the best applies to any  $|S|$ . This is the discrete-choice form of the learned policy (Section 5), and it is what lets the test set hold out worker counts that training never saw.
- *The choice is made online.* It runs once per request on the router’s request path, from information available at  $a_j$ , so it must be cheap and must not depend on the future.

**What the router observes.** At decision time a policy sees, through the router plugin API (Section 2.2):

- the request: its prompt length  $ISL_j$ , its prefix block hashes and an optional session identifier;
- per candidate, the router’s cache estimate: blocks of overlap with the request’s prefix and the estimated number of cached tokens;
- per candidate, load the router tracks from its own dispatches: active prefill tokens, decode blocks and active requests;
- per candidate, its configured KV capacity in blocks.

The learned policy reads nothing else. In particular it ignores the request’s expected output length, which offline replay fills with the true output length, and every performance-model estimate; engine-internal state such as waiting-queue length or KV usage is not available to it. Because it uses only these inputs, the artifact trained in replay is the one the live router runs (Section 5.1). Replay applies cache events to the router’s index at once, while a live router sees them late; this difference is a robustness check (Section 8.7), not part of the problem definition.

### 3.2 Load modes

A cell offers its trace to the workers in one of three ways, which differ in what a policy can change.

**Open loop.** Requests arrive at the trace’s timestamps, compressed or stretched by a speedup factor that sets the load. A policy cannot change when requests arrive, only how well they are served. In multi-turn session cells the session start times are fixed this way, while each follow-up turn is released a think time after the previous turn completes, as in a real conversation.

**Closed loop.** A fixed number of sessions is in flight; a session holds its slot through its think times, and a new session starts when one ends. Each Mooncake trace row is its own one-turn session. A policy that serves requests faster also receives more of them, so throughput depends on the policy.

**Agentic lanes.** A fixed number of lanes each run one recorded agent play at a time. Inside a play, requests follow the recorded dependency graph (sequential turns, subagent spawns and joins, tool delays), and a lane starts its next play when the previous one ends.

Mooncake, FAST25 and session cells run in open and closed loop, and AgentX cells in lanes; calibration sets each cell’s load (Section 4.5). Results are reported per mode, because open and closed systems respond to scheduling differently [45]. Session and agentic work is released causally because replaying follow-up turns at their recorded timestamps would fix a release schedule, and cache state, that a different policy is supposed to change [4, 41].

### 3.3 Objective

**Good requests.** The deployment’s service objective, set by the operator, concerns the pace of the token stream and the total request latency, not the time to first token. A request is *good* iff it completes and

$$g_j = \mathbf{1}[\overline{\text{ITL}}_j \leq I_{\max}] \cdot \mathbf{1}[\text{E2E}_j \leq s_{\max} E_0(\text{ISL}_j, \text{OSL}_j)], \quad (4)$$

where  $\overline{\text{ITL}}_j = (\text{E2E}_j - \text{TTFT}_j) / (\text{OSL}_j - 1)$  is the mean inter-token latency, the ITL clause is skipped when  $\text{OSL}_j \leq 1$ , and the slowdown clause allows a relative tolerance of  $10^{-6}$  for floating-point rounding. Rejected, errored and incomplete requests are never good.  $(I_{\max}, s_{\max})$  is one pair per workload family (Table 3).  $\text{TTFT}_j$  is reported but is not part of Equation (4); it enters only through  $\text{E2E}_j$ .

**The reference latency.**  $E_0(\text{ISL}, \text{OSL})$  is the request’s latency alone on an idle worker with no prefix reuse: a full chunked prefill of ISL tokens followed by  $\text{OSL} - 1$  decode steps at batch size 1, timed by AIS. It depends only on  $(\text{ISL}, \text{OSL})$  and is cached per pair. It agrees with isolated single-request replays to within  $2 \times 10^{-8}$  relative, and an independent audit’s own computation matched it on 880,139 pairs.  $E_0$  is used only for scoring; no policy sees it.

**Why a slowdown bound.** Request sizes vary widely. Mean prompt lengths are 8,590 tokens on Mooncake and 12,035 to 15,326 on the FAST25 traces, and AgentX plays were admitted as long as every request, prompt plus output, fits the 131,072-token context. In AIS’s timing, prefill alone on an idle worker takes 55.49 ms at 1,024 tokens, 475.97 ms at 8,192, 2,550 ms at 32,768 and 17,815.54 ms at 120,000. A single absolute bound on end-to-end latency would be unattainable for the longest requests even on an idle worker, or slack for the shortest. Fixed first-token deadlines have the same problem; SMetric’s authors note that providers typically set the TTFT deadline linear in the request length [54]. Dividing by  $E_0$  scales the bound with each request’s own unavoidable work: under the Mooncake bound  $s_{\max} = 3.46625$ , a request may take up to about 3.5 times its uncontended latency, whatever its size.

Two properties of  $E_0$  keep the bound fair across policies. It is policy independent, so every policy is held to the same per-request bound. And it assumes no prefix reuse, so a cache hit lowers  $E_2E_j$  but not  $E_0$ : a policy that earns more reuse finds the bound easier to meet, never harder.

**Why two clauses.** The clauses catch different failures. For a long prompt,  $s_{\max}E_0$  is dominated by prefill time, so a slow token stream after a fast prefill can pass the slowdown clause; the ITL clause catches it. Conversely, a request that waits a long time and then streams at an ordinary pace passes the ITL clause and fails the slowdown clause. At the calibrated bounds the slowdown clause does most of the work: at the calibration load nearest L2 on  $N = 8$ , the ITL clause alone fails 0.6% of  $\pi_0$ 's requests on sessions, 1.8% on AgentX and 4.5% on Mooncake, while 15% of requests are not good at L2 by construction.

**Windowed goodput.** A cell's score is its windowed goodput  $G_{c,k}(\pi)$ , good requests per simulated second inside a measurement window  $\mathcal{T}_c$  that starts after a warm-up. Its form depends on the load mode. In open-loop cells the window is a fixed interval of arrival times, and

$$G_{c,k}(\pi) = \frac{1}{|\mathcal{T}_c|} \sum_{j: a_j \in \mathcal{T}_c} g_j, \quad (5)$$

so a request that arrives in the window counts even if it finishes after it, and one that never completes counts as not good. In closed-loop and lane cells,  $G$  counts the good requests that *complete* inside a half-open window, divided by its length. That window ends at full occupancy, the last instant at which every session or lane slot is busy, so its end, and hence its length, depends on the policy.

The window removes three distortions.

- *Drain.* The replay's native goodput divides good requests by the makespan, which includes the drain after the last arrival, so it mixes SLO attainment with how fast a policy clears the tail. In closed loops and lanes, completions after full occupancy happen with fewer requests in flight, and in lane cells the lanes run out of plays at different times, so counting the tail would credit easy, under-loaded requests.
- *Cold start.* Caches start empty and session populations ramp up, so the first minutes of a replay are not the steady state the cell represents. The warm-up length matters: for open-loop session cells, a first calibration with a 180s warm-up gave  $\pi_0$  a good fraction that moved by up to 0.353 when the warm-up history was doubled (9 of 12 cells beyond 3 standard errors); with the adopted warm-up of two session lifetimes, doubling moves it by at most 0.014 (no cell beyond 3 standard errors).
- *Deferral.* An objective that ignored requests finishing after the horizon would reward deferring work past it; Decima's learned scheduler did exactly this under a fixed horizon [30]. Counting in-window arrivals that finish late removes the incentive.

When every arrival is set by the trace, as in the open-loop Mooncake and FAST25 cells, the number of in-window arrivals is the same for every policy on a replicate, so the ratio of two policies' goodputs equals the ratio of their in-window good fractions. In closed-loop and lane cells goodput also rewards throughput, which the policy legitimately controls there. SLO attainment as the primary serving metric follows Zhong et al. [65].

**SLO and load calibration.** The bounds  $(I_{\max}, s_{\max})$  are fixed per family from runs of  $\pi_0$  on train segments at  $N = 8$ , and each cell's load is one of three levels L1, L2, L3, at which  $\pi_0$ 's good

fraction is 0.95, 0.85 and 0.65. Levels are calibrated separately for every  $N$  and transform, on train segments only, and both bounds and levels were frozen before any policy was tuned (Section 4.5). The bounds differ widely by family, from  $s_{\max} = 1.16134$  on AgentX to 3.46625 on Mooncake, because each is anchored at its own family’s load knee. Every cell therefore sits where the default router is starting to miss its objective, the regime in which routing changes outcomes; that the regime is defined relative to one baseline is a threat discussed in Section 12.

**Normalized score.** Goodput levels differ by orders of magnitude across cells, so every comparison is a ratio on the same cell and replicate:

$$\rho_{c,k}(\pi, \pi') = \frac{G_{c,k}(\pi) + \varepsilon}{G_{c,k}(\pi') + \varepsilon}, \quad \Delta_{c,k}(\pi, \pi') = \text{clip}(\log \rho_{c,k}(\pi, \pi'), -\log 3, \log 3), \quad (6)$$

with  $\varepsilon = 10^{-3}$  good requests per second. Both policies run on the same workload replicate  $k$  with the same tie-breaking seed (common random numbers, Section 6.3). Training uses  $\pi' = \pi_0$  (Equation (10)); the headline uses  $\pi' = \pi_{\text{base}}^*$  (Equation (11)). The log makes the score symmetric, so doubling goodput on one cell and halving it on another nets to zero. The clip limits any one cell to a factor of 3 either way, which matters because a mean of raw ratios is dominated by the few cells where one policy collapses [2], and such cells occur here. On all 9 open-loop session cells of the train and validation splits, round-robin’s goodput is 0.111 to 0.425 times the default router’s, and below 1/3 on 5 of them. A policy that far past its stability limit has no steady state, so the ratio is not a stable quantity: on one cell, doubling the warm-up history halved it, from 0.118 to 0.055. The clip scores every ratio below 1/3 alike, so that unstable detail cannot drive training or the headline. The clipped log-ratio was chosen over the mean of ratios at calibration, before any training.

**What “better” means.** The headline compares the learned policy with the best baseline *on the same footing*: each policy, learned or heuristic, is one configuration tuned on the pooled train split with the same CMA-ES budget and selected on validation (Sections 6 and 8). The best baseline  $\pi_{\text{base}}^*$  is the baseline with the highest mean clipped log-ratio on fresh validation replicates. The learned side is chosen by the same rule among the learned variants that read only router-observable signals, and the number of variants compared is reported. The comparison is made once, on the frozen test set, by the pre-registered test of Section 8.4. Because the objective itself is a modeling choice, the same test records are also re-scored with scaled bounds and with alternative definitions of a good request: ITL alone, slowdown alone, and the ITL bound combined with either a length-scaled TTFT bound or an absolute latency bound. These secondary analyses never change the headline.

### 3.4 Notation

Table 3 fixes the symbols used throughout. The campaign’s contract, the policy’s specification (FEATURES.md) and the harness each grew their own names, and several collide. We resolve the collisions as follows.

- **The letter  $S$ .**  $S$  is the set of eligible candidate workers for one request, as in FEATURES.md ( $\bar{x}_S, z_{S,\ell}$ ). The contract’s end-to-end slowdown bound, also called  $S$  there, is written  $s_{\max}$  here. For symmetry the contract’s mean-ITL bound  $I$  is written  $I_{\max}$ .
- **The letter  $k$ .**  $k$  indexes common-random-number (CRN) workload replicates, as in the contract (Amendment A1) and the harness (repeat). FEATURES.md indexes context terms by  $k$ ; here they are indexed by  $\ell = 1, \dots, r$ .

- **The letter  $B$ .**  $B$  is the CMA-ES evaluation budget per run. The request’s block count, which `FEATURES.md` calls  $B$ , is  $b_j$  here.
- **The letter  $T$ .** `FEATURES.md`’s token normalizer  $T = 8192$  is  $T_{\text{ch}}$ . The plan’s TTFT threshold  $T$  no longer exists (Amendment A2 dropped the TTFT SLO).
- **$e$  and  $E$ .**  $\mathbf{e}_f$  (bold) is a standard basis vector, so the parity anchor is  $\theta = -\mathbf{e}_0$ ;  $E_0$  is a latency.
- **$\tau$ .**  $\tau$  is the policy temperature; Kendall’s rank correlation is  $\tau_K$ .
- **$\sigma$  and segments.**  $\sigma$  belongs to CMA-ES (step size  $\sigma$ , initial  $\sigma_0$ ); workload segments are  $\omega \in \Omega$ . Ratios are  $\rho$ , restarts  $R$ .

**Table 3:** Notation. “Value” is the setting in this campaign; code and file names are in typewriter. Pilot values are *preliminary (pilot, validation split)*.

| Symbol                         | Meaning                                                                                                                                                             | Value / code name                                                                                    |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <i>Deployment and requests</i> |                                                                                                                                                                     |                                                                                                      |
| $N$                            | number of identical workers (one Qwen3-32B TP2 vLLM engine each)                                                                                                    | $\{2, 4, 6, 8, 16, 32\}$ ; train $\{4, 8\}$ , val $\{4, 6, 8\}$ , test all; <code>num_workers</code> |
| $j$                            | request index                                                                                                                                                       |                                                                                                      |
| $a_j$                          | arrival time of request $j$ (simulated)                                                                                                                             |                                                                                                      |
| $\text{ISL}_j, \text{OSL}_j$   | prompt and output length (tokens)                                                                                                                                   | <code>input_length</code> , <code>output_length</code>                                               |
| $b_j$                          | request blocks, $\max(\lceil \text{ISL}_j/16 \rceil, 1)$ ( <code>FEATURES.md</code> : $B$ )                                                                         | KV block size 16; <code>request_blocks</code>                                                        |
| $\text{TTFT}_j, \text{E2E}_j$  | time to first token; end-to-end latency (simulated)                                                                                                                 | <code>per_request</code> rows                                                                        |
| $\overline{\text{ITL}}_j$      | mean inter-token latency,<br>$(\text{E2E}_j - \text{TTFT}_j)/(\text{OSL}_j - 1)$ , for $\text{OSL}_j > 1$                                                           |                                                                                                      |
| $E_0(\text{ISL}, \text{OSL})$  | uncontended no-reuse latency: the request alone on an idle worker, full prefill, then OSL decode steps at batch 1, timed by AIS. Evaluation only; no policy sees it | <code>ais-chunked-estimator-v2</code> ;<br><code>learned_routing/e0.py</code>                        |
| $I_{\max}$                     | mean-ITL bound of the SLO (contract: $I$ )                                                                                                                          | Mooncake and FAST25 252.117 ms;<br>sessions 46.0478 ms; AgentX 38.6205 ms                            |
| $s_{\max}$                     | end-to-end slowdown bound (contract: $S$ )                                                                                                                          | Mooncake and FAST25 3.46625;<br>sessions 2.11784; AgentX 1.16134                                     |
| $g_j$                          | good-request indicator, Equation (4)                                                                                                                                | <code>learned_routing/goodput.py</code>                                                              |
| <i>Policy</i>                  |                                                                                                                                                                     |                                                                                                      |
| $S$                            | eligible candidate set of one request; $ S  \leq N$                                                                                                                 | host eligibility                                                                                     |
| $i$                            | a candidate worker, $i \in S$                                                                                                                                       |                                                                                                      |
| $x_i \in \mathbb{R}^d$         | feature vector of candidate $i$ , fixed order (Table 8)                                                                                                             | $d = 8$ (v1) or 23 (v2); <code>feature_set</code>                                                    |
| $\bar{x}_S$                    | candidate mean of $x$ over $S$                                                                                                                                      |                                                                                                      |
| $\mathbf{e}_f$                 | standard basis vector; $\theta = -\mathbf{e}_0$ reproduces the default router                                                                                       |                                                                                                      |
| $T_{\text{ch}}$                | token normalizer of the prefill features ( <code>FEATURES.md</code> : $T$ )                                                                                         | 8192                                                                                                 |
| $\theta \in \mathbb{R}^d$      | linear coefficients                                                                                                                                                 | <code>parameters.theta</code> ; $\theta_0 = -1$<br>pinned in training                                |
| $r, \ell$                      | context rank; context-term index $\ell = 1, \dots, r$ ( <code>FEATURES.md</code> : $k$ )                                                                            | $r = 0$ (M1), $r = 2$ (M2);<br><code>len(context.p)</code>                                           |

| Symbol                                                                            | Meaning                                                                                                                   | Value / code name                                                                             |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| $p_\ell \in \mathbb{R}^d$                                                         | context direction of term $\ell$                                                                                          | <code>context.p</code>                                                                        |
| $z_{S,\ell}$                                                                      | context value of term $\ell$ : a named source (Table 10) or, in the pooled form, $q_\ell^\top \bar{x}_S$                  | <code>context.sources</code> or <code>context.q</code>                                        |
| $w_S$                                                                             | effective weights, $\theta + \sum_\ell z_{S,\ell} p_\ell$                                                                 | <code>features.rs</code>                                                                      |
| $u_i$                                                                             | utility of candidate $i$ , $w_S^\top x_i$                                                                                 |                                                                                               |
| $\tau$                                                                            | temperature: $\tau = 0$ is argmax with a seeded tie-break, $\tau > 0$ samples $\text{softmax}(u/\tau)$                    | <code>temperature</code> ; fixed at 0 for M1 and M2                                           |
| $\pi, \pi_\theta$                                                                 | a routing policy; the learned policy                                                                                      | one policy YAML each                                                                          |
| $\pi_0$                                                                           | <code>default@defaults</code> : <code>dynamo-default-cost-fn</code> at shipped parameters; the normalizer of every ratio  | harness name <code>default</code>                                                             |
| $\pi_{\text{base}}^*$                                                             | the validation-selected best baseline (Section 8)                                                                         | tuned <code>ramjet</code>                                                                     |
| <i>Workloads, metric and tests</i>                                                |                                                                                                                           |                                                                                               |
| $c$                                                                               | cell: one trace segment, transform, load mode and value, $N$ and SLO                                                      | <code>cell_id</code>                                                                          |
| $\mathcal{C}_{\text{train}}, \mathcal{C}_{\text{val}}, \mathcal{C}_{\text{test}}$ | cells of each split                                                                                                       | 34, 14, 60                                                                                    |
| $\omega, \Omega_{\text{split}}$                                                   | segment (unit of statistical independence); segments of a split                                                           | 10, 6, 12                                                                                     |
| $\mathcal{T}_c$                                                                   | measurement window of cell $c$ (Section 3.3)                                                                              | <code>measure</code>                                                                          |
| L1, L2, L3                                                                        | load levels at which $\pi_0$ 's good fraction is 0.95, 0.85, 0.65                                                         | <code>load.level</code>                                                                       |
| $k$                                                                               | CRN replicate index; replicate $k$ also sets policy seed $k + 1$                                                          | <code>repeat</code>                                                                           |
| $G_{c,k}(\pi)$                                                                    | windowed goodput (good requests per simulated second)                                                                     | <code>goodput_rps_window</code>                                                               |
| $\varepsilon$                                                                     | offset in goodput ratios                                                                                                  | $10^{-3}$ ; <code>--eps</code>                                                                |
| $\rho_{c,k}(\pi, \pi')$                                                           | $(G_{c,k}(\pi) + \varepsilon) / (G_{c,k}(\pi') + \varepsilon)$ ; $\pi' = \pi_0$ unless stated                             | <code>pair_score</code>                                                                       |
| $\Delta_{c,k}(\pi, \pi')$                                                         | clipped log-ratio, $\text{clip}(\log \rho, -\log 3, \log 3)$                                                              | <code>clipped_log_ratio</code>                                                                |
| $D_\omega$                                                                        | per-segment headline statistic, Equation (11)                                                                             | <code>facts/HEADLINE_TEST.json</code>                                                         |
| MDE                                                                               | minimum detectable effect, $\max(1\%, 2 \times \text{validation segment SE})$                                             | 0.038, preliminary (pilot, validation split)                                                  |
| $\delta$                                                                          | router-state lag, robustness check only                                                                                   | <code>{10, 50, 200} ms</code> on test; <code>router_state_lag_ms</code>                       |
| $\tau_K$                                                                          | Kendall rank correlation of two policy rankings (a perturbed against the nominal condition; simulation against live GPUs) | <code><math>\tau</math>-b</code>                                                              |
| <i>Training</i>                                                                   |                                                                                                                           |                                                                                               |
| $J(\pi)$                                                                          | training objective, Equation (10)                                                                                         | <code>--objective clipped_log_ratio</code>                                                    |
| $\phi$                                                                            | CMA-ES internal coordinates of the free parameters (bounded ones in $[0, 1]$ )                                            | <code>space.yaml</code>                                                                       |
| $t$                                                                               | CMA-ES generation                                                                                                         | <code>0, ..., 24</code>                                                                       |
| $\lambda$                                                                         | population size                                                                                                           | <code>16</code> ; <code>--popsize</code>                                                      |
| $B$                                                                               | objective evaluations per run (FEATURES.md's $B$ is $b_j$ )                                                               | <code>400 = 25 \times 16</code> ; <code>--budget-evals</code>                                 |
| $K, K_{\text{pool}}$                                                              | replicates per objective evaluation; the pool they rotate through                                                         | <code>2</code> and <code>8</code> ; <code>--replicates</code> , <code>--replicate-pool</code> |
| $\mathcal{K}_t$                                                                   | replicates of generation $t$ , $\{(tK + i) \bmod K_{\text{pool}} : i < K\}$                                               | <code>Trainer.eval_set</code>                                                                 |

| Symbol           | Meaning                                                      | Value / code name      |
|------------------|--------------------------------------------------------------|------------------------|
| $K_{\text{val}}$ | replicates per validation score inside <code>lr-train</code> | 3 ( $k = 0, 1, 2$ )    |
| $\sigma_0$       | initial CMA-ES step size, in internal units                  | per policy (Section 6) |
| $R$              | restarts per policy                                          | 3                      |

## 4 Simulator and workloads

Every number in this paper comes from offline replay: AISim runs Dynamo’s KV router in front of  $N$  simulated vLLM engines, in simulated time, on recorded or generated request traces. This section describes that environment and the workloads it replays, so that a reader can judge which results could change on a live deployment. Three properties carry the rest of the paper. Engine timing comes from AIS, AISimulate’s performance model, and not from the mock engine’s fallback model (Section 4.1). A replay is a deterministic function of the policy, the cell and a replicate index, so every comparison between two policies is paired on identical workloads (Section 4.3). And the cells, their service objective and their loads were fixed on train segments with the default router alone, then frozen with SHA-256 hashes before any policy was tuned (Sections 4.5 and 4.6). What replay leaves out is listed in Section 4.2.

### 4.1 Offline replay with AIS timing

AISim’s offline replay is a discrete-event simulation in virtual time [37, 38]. The event loop, the mock engines and the report collector come from the `aisimulate-core` crate; Dynamo contributes the adapter that places each arriving request through its KV router and feeds the engines’ KV-cache events back into the router’s index. Simulated time jumps from one event to the next (an arrival, an engine step, a completion), so the outcome of a replay does not depend on how fast the host runs it. Each mock engine runs `aisimulate-core`’s model of the vLLM scheduler: continuous batching, chunked prefill under a per-step token budget, and prefix caching over a paged KV cache. The duration of each forward pass comes from AIS, an operator-level performance model of the configured model, GPU, backend and parallelism. Batch-level simulators of this kind are an established tool for sizing LLM serving [3].

Worker selection in replay goes through the same Rust policy catalog as in the live router, and a policy is configured by the same policy YAML. Replay rejects unknown policy types and unknown parameters by name, and the harness writes every policy YAML in one canonical form. Running catalog policies inside offline replay is a feature of the branch this campaign builds on; on Dynamo’s main branch, replay rejects custom policies.

AIS times the engines only. The router-side AIS estimator, which some router modes use to model prefill load, is off: the policies under study see the same router state a live router without a performance model would see (Section 3.1). Amendment A16 later added a separate, secondary league of policies that may read AIS-derived estimates at runtime; it runs on its own build, with the router-side estimator on, and is described in Sections 7.3 and 9.10.

**Engine facts.** Table 4 lists the deployment. Setup confirmed that replay uses AIS timing: a single 1,024-token request replayed alone has a TTFT of 55.49 ms, equal to the AIS estimator’s prediction to within 1 ns, and an 8,192-token request 475.97 ms. The fallback model underestimates long prompts badly (7.2 times at 120,000 tokens), so a replay without AIS would mislead about exactly the long-context traffic this study cares about. Prefill cost per token grows with the context already in the KV cache: an 8,192-token chunk costs 59.5 ms per thousand tokens with no prefix and

**Table 4:** The simulated deployment. Top: engine configuration of every worker (single source of truth: `config/engine.json`). Bottom: time to first token of one request replayed alone on an idle worker, with AIS timing and with the mock engine’s built-in polynomial model, which replay falls back to when the bindings lack AIS support. Every replay in this paper uses AIS timing.

|                |                                                                                                                             |  |  |  |  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
| Model, backend | Qwen/Qwen3-32B on vLLM 0.24.0, aggregated (prefill and decode on one worker)                                                |  |  |  |  |
| Hardware       | H100 SXM ( <code>h100_sxm</code> ), tensor parallelism 2                                                                    |  |  |  |  |
| Timing         | AIS, operator-level estimation; router-side AIS off                                                                         |  |  |  |  |
| KV cache       | block size 16; 18,863 blocks (301,808 tokens) per worker, pinned to the AIS estimate so every replay uses the same capacity |  |  |  |  |
| Context        | <code>max_model_len</code> 131,072 (YaRN)                                                                                   |  |  |  |  |
| Batching       | <code>max_num_seqs</code> 1,024; <code>max_num_batched_tokens</code> 8,192; chunked prefill and prefix caching on           |  |  |  |  |
| Workers        | $N \in \{2, 4, 6, 8, 16, 32\}$ identical workers                                                                            |  |  |  |  |
| Versions       | <code>ai-dynamo-runtime</code> 1.6.0; <code>aisimulate</code> 0.13.0.dev2026093000000000061                                 |  |  |  |  |

| Prompt length (tokens)         | 1,024 | 8,192  | 32,768   | 65,536   | 120,000   |
|--------------------------------|-------|--------|----------|----------|-----------|
| TTFT, AIS (ms)                 | 55.49 | 475.97 | 2,550.45 | 6,838.33 | 17,815.54 |
| TTFT, polynomial fallback (ms) | 32.49 | 169.14 | 676.55   | 1,353.10 | 2,476.95  |

249.6 ms per thousand after a 114,688-token prefix. A decode step takes 15.1 ms for one 2,048-token sequence and 26.5–29.2 ms for about 262,000 tokens of context in the batch, whether they are 128 sequences of 2,048 tokens, 32 of 8,192 or 8 of 32,768. One request at the 131,072-token context limit would occupy 43% of a worker’s KV cache, and the mean AgentX prompt (66,313 tokens) 22%, so four or five AgentX contexts fill a worker. Replay rejects a request whose prompt reaches the context limit and silently truncates the output of one whose prompt plus output exceeds it; the trace transforms therefore drop or clamp such rows explicitly (Section 4.4).

## 4.2 What replay does not model

Each item below is a difference between replay and a live deployment that could change which policy wins; Table 2 gives their status. Section 12 weighs them, and the robustness and live checks of Sections 8.7 and 11 probe the first and the last directly.

- **Router state is perfectly fresh.** Replay applies the engines’ KV events to the router’s index synchronously; a live router sees them late. Load counters come from the router’s own request tracking in both settings, but whether their update timing matches is an assumption. A replay patch that delays these updates is used only as a test-time robustness check (Section 8.7).
- **One cache tier.** Replay models only the GPU (device) KV cache. A live router with host or disk tiers blends them into its overlap estimate, a signal no policy saw in training.
- **Fields a live router would not have.** Replay fills the request’s expected output length with the true output length, and in closed-loop mode it gives every flat trace row a single-use session identifier and passes it to the router, which open-loop mode and a live client do not. No policy in the headline reads the output length (Section 5.1); a single-use session identifier never has a previous request, so the session-affinity feature is unaffected, and the one experimental feature that keyed on session identifiers was changed to key on the prompt prefix.
- **AIS fidelity** [hypothesis about the size of the effect]. AIS times a prefill pass that mixes requests from the mean new and mean cached length of the batch, so adding a short prompt

to a long-context chunk can shorten the predicted pass (from 1,744 ms to 1,150 ms in one probe). Its attention table for this configuration covers prompts up to 16,384 tokens with no cached prefix, so longer prefills are extrapolated. The likely effect is a small optimism when short prompts share a worker with long-context prefills; a search procedure could exploit it, which matters most for AgentX.

- **Errors near capacity.** Calibration places every cell near the default router’s throughput knee (Section 4.5), which is where routing matters and also where small timing errors are most amplified [3]. This study does not validate AIS’s absolute accuracy; it relies on replay to rank policies, which the live check tests.
- **No network, no failures, identical workers,** and, in open-loop cells, arrivals that do not react to latency (Section 4.4.2).

### 4.3 Determinism and common-random-number replicates

**Determinism.** For a fixed policy, workload and policy seed, a replay reproduces every per-request row exactly, across repeated runs in one process and across fresh processes; request UUIDs only reorder the output. The one source of run-to-run variation found at setup was the default picker’s unseeded tie-break. This branch adds a `seed` parameter to `dynamo-default-cost-fn`, and the harness always passes one, so every policy in this study is deterministic given its seed.

Bit-exact results also need the same operating-system image, because replay calls the host’s math library. Each distinct remote CPU node image (operating system and C library) was checked against the workstation before use: on the first check, 28 of 28 remote records and all 186,548 per-request rows matched a fresh local run.

**Where noise comes from.** Because a replay is deterministic, running it twice measures nothing; the setup audit rejected the original plan to estimate noise that way. What is arbitrary in a recorded workload is the order of requests logged at the same instant. The Mooncake trace has 1,180 distinct timestamps in its hour, a grid of about 3,053 ms, and up to 47 requests share one; the generated sessions use a 1 s grid. Replay delivers each burst at once and places it in file order.

**Replicates.** Replicate  $k$  of a cell re-draws only that arbitrary part of the trace, and the same replicate is used for every policy (common random numbers):

- **Arrival spread** (`crn-spread-v1`; Mooncake, FAST25 conversation and sessions). Each session’s first arrival moves to a uniform random point inside its logging slot (3,053 ms, 3,000 ms and 1,000 ms for the three traces); sessions are re-sorted by the new times and later turns keep their completion-relative delays.
- **Order permutation** (`crn-order-v1`). FAST25 synthetic, whose timestamps are nearly unique, permutes the sessions that tie; AgentX permutes the order of its recycled play copies, which re-deals them to lanes (Section 4.4.1).
- **Policy seed**  $k + 1$  for every policy that takes a seed.

The replicate seed is a hash of the protocol, the source trace’s SHA-256 and  $k$ . Cells that share a trace (other  $N$ , other load level) therefore share their replicates, and every replicate, including  $k = 0$ , is a fresh draw, so the recorded file order has no special status.

Spreading arrivals inside their slot is a modeling choice, not only a noise source. Replay delivers a logged burst at once whatever the arrival speedup, and with load matched per worker a worker’s

**Table 5:** Source traces as recorded or generated, before windows, transforms or recycling. Reuse is the mean fraction of a request’s leading blocks already seen earlier in the trace, an infinite-cache hit rate; for AgentX it is computed within each play, since plays never share prefixes. Sessions are pooled over the ten 720 s generator seeds.

| Source              | Requests | Span        | Prompt tokens |        |         | Output tokens |     | Reuse |
|---------------------|----------|-------------|---------------|--------|---------|---------------|-----|-------|
|                     |          |             | mean          | p50    | p90     | mean          | p50 |       |
| Mooncake            | 23,608   | 60 min      | 8,590         | 6,345  | 16,787  | 182           | 30  | 0.63  |
| FAST25 conversation | 12,031   | 59 min      | 12,035        | 6,909  | 27,367  | 343           | 350 | 0.38  |
| FAST25 synthetic    | 3,993    | 17 min      | 15,325        | 11,587 | 38,615  | 149           | 69  | 0.42  |
| Sessions (10 seeds) | 36,170   | 12 min each | 6,542         | 5,539  | 11,932  | 341           | 249 | 0.83  |
| AgentX (82 plays)   | 3,132    | —           | 66,313        | 66,944 | 103,488 | 749           | 314 | 0.87  |

share of each burst grows as  $N$  shrinks. With bursts kept, the default router’s 85th percentile slowdown at  $N = 2$  and 4,000 input tokens per second per worker was 8.32; with arrivals spread it was 1.46. Without the spread, small- $N$  cells would measure burst queueing that the logging grid created.

Replicates move goodput by amounts comparable to the effects of interest. On three uncalibrated 2,000-request Mooncake cells, goodput varied across 8 replicates with a coefficient of variation of 1.2–3.9% for every policy tested. On the calibrated cells, the standard deviation of round robin’s per-replicate goodput ratio against  $\pi_0$ , pooled by family,  $N$  and load mode, ranges from 0.004 to 0.045. Replicates re-draw one segment; they do not create new independent workloads (Section 8).

**Cost.** Replay is cheap enough for black-box search. A 2,000-request Mooncake replay takes about 1.5 s of one CPU core in a fresh process, nearly flat in  $N$  from 2 to 32, and the full 23,608-request trace 7.6–8.8 s. AgentX traffic costs 2.9–4.9 ms per request, so an AgentX cell at  $N = 32$  takes 2–4 minutes. Results are cached under a key built from the policy, the cell’s content, the replicate, the harness version and the replay build, so a rebuilt binding or an edited cell can never return a stale result. Section 6.9 gives the campaign’s compute.

#### 4.4 Workloads

The workloads differ in where prefix reuse comes from, because that is what a cache-aware router exploits: shared system prompts across users (Mooncake, FAST25), a conversation’s own growing history (sessions), or long per-agent contexts carried across tool calls (AgentX). Table 5 summarizes the sources and Table 6 how they are split. A *segment* is a disjoint piece of one source, such as a time window, a generator seed or a subset of agent plays. Segments are the unit of statistical independence: cells cut from one segment at different  $N$ , loads or transforms count as one workload (Section 8).

**Mooncake.** The public Mooncake trace [21] holds 23,608 single-turn requests over one hour, with prefixes given as 512-token hash blocks under four distinct first blocks. It is cut into six disjoint 10-minute slices, each a 4-minute warm-up followed by a 6-minute measurement window: train w0, w2 and w4, validation w1, test w3 and w5. An earlier design used overlapping 12-minute slices, in which each window’s warm-up was the previous window’s measurement; an audit found that half of test window w3’s scored rows were also replayed in train cells, and the slices were redesigned before calibration. No row now appears in two slices.

**Table 6:** Segments and cells per split. Open: open-loop arrivals; closed: fixed concurrency; lanes: fixed number of concurrently running agent plays. FAST25 conversation shares Mooncake’s arrival and length skeleton, so its windows are aligned to Mooncake’s w3 and w5 and counted as those segments.

| Family              | Segments                 | Load mode | Cells |     |      |
|---------------------|--------------------------|-----------|-------|-----|------|
|                     | train / val / test       |           | Train | Val | Test |
| Mooncake            | w0, w2, w4 / w1 / w3, w5 | open      | 12    | 3   | 18   |
|                     |                          | closed    | 6     | 2   | 7    |
| Sessions            | s0–s3 / s4, s5 / s6–s9   | open      | 6     | 3   | 9    |
|                     |                          | closed    | 2     | 2   | 6    |
| AgentX              | A1–A3 / V1–V3 / T1–T4    | lanes     | 8     | 4   | 14   |
| FAST25 conversation | – / – / (w3, w5)         | open      | –     | –   | 2    |
|                     |                          | closed    | –     | –   | 1    |
| FAST25 synthetic    | – / – / x0, x1           | open      | –     | –   | 2    |
|                     |                          | closed    | –     | –   | 1    |
| All                 | 10 / 6 / 12              |           | 34    | 14  | 60   |

**FAST25 conversation and synthetic.** Two traces from the Mooncake project’s FAST’25 release [21] are test-only families. The conversation trace shares Mooncake’s arrival grid and its (timestamp, prompt length, output length) skeleton on 91% of rows but has a different prefix structure (reuse 0.38 against 0.63, one shared first block); its windows are aligned to Mooncake’s w3 and w5 and are counted as those segments. The synthetic trace is independent of both and has nearly unique timestamps; it is cut into two windows of 8.5 minutes, each with a 3-minute warm-up. Its longest prompts exceed the context limit and are dropped by the context cap (Section 4.4.3).

**A trace we excluded.** A further trace from that release, the tool-agent trace, is excluded from every split. It is a relabeled copy of the Mooncake trace: its output lengths match on all 23,608 rows, its hash blocks map one-to-one onto Mooncake’s over 409,356 aligned blocks with no conflict, and its timestamps are Mooncake’s scaled by 0.9825. Including it would have put the same workload in two families, and possibly in two splits.

**Synthetic sessions.** Replay’s built-in synthetic-session source cannot express a conversation: every request has the same prompt and output length, every inter-turn delay is the same constant, and a later turn does not extend the earlier turns’ context, which leaves session affinity nothing to exploit. We therefore generate multi-turn sessions and write them as a Mooncake session trace. Sessions start as a Poisson process at one per second, floored to a 1 s grid. Each draws one of 16 system prompts with Zipf popularity (exponent 1.1; lognormal length, median 2,000 tokens). Every turn resends the whole conversation so far plus a new user message, so its hash blocks extend the previous turn’s. Turns per session are geometric with mean 5 (at most 24); user messages are lognormal (median 600 tokens for the first, 250 later) and outputs lognormal (median 250, at most 2,000), and a session ends before its context would exceed 49,152 tokens. Each later turn is released an exponential think time (mean 8 s) after the previous turn *completes*, so release is causal: a slow policy delays the session’s next request, as it would with a real user. One generator seed defines one segment (s0–s3 train, s4 and s5 validation, s6–s9 test). For the calibrated cells each seed was regenerated with a 48,000 s duration; the generator is sequential, so the 720 s trace is a byte prefix of the long one, which adds independent sessions from the same seed rather than copies.

**AgentX.** AgentX plays come from a public corpus of recorded agent sessions [46] (`semianalysis ai/cc-traces-weka-062126`, revision `23f152f6`; 393 plays, 98,827 requests). A play is one agent run: a dependency graph of requests, including subagents that run in parallel and join, with recorded tool and think gaps between them. We keep the 82 plays in which every request satisfies  $\text{prompt} + \max(\text{output}, 1) \leq 131,072$ , whole and unchanged: 3,132 requests with 39 explicit subagent groups and a mean prompt of 66,313 tokens. The plays were recorded with other models and are replayed as Qwen3-32B traffic, so they supply realistic structure (lengths, reuse, dependencies, gaps), not Qwen3-32B’s behavior. Recorded gaps include users stepping away for hours: 140 of 2,666 gaps (in 56 of 82 plays) exceed 300s, the longest 355,532s. Every AgentX trace caps idle gaps at 300s with a monotone time warp that keeps every busy period, request duration and ordering; uncapped, the 82 plays would simulate for 399,658s instead of 11,942s, mostly idle. The plays are dealt, by a seeded round robin that places plays with subagents first, into ten disjoint segments: A1–A3 (11 plays each) for training, V1–V3 and T1–T4 (7 each) for validation and test.

#### 4.4.1 Recycling AgentX plays at any worker count

Replaying the plays natively cannot hold a steady AgentX load on many workers. In timestamp mode every play’s root starts at time zero; in lanes mode the plays are dealt to lanes once and never reused, so 82 plays cannot keep 16 or 32 workers busy, and the first design had no AgentX cell above  $N = 8$ . We therefore convert each play once into AISim’s Agentic Mooncake v2 trace format and recycle copies of it (Amendment A3).

- **Lowering.** The conversion is AISim’s own: a small Rust tool calls the public Weka importer of the `aisimulate-core` version the bindings link, so the rows are the importer’s output, not a reimplemention.
- **Copy isolation.** Each copy of a play gets fresh play, session and request identifiers and its own consecutive range of hash identifiers. Replay interns hash identifiers globally across a file, so without the new range two copies of one play would share KV prefixes; with it, copies never do. Dependency edges, delays and lengths are copied unchanged.
- **Draws.** Each cell draws its copies with replacement from its own segment’s plays, so segments remain independent workloads.
- **Lanes.** Every AgentX cell runs closed: a fixed number of lanes per worker, each lane running its dealt copies back to back, with many more copies than lanes so that every lane stays busy through the measurement window. The lowering also supports open-loop Poisson play arrivals, but open-loop AgentX has a cliff between 0.002 and 0.0025 plays per second per worker, past which the number of plays in flight keeps growing; no stationary open-loop load reaches the L3 band, so no cell uses it.

The conversion preserves replay semantics exactly where it can be compared with native replay. For all 82 plays and each of the six play transforms the cells use, the native graph and the converted file compile to the same canonical graph digest. Replaying six plays that cover subagents, joins and concurrent spawns (alone, as one corpus, and as a 12-copy recycled corpus, at  $N = 2$  and 4, under the seeded default and round robin) gave identical per-request records in 84 of 84 cases for every transform. The order of copies within a file is a free choice: the same recycled corpus in a different copy order changed mean end-to-end latency by  $-7.4\%$  to  $+4.7\%$  for the untransformed plays, and by up to  $-15.6\%$  and  $+18.0\%$  under the output-length transforms, in a synchronized-start worst case at  $N = 2$ . That is the same kind of variation as a replicate, and the replicate protocol re-draws it (Section 4.3).

#### 4.4.2 Load modes

Each cell uses one of three load modes. *Open loop* replays the trace’s arrival times divided by a speedup factor. *Closed loop* keeps a fixed number of sessions in flight, starting the next one when one finishes. *Agentic lanes* keep a fixed number of agent plays running. Each family has a primary mode, used to set its service objective: open loop for Mooncake and sessions, lanes for AgentX. Open and closed systems respond to scheduling differently [45], so every result is reported per mode. AgentX plays average 38 requests each, well past the point at which a partly-open system behaves as a closed one, which is one more reason lanes are its natural mode.

Replay’s speedup divides inter-turn delays as well as arrival times. In the first session cells this meant that think time shrank with load and with  $N$ : at matched per-worker load the mean think time was 8s at  $N = 2$  and about 0.3s at  $N = 32$ . Session cells now pre-multiply each delay by the speedup, so a user’s think time is the same at every load and every  $N$ ; only the session arrival rate scales. Release is causal in both multi-turn families: a session’s next turn waits for the previous turn to complete, and an AgentX request for the requests it depends on [41]. Open-loop arrivals of flat traces do not react to latency, a known bias of trace-driven evaluation [4].

#### 4.4.3 Transforms

Transforms vary the workload along axes a deployment might drift on, so that the test set can ask whether a policy extrapolates. They are deterministic and seeded, applied in a fixed order, and each derived trace is stored under the SHA-256 of its bytes with a record of its specification, sources and realized statistics.

- **Prompt length, shared and unique parts separately.** A block that occurs in two or more requests is *shared*; it is expanded by the same seeded per-hash draw wherever it occurs, so hash identity stays consistent. Each request’s *unique* suffix is scaled on its own.
- **Output length.** Each OSL is multiplied, rounded and kept at least 1.
- **Prefix roots.** The shared root segment of the requests that start with the same block (a system prompt) is split into several disjoint hash-space copies (two to four here), and each request family below it goes to one copy; sharing within a family is unchanged, but there are proportionally more distinct system prompts.
- **Think time.** Session delays are scaled; AgentX plays are dilated in time about their first request, which scales every recorded gap and keeps every ordering.
- **Context cap.** Rows whose prompt reaches 131,072 tokens are dropped and outputs are clamped so that prompt plus output fits (AgentX plays, which cannot lose a request without breaking their graph, clamp outputs only).

Train cells stay within  $[1, 1.5]$  for both prompt multipliers,  $[0.5, 2]$  for output and think time, and  $[1, 2]$  for prefix roots. Ten test cells go beyond these ranges: prompt multipliers of 2.5, output multipliers of 0.25, 2.5 and 4, four prefix roots, and think-time multipliers of 0.25 and 4.

### 4.5 SLO and load calibration

Before any policy was tuned, calibration fixed each family’s SLO pair  $(I_{\max}, s_{\max})$  and each cell’s load, using only  $\pi_0$  (`default@defaults`) and round robin, and only train segments. Calibration has to solve two problems at once. Routing only matters in a band of load: far below capacity every policy meets the SLO, far above it every policy fails. And the SLO and the band depend on each

other, because whether a load is “heavy” depends on the SLO it is scored against. The rules below fix both, and Figure 3 shows the result at  $N = 8$ .

**Load levels.** L1, L2 and L3 are the per-worker loads at which  $\pi_0$ ’s windowed good fraction, averaged over the train segments and two replicates, falls to 0.95, 0.85 and 0.65 under the family’s SLO. Each level is interpolated linearly in log-load between the two grid points that bracket it. Loads are expressed per worker (offered input tokens per second for Mooncake-format open loop, sessions per second for session open loop, concurrent sessions for closed loop, lanes for AgentX) and multiplied by  $N$ , rounding to an integer for closed loop and lanes.

Matching load per worker does not hold the routing regime fixed across worker counts. Under join-the-shortest-queue, waiting vanishes as the number of servers grows at a fixed per-server load [53], and one measured learned router tied a simple baseline, led it and tied it again as the pool widened [52]. Levels are therefore calibrated separately at every  $N \in \{2, 4, 6, 8, 16, 32\}$ . They differ: Mooncake’s open-loop L2, for example, is 7,477 input tokens per second per worker at  $N = 2$  and 8,993 at  $N = 16$ . Calibration therefore ran  $\pi_0$  and round robin at the held-out worker counts, but only on train segments, and it replayed no test trace. Transformed cells get their own levels, calibrated the same way with the transform applied to the train segments, because a transform changes how much load a worker can carry. The held-out FAST25 families inherit Mooncake’s per-worker levels and SLO. At  $N = 16$  and 32 the AgentX grid starts at 3 lanes per worker, where  $\pi_0$ ’s good fraction is already 0.916 and 0.934, so AgentX has no L1 there and no cell uses one.

**The SLO as a fixed point.** The SLO of Equation (4) has two bounds per family, set at  $N = 8$  in the family’s primary mode:

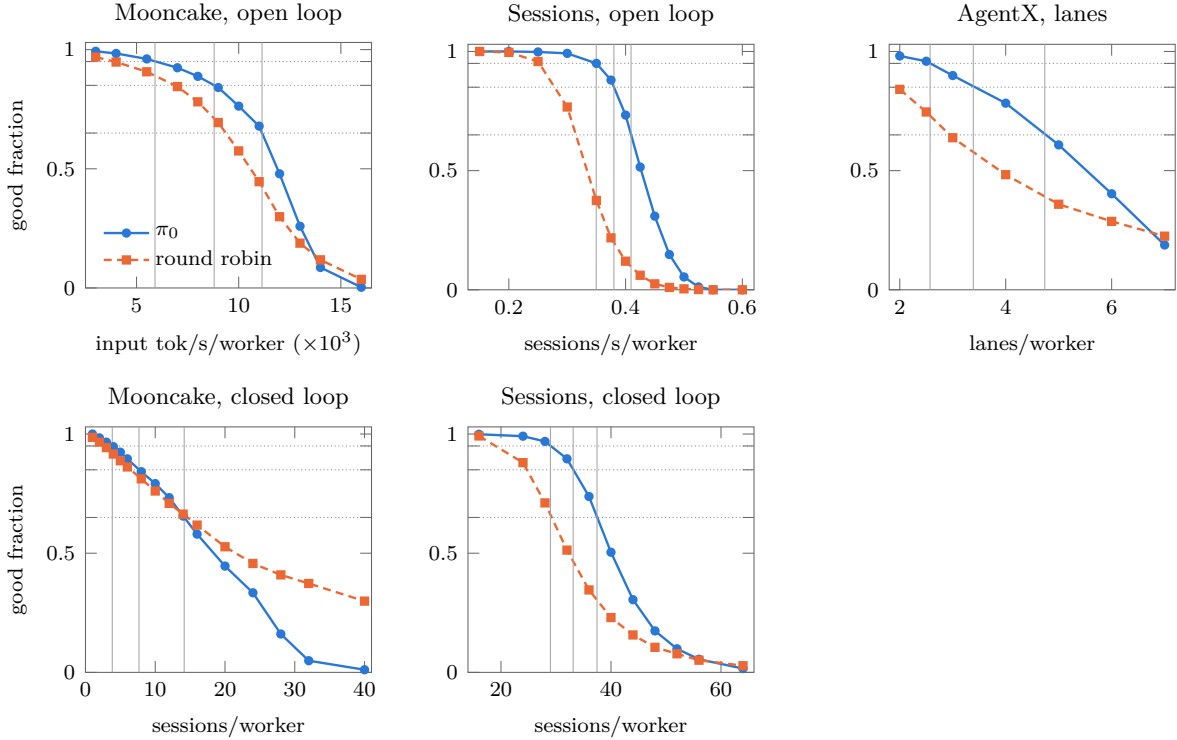
- the *knee* marks where  $\pi_0$  starts to saturate: in open loop, it is the load at which the mean slowdown of the last third of in-window arrivals exceeds the first third’s by more than 15%; in lanes, the load at which completions per second reach 0.95 of the grid’s maximum (sessions use a different rule, below);
- $s_{\max}$  is chosen so that  $\pi_0$ ’s good fraction is 0.65 at 0.9 times the knee, which makes that load L3;
- $I_{\max}$  is 1.2 times the 95th percentile of  $\pi_0$ ’s per-request mean ITL at L2.

L2 depends on  $(I_{\max}, s_{\max})$  and  $I_{\max}$  depends on L2, so the three are iterated to a fixed point; for Mooncake,  $I_{\max}$  went from 260.31 to 253.78 to 252.12 ms. The 10% headroom below the knee matters. A first version anchored L3 at the knee itself, which gave Mooncake  $s_{\max} = 5.69$  and left two train cells near the open-loop cliff with good fractions of 0.243 and 0.018; with the headroom they are 0.548 and 0.817. The resulting values are in Table 7. At L2 the ITL bound alone fails only 0.6–4.5% of  $\pi_0$ ’s requests, so most misses come from the end-to-end slowdown bound.

**Reading the curves.** Figure 3 shows what the levels mean. In the primary modes the bands are steep: on sessions,  $\pi_0$ ’s good fraction falls from 0.95 to 0.65 between 0.350 and 0.409 sessions per second per worker, and round robin, which ignores where a conversation’s history is cached, is already at 0.198 at L2. On Mooncake’s closed loop the two curves stay within 0.04 of each other up to L3 and cross just below it. Calibration fixes where the cells sit; it does not guarantee that every cell separates policies. The bands are targets for the mean over train segments, not for each cell: on individual train and validation cells,  $\pi_0$ ’s realized good fraction ranges from 0.423 to 0.965. One intended spread was not achieved. Measured as the distinct prompt tokens touched in 100s of

**Table 7:** Calibrated service objective per family and per-worker loads at  $N = 8$  (rounded). The SLO applies to every split, load mode and  $N$  of its family; the FAST25 families inherit Mooncake’s. The last column is round robin’s good fraction at  $\pi_0$ ’s L2 in the primary mode, a measure of how much routing matters at that load.

| Family   | $I_{\max}$ (ms) | $s_{\max}$ | Mode   | Unit        | Per-worker load at $N = 8$ |       |        | RR good<br>frac. at L2 |
|----------|-----------------|------------|--------|-------------|----------------------------|-------|--------|------------------------|
|          |                 |            |        |             | L1                         | L2    | L3     |                        |
| Mooncake | 252.117         | 3.46625    | open   | input tok/s | 5,903                      | 8,798 | 11,142 | 0.711                  |
|          |                 |            | closed | sessions    | 3.84                       | 7.67  | 14.15  |                        |
| Sessions | 46.0478         | 2.11784    | open   | sessions/s  | 0.350                      | 0.380 | 0.409  | 0.198                  |
|          |                 |            | closed | sessions    | 29.0                       | 33.1  | 37.5   |                        |
| AgentX   | 38.6205         | 1.16134    | lanes  | lanes       | 2.57                       | 3.39  | 4.74   | 0.572                  |



**Figure 3:** Calibration load curves at  $N = 8$  on train segments: the fraction of in-window requests that are good (Equation (4)) under the family’s final SLO, for  $\pi_0$  (blue, solid) and round robin (orange, dashed), averaged over segments and two replicates each (6 runs per point for Mooncake and AgentX, 8 for sessions). Dotted horizontal lines mark 0.95, 0.85 and 0.65; vertical lines mark the calibrated L1, L2 and L3, where  $\pi_0$ ’s curve crosses them. Top row: each family’s primary mode, which also anchors its SLO.

replay divided by the workers’ total KV capacity, every one of the 55 open-loop cells has a cache pressure between 0.62 and 2.75, so no split contains a cell with little cache pressure.

**Warm-up and measurement windows.** Each cell’s measurement window  $\mathcal{T}_c$  (Section 3.3) starts after a warm-up that lets caches and queues fill:

- **Open loop, flat traces:** the slice’s 4-minute warm-up and 6-minute window (FAST25 synthetic: 3 minutes and 5.5), divided by the cell’s speedup; scored by arrival time.
- **Closed loop:** sessions that arrive during the warm-up are excluded by identity, and the window ends at full occupancy, the last instant all sessions are busy; for generated sessions, the first twice-the-concurrency sessions are warm-up and the next ten times the concurrency are measured.
- **AgentX lanes:** the warm-up is the 90th percentile of the recorded span of the cell’s plays (1,825 s to 13,191 s of simulated time across cells), and each lane gets the fewest copies among 8, 10, 12, 16 and 24 for which its least-loaded lane still covers the warm-up plus two mean play spans in every replicate  $k < 16$ .
- **Open loop, generated sessions:** a 540 s window after a warm-up long enough for the population of open sessions to settle, described next.

**The session warm-up fix.** An independent re-run of the first calibration found one major problem. Open-loop session cells opened their window after a fixed 180 s warm-up, while the session population was still growing: by a policy-free lifetime model, at 180 s it had reached 0.78 of its plateau at the base think time and 0.54 at four times the think time. The windows therefore mixed a lightly loaded ramp with saturation. With a 900 s warm-up, one train cell labeled L3 had a  $\pi_0$  good fraction of 0.435 instead of 0.814, and the light ramp had also biased the sessions knee and so the sessions SLO. The fix sets the warm-up from session lifetimes. A session’s envelope lifetime is its summed think time (scaled by the cell’s think multiplier) plus three times its summed uncontended latency  $E_0$ ; the warm-up is twice the 99th percentile of that lifetime over the train sessions, rounded up to a minute and at least 180 s. That gives 960 to 2,160 s, growing with the think multiplier. One lifetime (600 s at the base think time) was tried first and failed the acceptance check. The check replays the same in-window sessions with twice the warm-up history: across the 12 open-loop session cells that calibration replays (train and validation cells, plus calibration-only cells on train segments at other  $N$ ),  $\pi_0$ ’s good fraction changed by  $-0.0006$  (standard error 0.0019, no cell beyond three standard errors) and round robin’s goodput ratio to  $\pi_0$  by  $+0.0074$ , against  $+0.061$  and  $+0.217$  for the original 180 s warm-up on the same sessions. With the population settled, the within-window drift test no longer detected the knee: it placed the sessions knee at 0.510 sessions per second per worker, although at 0.475 and 0.5 doubling the history still moved  $\pi_0$ ’s good fraction by 15–17%. The sessions knee is therefore the load at which doubling the warm-up history lowers  $\pi_0$ ’s good fraction by more than 5%, solved jointly with the SLO fixed point (0.4548 at  $N = 8$ ). The sessions SLO moved from  $I_{\max} = 40.4165$  ms,  $s_{\max} = 1.90627$  to the values in Table 7. Only session cells changed (8 train, 5 validation and 15 test cells); the other families’ SLOs, levels and cells are byte-identical to the first calibration. A second independent re-run then passed.

Calibration took 13,532 replays (about 3.3 hours on one 24-core host), and the session fix another 9,792.

## 4.6 Splits and the frozen test set

The splits follow the segments of Table 6. Training uses  $N \in \{4, 8\}$ , validation adds  $N = 6$ , and the test set covers all six worker counts. Every test cell lies on a test segment, so it is held out at least by workload; each also has a primary hold-out axis:

- **time window:** Mooncake w3 and w5 at  $N \in \{4, 8\}$ ;
- **workload draw:** session seeds s6–s9;
- **agent plays:** AgentX segments T1–T4;
- **family:** FAST25 conversation and synthetic, never trained on;
- **transform extrapolation:** the ten cells beyond the train ranges (Section 4.4.3);
- **worker count:**  $N \in \{2, 6, 16, 32\}$ , where  $N = 2, 16$  and  $32$  are extrapolation and  $N = 6$ , which validation also uses, is labeled selection-exposed.

Section 8.1 gives the number of test cells per axis and per  $N$ . Calibration added four AgentX test cells at  $N = 16$  and  $32$ , which recycling made possible.

**Freeze.** After calibration the cells were frozen as `lr-cells-v5`. The freeze record holds the SHA-256 of the test cell file (`7b998b8e...`, 60 cells), of each of the 60 test cells’ contents, of the 39 trace files they replay (2.13 GB) and the 7 sources those were derived from, of the engine configuration and of the split manifest, together with the replay build’s identifier. The rule is that nobody evaluates a test cell before the test stage, and that stage verifies every hash before its single pass. The test set was frozen twice. The first freeze (`b8648ac5...`) was replaced after the session warm-up fix; that was allowed because none of the 35,735 result records then on disk touched a test cell, trace or split. At the pilot gate the pilot had produced no record that touches the test set, and the test file still matched the freeze. The test stage verified all 149 freeze hashes before its single pass.

## 5 The learned policy

`learned-choice` takes over the final step of Dynamo’s worker selection. Given the eligible candidates for one request and the signals the router already tracks for them, it scores each candidate with a linear utility over a handful of features and routes to the highest score. Three constraints shape the design. It reads only inputs the live router has, so the coefficients tuned in replay can run unchanged in a live router (Section 5.1). Its coefficients are shared by all candidates and its features do not change when the candidate set is duplicated, so one coefficient vector applies at every worker count, including counts that training never sees (Section 5.5). And one coefficient vector,  $\theta = -\mathbf{e}_0$ , reproduces the default router, so tuning starts from the status quo and every learned coefficient reads as a deviation from it (Section 5.6). The normative specification is `FEATURES.md` next to the Rust code; this section follows it.

### 5.1 Router integration

In the API of Section 2, `learned-choice` registers under the type string `learned-choice` as one scorer and one picker. The scorer is the default router’s own cost function evaluated at default weights; its output becomes feature 0 (Section 5.2). The picker (a `WorkerPicker`) computes the other features, the utilities and the choice. A provider deserializes and validates the parameters

```

worker_selection:
  aggregated: candidate
  instances:
    - name: candidate
      type: learned-choice
      parameters:
        feature_set: v1
        theta: [-1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0] # = default router
        temperature: 0.0
        tie_break: one_draw
        seed: 1 # replay sets k + 1; omit in production
        # M2 adds a named context term:
        # context:
        #   sources: [<source 1>, <source 2>]
        #   p: [[8 floats], [8 floats]]

```

**Figure 4:** A learned-choice policy YAML at the parity anchor  $\theta = -\mathbf{e}_0$  (restart s1). The same file is a replay input and a live router configuration; unknown keys and wrong vector lengths fail at startup. The selected M1-v2 file has 23 coefficients (Section 9.8).

once at startup and returns a factory that builds one policy instance per routing partition and worker role; each instance owns its random generator and its session map.

The picker declares the plugin inputs **CACHE** and **LOAD**. Per request it reads the prompt length  $ISL_j$ , the request block count  $b_j$ , the block size, the prefix block hashes (feature set v2 only) and the session identifier, if any. Per candidate  $i \in S$  it reads the router’s cache estimate (overlap blocks  $o_i$  and estimated cached tokens  $h_i$ ), the router’s own load tracking (active prefill tokens  $L_i^{\text{pf}}$ , decode cost blocks  $L_i^{\text{dec}}$  including this request’s new blocks, active requests  $L_i^{\text{req}}$ ) and the configured KV capacity  $\text{cap}_i$  in blocks. Host pins and eligibility always win: a pinned request goes to its pinned worker, and only eligible workers are candidates.

The coefficients are ordinary parameters in the policy YAML (Figure 4), so a new coefficient vector is a new file rather than a rebuild. Each CMA-ES candidate in training is one such file, and the file that training selects is the file a live router would load. Parsing is strict: unknown keys at any level, a vector of the wrong length, a non-finite coefficient, a negative or non-finite temperature, a context with both pooled and named forms, a context direction without either, an unknown or repeated context source, and **max\_sessions**: 0 all fail at startup with a message naming the problem. The unit tests check 21 invalid and 5 valid specifications through the same registry path the Python bindings use, and the build audit checked 67 invalid and 15 valid ones independently.

**Cost per decision.** The picker fills an  $|S| \times d$  feature table, folds the context into one effective weight vector ( $rd$  operations), takes one dot product per candidate and makes one pass for the maximum. Before that it sorts the candidates by worker ID, which fixes the visiting order (Section 5.3). A decision therefore costs  $O(Nd + rd)$  arithmetic plus an  $O(N \log N)$  sort; feature set v2 adds three more sorts and  $N$  rendezvous hashes. Feature 0 is the default scorer’s output, which the default router computes anyway. Apart from scratch buffers that are reused across decisions, the only state is the session map: at most **max\_sessions** entries (default 65,536), updated in logarithmic time. The campaign did not measure the per-decision latency of the Rust policy against **dynamo-default-cost-fn**; the bound above is from the code.

**Table 8:** Feature set `v1`, in order.  $T_{\text{ch}} = 8192$  tokens, the engine’s chunked-prefill batch;  $b_j = \max(\lceil \text{ISL}_j / 16 \rceil, 1)$ . Feature 0 is computed by the default scorer itself at default weights, so  $\theta = -\mathbf{e}_0$  reproduces the default router.

| $f$ | Name                                 | $x_{i,f}$                                                      |
|-----|--------------------------------------|----------------------------------------------------------------|
| 0   | <code>default_logit_scaled</code>    | $c_i^{\text{def}}/b_j$ at default weights                      |
| 1   | <code>overlap_frac</code>            | $o_i/b_j$                                                      |
| 2   | <code>new_prefill_tokens_k</code>    | $\max(\text{ISL}_j - h_i, 0)/T_{\text{ch}}$                    |
| 3   | <code>active_prefill_tokens_k</code> | $L_i^{\text{pf}}/T_{\text{ch}}$                                |
| 4   | <code>kv_load_frac</code>            | $L_i^{\text{dec}}/\text{cap}_i$                                |
| 5   | <code>active_requests_s</code>       | $L_i^{\text{req}}/32$                                          |
| 6   | <code>session_affinity</code>        | 1 if $i$ served the previous request of $j$ ’s session, else 0 |
| 7   | <code>isl_x_prefill_load</code>      | $(\text{ISL}_j/T_{\text{ch}})x_{i,3}$                          |

**Excluded inputs.** The policy never reads the request’s expected output length, which offline replay fills with the true output length, so reading it would give the policy future information that no live router has. It never reads the modeled prefill backlog (the router’s `PREFILL_TIME` input) or any other estimate from AIS, the performance model that times the simulated engines; that keeps the policy independent of the simulator it is trained in, so at runtime it needs nothing a live router lacks. Two independent audits checked these exclusions. The first replayed the first 2,000 requests of a Mooncake trace at  $N = 4$  and, from a probe arrival burst onward, either multiplied every output length by four or deleted the requests. None of the 965 decisions made up to and including the probe burst changed, for three **learned-choice** configurations (`v1` with a named context, `v1` with a pooled context at  $\tau = 0.3$ , and `v2`) as well as for the default and sticky-session policies. The same perturbation started 26 bursts earlier, as a positive control, changed 210 to 286 of those decisions. The second audit reimplemented the feature map from the specification and matched the policy’s choice in all of 18,000 random-coefficient trials across the three context forms, detected 30 of 31 deliberately mis-specified feature maps, and found that 16,000 pairs of decisions that differed only in excluded inputs never differed in outcome (positive controls changed 272 to 744).

**Porting notes.** Three properties matter when the YAML leaves replay. First, like the default policy, **learned-choice** asks the host to treat an eligible session-affinity target as exclusive. Replay never sets such a target, but a live router with session affinity enabled pins every later turn of a session to its target, so the decision never reaches the picker and the learned session coefficient  $\theta_6$  is replaced by hard stickiness. Shipping the learned behavior requires host session affinity to be off, or accepting hard stickiness in its place. Second, the router settings that track prefill tokens and assume KV reuse change what features 0, 3, 4 and 7 mean (and the `v2` prefix key), so the live router must keep the values used in training. Third, without a `seed` each policy instance draws fresh entropy, so router replicas do not break ties in lockstep. How replay’s information differs from a live router’s (fresher cache state, one cache tier) is discussed in Section 4.1.

**The default cost it starts from.** Feature 0 is the default cost function’s own cost (Equation (1), Section 2.3) at the default weights, computed by the default scorer that **learned-choice** composes as its one scorer.

## 5.2 Features

Table 8 defines the eight v1 features, which have been frozen since calibration began. They are built only from the inputs listed in Section 5.1.

- **The anchor.** Feature 0 is the default cost divided by the request’s block count. The division does not change the default’s argmin, because  $b_j$  is the same for every candidate. Because the default scorer is composed into the policy, the cases it distinguishes (missing cache tiers, unavailable load, untracked prefill) are handled exactly as the default handles them, and the host’s own score-weight settings do not change feature 0.
- **Cache.** Features 1 and 2 come from the router’s accounting estimate. In replay the effective overlap equals the device-tier overlap and cached tokens are whole blocks. Feature 1 is not clamped; it stays at most 1 whenever the estimate does not exceed the prompt.
- **Load.** Features 3 to 5 come from the router’s own request tracking. Feature 4 divides by the capacity a worker advertises, or by 65,536 blocks when it advertises none; replay advertises the engine’s 18,863 blocks, so the fallback is unused here.
- **Session.** Feature 6 needs state. Replay never sets the router’s own affinity target, so the policy keeps a bounded LRU map from session identifier to the worker it last chose for that session, pinned choices included. It is 0 for every candidate when the request has no session or the session is new or evicted.
- **Prompt length.** A request quantity alone, such as  $ISL_j$ , is the same for every candidate and cannot change an argmax. Prompt length matters only through features that combine it with a worker quantity (features 0, 1, 2 and 7; feature 7 is the explicit product of prompt length and queued prefill) or through the context term (Section 5.4).

The features are dimensionless, but they are not on a common scale. On candidate tables reconstructed from the default router’s train replays in the pilot, the within-set standard deviation ranged from 0.060 (`active_requests_s`) to 6.6 (`default_logit_scaled`); the tables are approximate, because replay does not log candidate tables. Training therefore bounds each coefficient relative to its feature’s spread (Section 6.2).

**Reading the coefficients.** With  $\theta_0 = -1$ , as in M1 and its variants,  $u_i = -c_i^{\text{def}}/b_j + \sum_{f \geq 1} \theta_f x_{i,f}$ , so every term is measured in default-cost blocks per request block. Two consequences help when reading tuned values. When the clip in Equation (1) is inactive, that is, when the overlap credit does not exceed the prompt’s blocks plus the queued prefill blocks,  $c_i^{\text{def}}$  contains  $-o_i$ , so  $\theta_1 x_{i,1} = \theta_1 o_i/b_j$  adds  $\theta_1$  blocks of credit per overlapping block to the default’s one:  $\theta_1 > 0$  is more cache-seeking than the default. And features 0 and 1 are per request block while features 2–7 are not, so at fixed  $\theta$  the anchor’s differences between candidates shrink as  $1/b_j$  relative to the load features: the balance between the default cost and the learned load terms depends on prompt length by construction.

**Feature set v2.** Table 9 lists 15 features that v2 appends to the eight of v1, which keep their indices. They are used by the M1-v2 arm (Section 5.9). They add three things v1 cannot express. Log-scale load terms make products of loads linear in the utility, so LMetric’s multiplicative rule [63] lies inside the model class:  $\theta = -(\mathbf{e}_8 + \mathbf{e}_{10})$  routes to the minimum of  $(L_i^{\text{pf}} + (ISL_j - h_i)^+)(L_i^{\text{req}} + 1)$ , LMetric’s product with queued prefill, and  $\theta = -(\mathbf{e}_9 + \mathbf{e}_{10})$  reproduces the branch’s `lmetric` port without its hot-spot filter when cached tokens are whole overlapping blocks, as in replay; a unit test checks both on 1,000 random tables. A prefix-hash home indicator marks two hash-chosen

**Table 9:** Features that v2 appends to v1.  $(y)^+ = \max(y, 0)$ ;  $(\bar{x}_S)_f$  is the candidate mean of feature  $f$ . Indices 12–20 apply one transform to each of features 4, 3 and 5, in that order; the ratio floors are 0.01,  $512/T_{\text{ch}}$  and  $1/32$ , about one unit of each load.

| $f$   | Name                   | $x_{i,f}$                                                                                                                                                                                                 |
|-------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8     | log_ptok               | $\ln(\max(L_i^{\text{pf}} + (\text{ISL}_j - h_i)^+, 1)/1024)$                                                                                                                                             |
| 9     | log_new_prefill        | $\ln(\max((\text{ISL}_j - h_i)^+, 1)/1024)$                                                                                                                                                               |
| 10    | log_bs                 | $\ln(1 + L_i^{\text{req}})$                                                                                                                                                                               |
| 11    | hash_home              | 1 if $i$ has one of the two highest rendezvous weights for the request’s prefix key (the hash of the prompt’s first 256 tokens); 1 for all candidates when $ S  \leq 2$ ; 0 for all without prefix hashes |
| 12–14 | *_ratio                | $x_{i,f}/(\text{floor}_f + (\bar{x}_S)_f)$                                                                                                                                                                |
| 15–17 | *_below_frac           | $ \{i' \in S : x_{i',f} < x_{i,f}\} / S $                                                                                                                                                                 |
| 18–20 | *_excess               | $(x_{i,f} - (\bar{x}_S)_f)^+$                                                                                                                                                                             |
| 21    | new_prefill_x_requests | $x_{i,2} x_{i,5}$                                                                                                                                                                                         |
| 22    | prefill_attn           | $x_{i,2} (\text{ISL}_j/T_{\text{ch}} - x_{i,2}/2)$ , the attention work of prefilling the uncached suffix, in units of $T_{\text{ch}}^2$                                                                  |

candidates per prompt prefix, the consistent-hash candidate membership of DualMap and Lodestar [23, 61]. Set-relative load terms make a worker’s load count relative to the other candidates; they enter nonlinearly, because a linear set-relative term would be a no-op (Section 5.4). The ratio form follows divisive normalization [56]. The product of new prefill and active requests stands for the prefill–decode interference that two routers model [17, 48], and the attention term is SMetric’s prefill cost [54].

The home indicator keys on the prompt prefix only. An earlier version keyed on the session identifier when one existed, but replay gives every request of a closed-loop flat trace a synthetic single-use session identifier that a live client would not send, so the feature meant different things in open and closed loop on the same trace; the build audit caught this. Session continuity is feature 6’s job. On Mooncake the prefix key takes only four distinct values, with about 85% of requests on two of them, so a large positive  $\theta_{11}$  concentrates load; the evaluation audits worker shares (Section 8.8).

A later operator decision (contract Amendment A16) admits a third set, v3, built on a separate branch and build: the eight v1 features plus five per-worker estimates from AIS, in seconds. They are this request’s predicted prefill time for its uncached tokens on the worker (`est_prefill_ms_s`); the worker’s remaining modeled prefill backlog, which counts only the unfinished part of in-progress prefills (`prefill_backlog_ms_s`); their sum, an estimated TTFT (`est_ttft_ms_s`); the decode-step time once the request joins (`est_decode_step_ms_s`); and the per-step increase the request imposes on the requests already on the worker (`itl_externality_ms_s`). The backlog is the host’s PREFILL\_TIME input, which a live router fills when it runs the AIS prefill-load model. The request’s own prefill time needed a new research input that only offline replay fills; live, a closed form fitted to AIS offline stands in for it, at 6.6% mean error on a grid up to 128K tokens. The decode terms use a closed form in the batch size and the router’s decode blocks, which undercount context shared by co-resident requests. Arms on v3 form a separate, secondary league that is not part of the headline (Section 9.10).

### 5.3 Utility and decision rule

For request  $j$  with candidates  $S$  and features  $x_i$ , the policy scores

$$u_i = \theta^\top x_i + \sum_{\ell=1}^r z_{S,\ell} p_\ell^\top x_i = w_S^\top x_i, \quad w_S = \theta + \sum_{\ell=1}^r z_{S,\ell} p_\ell, \quad (7)$$

and decides

$$\tau = 0 : \quad i^* \in \arg \max_{i \in S} u_i \text{ (seeded tie-break);} \quad \tau > 0 : \quad \Pr[i^* = i] = \frac{\exp(u_i/\tau)}{\sum_{i' \in S} \exp(u_{i'}/\tau)}. \quad (8)$$

With  $r = 0$  this is a conditional logit, the simplest member of the generalized extreme-value family of choice models [51]: one coefficient vector shared by every candidate, so the same parameters apply at any  $N$ . The context values  $z_{S,\ell}$  are scalars computed once per decision (Section 5.4); folding them into  $w_S$  is what keeps a decision at  $O(Nd)$ . A non-finite utility fails the selection rather than routing on it.

The temperature  $\tau$  is not range-normalized, unlike the default’s `router_temperature`, which divides costs by their range before scaling; the two are different parameters. At a fixed  $\tau > 0$  the probability mass on workers other than the best grows with  $N$ , so a sampled policy does not mean the same thing at every worker count. M1, M2 and their variants are trained and evaluated at  $\tau = 0$ , and a  $\tau > 0$  variant is only a conditional robustness extra. At  $\tau = 0$ , scaling  $\theta$  and the  $p_\ell$  together never changes a decision, which is why training pins  $\theta_0 = -1$  (Section 6.2). Algorithm 1 gives one decision in full.

**Determinism and tie-breaking.** Candidates are visited in worker-ID order, never in the host’s unspecified row order, and set means are summed in that order, so a seeded policy decides identically across processes. Each policy instance owns one random generator. Under the default discipline, `one_draw`, every decision consumes exactly one 64-bit draw, whether it ends in a tie, a sample, a unique best or a pin. Two policies with the same seed therefore stay on the same random stream even after one early decision differs, which keeps common random numbers aligned across the CMA-ES candidates of a generation (Section 6.3); a unit test checks that two policies differing in their first decision break 40 later ties identically under `one_draw` and not under `reservoir`. The `reservoir` discipline reproduces the seeded default picker draw for draw and exists for parity checks (Section 5.6). In replay, the harness sets the seed to  $k + 1$  for replicate  $k$  (Table 3).

### 5.4 Context terms and the IIA restriction

With  $r = 0$ , the policy inherits the conditional logit’s independence of irrelevant alternatives (IIA). At  $\tau > 0$ ,

$$\frac{\Pr[i^* = i]}{\Pr[i^* = i']} = \exp(\theta^\top (x_i - x_{i'})/\tau), \quad (9)$$

which depends only on the two workers’ own features, and at  $\tau = 0$  the same holds for which of the two ranks higher. The exchange rate between features, for example how many queued prefill tokens one cached block is worth ( $\theta_1$  against  $\theta_3$ ), is then the same whether the other workers are idle or saturated. A router plausibly wants it to change: when every candidate is busy, waiting behind the worker that holds the prefix may beat recomputing the prefix elsewhere, and when idle workers exist it may not. This is a hypothesis about the best policy, not an established property; the drift check described below measures it.

---

**Algorithm 1** One learned-choice decision. Before the call, the host has fixed the eligible set  $S$  and written each candidate’s default cost  $c_i^{\text{def}}$  at default weights. The draw discipline is `one_draw`; under `reservoir` there is no up-front draw, a tie draws once per tied candidate during the scan, a sample draws one uniform, and a pin draws nothing, as in `dynamo-default-cost-fn`.

---

**Require:** parameters  $\theta$ , sources and directions  $p_1, \dots, p_r$ , temperature  $\tau$ ; per-instance random generator and session map

**Require:** request  $j$ ; candidates  $S \neq \emptyset$  with their cache, load and capacity signals

```

1:  $draw \leftarrow$  next 64-bit value of the generator ▷ exactly one per decision
2: sort  $S$  by worker ID ▷ all later loops and sums use this order
3: if the host pins  $j$  to a worker then
4:    $i^* \leftarrow$  the pinned worker
5: else
6:    $prev \leftarrow$  the worker the session map holds for  $j$ ’s session, or none
7:   compute  $x_i$  for every  $i \in S$  (Table 8);  $x_{i,6} = 1$  iff  $i = prev$ 
8:    $w_S \leftarrow \theta + \sum_{\ell} z_{S,\ell} p_{\ell}$ 
9:    $u_i \leftarrow w_S^{\top} x_i$  for every  $i \in S$ ; fail if any is not finite
10:  if  $\tau = 0$  then
11:     $tied \leftarrow \{i \in S : u_i = \max_{i'} u_{i'}\}$ , in sorted order
12:     $i^* \leftarrow tied[\lfloor draw \cdot |tied|/2^{64} \rfloor]$ 
13:  else
14:     $target \leftarrow \lfloor draw/2^{11} \rfloor 2^{-53} \cdot \sum_i e^{(u_i - \max_{i'} u_{i'})/\tau}$ 
15:     $i^* \leftarrow$  first  $i$  whose running sum of  $e^{(u_i - \max_{i'} u_{i'})/\tau}$  exceeds  $target$ 
16:  end if
17: end if
18: if  $j$  has a session then
19:   session map  $\leftarrow (j$ ’s session  $\mapsto i^*)$ ; evict the least recently used entry beyond max_sessions
20: end if
21: return  $i^*$ 

```

---

A set statistic added directly to the utility cannot express this. A term that is the same for every candidate, or a linear set-relative feature such as  $x_{i,f} - (\bar{x}_S)_f$ , shifts every utility in  $S$  by the same amount and leaves the argmax unchanged. In the linear context logit, choice likewise depends only on each item’s deviation from the set mean [50, Lemma 4.3]. Set information has to change the coefficients instead, and the context term in Equation (7) does exactly that:  $w_S = \theta + \sum_{\ell} z_{S,\ell} p_{\ell}$ , so the exchange rate between two features becomes a linear function of the context values. This is the linear context logit (LCL) of Tomlinson and Benson [50],  $u_i = (\theta + A z_S)^{\top} x_i$ , with two changes: the mean feature vector is replaced by a short vector of named set and request statistics, and  $A$  is restricted to one free column  $p_{\ell}$  per statistic. It is not their decomposed variant (DLCL), which is a mixture of logits.

**Named rather than pooled.** The original contract specified a *pooled* form,  $z_{S,\ell} = q_{\ell}^{\top} \bar{x}_S$  with a free  $q_{\ell}$ . The policy still accepts it, but the campaign uses the *named* form of Table 10, for two reasons. The pooled form has a gauge freedom ( $p_{\ell} \rightarrow c p_{\ell}$ ,  $q_{\ell} \rightarrow q_{\ell}/c$ , including the sign), a flat direction in which CMA-ES spends budget for nothing; low-rank choice models have the same non-uniqueness [20]. The named form is linear in the free  $p_{\ell}$  and has none. And mean-pooling a one-hot feature leaks the set size: the candidate mean of `session_affinity` is  $1/N$  or 0. The named sources are chosen to avoid that (Section 5.5). Two redundancies remain and are handled

**Table 10:** Named context sources  $z_{S,\ell}$ . Each is unchanged when the candidate set is duplicated (Section 5.5).

| Source                                    | Value                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>mean_kv_load_frac</code>            | $(\bar{x}_S)_4$ , mean over $S$ of $x_{i,4}$                                                   |
| <code>mean_active_prefill_tokens_k</code> | $(\bar{x}_S)_3$                                                                                |
| <code>mean_active_requests_s</code>       | $(\bar{x}_S)_5$                                                                                |
| <code>max_overlap_frac</code>             | $\max_{i \in S} x_{i,1}$                                                                       |
| <code>isl_k</code>                        | $\text{ISL}_j / T_{\text{ch}}$                                                                 |
| <code>is_first_turn</code>                | 1 if the policy remembers no previous worker for $j$ 's session (new, evicted or none), else 0 |

in the search space rather than in the policy: when `isl_k` is a source,  $\theta_7$  duplicates the (`isl_k`, feature 3) entry of the context term and is frozen at 0; and on traffic without sessions `is_first_turn` is always 1, so its direction duplicates  $\theta$ . A unit test checks both context forms against the literal bilinear formula on 100 random tables.

**When context should pay.** A context term helps only where the best coefficients drift with the set statistics, and a larger model that nests a smaller one can still lose to it on a fixed optimizer budget; Tomlinson and Benson note that LCL, although it nests the conditional logit, can fail to beat it within a training budget of 500 epochs [50]. The campaign therefore measures drift before relying on context: M1 is tuned at 96 evaluations separately on L1-only, L3-only,  $N=4$ -only and  $N=8$ -only train cells, and each result is scored on the others. Cross-play losses beyond the minimum detectable effect argue for context, and the drifting statistic is the natural source. The check came out inconclusive. The one cross-play loss above the MDE, 0.087 for the model tuned on L1 cells evaluated on L3 cells, rests on 3 cells and is 1.3 standard errors from zero, and a single model tuned on all cells beat every regime specialist on the specialist's own cells; the specialists, at a quarter of the budget, had moved only a third as far from their start as the full M1. It is therefore read as an underpowered check with no evidence of drift, and it was not used to choose M2's sources. Those were fixed before any M2 evaluation: `isl_k`, because the documented M1 failure axis is prompt length (Section 10), and `mean_active_prefill_tokens_k`, the candidate set's prefill pressure.

## 5.5 Set-size invariance

Training sees  $N \in \{4, 8\}$ ; the test also scores  $N = 2, 16$  and 32 (Section 8.1). Two properties make one coefficient vector meaningful at any worker count. First, the coefficients are shared by all candidates and no feature is a worker's identity, so the policy is defined for any number of candidates; Lodestar's router uses the same principle [23]. Second, every feature and context source is designed to be invariant under duplicating the candidate set: if every worker is replaced by two identical copies ( $N \rightarrow 2N$ ), each candidate's features, every context value and the chosen worker (up to which copy) stay the same. Means and maxima pass this test. Sums and counts grow with  $N$ ; a mean of a one-hot feature is  $1/N$ ; a z-score is  $\pm 1$  at  $N = 2$  whatever the gap; and a leave-one-out rank fraction of 1 at  $N = 2$  becomes  $2/3$  after duplication. For that reason `v2`'s below-fraction counts strictly smaller loads over  $|S|$ , which keeps its value ( $1/2$  in the same example). A unit test duplicates 50 random candidate tables under random `v2` coefficients and checks that features, sources and the chosen worker are unchanged. Two features are excluded from the property because they are relations to particular workers: `hash_home` (its coefficient is zero in the test) and `session_affinity`, since a copy of a session's previous worker did not serve the session (the test uses requests without a session).

Duplication invariance is necessary for transfer across worker counts, not sufficient. At a different  $N$ , the same load level produces different per-worker states, so the held-out worker counts in the test measure transfer directly (Section 8).

## 5.6 Parity with the default router

With  $\theta = -\mathbf{e}_0$ ,  $r = 0$  and  $\tau = 0$ , Equations (7) and (8) pick  $\arg \max_i (-c_i^{\text{def}}/b_j) = \arg \min_i c_i^{\text{def}}$ , the default router’s choice. Three unit tests check this through the same YAML path a deployment uses. On 2,000 random candidate tables, spread over the four input situations the default scorer distinguishes (complete inputs, no per-tier cache matches, no load reports, prefill not tracked), **learned-choice** with **v1** and with **v2** picks the default’s unique minimum on every table that has one (at least 1,500 are required). On 500 tables built to tie (more than 250 do), the **reservoir** discipline with the same seed picks exactly what the seeded default picks. And on 200 tables, feature 0 stays at default weights when the host is configured with other score weights. The one known exception is two costs that differ by less than the rounding error of the division by  $b_j$ .

In replay, on 11 cells with 8 replicates each, **learned-choice** at  $-\mathbf{e}_0$  with **reservoir** ties produced per-request rows identical to **dynamo-default-cost-fn** on all 88 (cell, replicate) pairs. With the **one\_draw** discipline used in training, the rows differ on all 88, because tied decisions are broken by a different random draw, but goodput does not differ beyond replicate noise. The largest deviation of a cell’s mean goodput ratio from 1 was 0.0088, below three times the pooled replicate standard deviation of the ratio (0.0102), and over 7 segments the mean deviation was 0.00055, below twice its standard error (0.00117). In this check the session and AgentX cells sat near a good fraction of 1.0 under the provisional SLA used at integration, so the Mooncake cells carry the goodput comparison. In an earlier smoke replay of 1,000 Mooncake requests at  $N = 4$ , the two disciplines gave the same goodput with 733 of 1,000 assignments different, and the per-worker request counts were a permutation of each other: ties among idle workers at the start relabel the workers.

This parity is what makes  $-\mathbf{e}_0$  a meaningful start for training (restart s1 in Section 6.4) and lets every learned coefficient be read as a deviation from the default router.

## 5.7 The nested-logit option

The plan’s third rung, M3, is a nested logit over groups of candidates, such as the data-parallel ranks of one engine or the workers on one node. A nested logit lets unobserved utility be correlated within a nest, so IIA holds within each nest but not across nests, and the choice factors into a choice of nest, driven by the nest’s inclusive value, times a logit within the nest [51]. Two observations limit its value here. Every worker in this deployment is one engine with data-parallel size 1, so there are no nontrivial nests. And the nest parameters only shape the distribution of the unobserved utility; in the deterministic limit the choice is the argmax of the same systematic utilities, so at  $\tau = 0$  they would not change a decision. This corollary is ours, not Train’s. Where nests exist, the useful part of M3 at  $\tau = 0$  is nest-level features, such as the aggregate load of a node or a data-parallel group, added to  $x_i$ . M3 was not run.

## 5.8 Sticky-session policies

Session affinity is the simplest alternative to learning a session coefficient, so the campaign added **sticky-session** (Section 2.4), built from the same parts as **learned-choice**: the canonical-order chooser with both draw disciplines and the bounded LRU session map. A pinned request rebinds its session to the pinned worker. Without sessions, **sticky-session** with **reservoir** ties picks

**Table 11:** The learned arms and the tuned default. “Free” counts searched coefficients. Every learned arm runs at  $\tau = 0$  with  $\theta_0$  pinned at  $-1$  and, after the gaming finding, with load coefficients clamped so that load cannot attract (Section 6.5). Only router-observable arms can be the headline learned model; results are in Section 9.7.

| Arm                  | Features                                                | Free   | Role and outcome                                                                                                                    |
|----------------------|---------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------|
| M0                   | <code>dynamo-default-cost-fn</code>                     | 5      | tuned default: $\beta_{\text{ov}}$ , $\beta_{\text{decay}}$ , $\beta_{\text{pf}}$ , $\beta_{\text{req}}$ and the router temperature |
| M1                   | <code>v1</code> , $r = 0$                               | 7      | headline candidate; sign-constrained re-run of tier A                                                                               |
| M1 unconstrained     | <code>v1</code> , $r = 0$                               | 7      | the original tier-A M1; gaming-flagged, reported as a secondary row                                                                 |
| M1 continued         | <code>v1</code> , $r = 0$                               | 7      | 400 more evaluations from the selected M1; the comparator for M2                                                                    |
| M2 rank 2            | <code>v1</code> , two named sources                     | 20     | headline candidate; sources <code>isl_k</code> and <code>mean_active_prefill_tokens_k</code> ; ranks 3–4 not run (gate failed)      |
| M1-noaff             | <code>v1</code> , $\theta_6 = 0$                        | 6      | headline candidate (A15)                                                                                                            |
| M1 + queue threshold | <code>v1</code> and <code>router_queue_threshold</code> | 8      | headline candidate (ablation a)                                                                                                     |
| M1-v2                | <code>v2</code> , $r = 0$                               | 22     | headline candidate (A17.1); selected                                                                                                |
| M3                   | nested logit                                            | –      | not run: no nontrivial nests (Section 5.7)                                                                                          |
| M1-ais, M2-ais       | <code>v3</code> (and two named sources)                 | 11, 33 | secondary AIS league (A16)                                                                                                          |

exactly what the seeded default picks, and the build audit’s independent model matched it on 20,000 decisions. In the campaign both modes are tuned baselines (Section 7), tuning the default cost’s knobs as well.

`learned-choice` can express hard stickiness: with  $\theta = -\mathbf{e}_0 + 10^6 \mathbf{e}_6$  a session’s later requests stay on its previous worker whenever that worker is a candidate, and otherwise feature 6 is 0 for every candidate and the default cost decides. This matches `hard` mode with default weights. In a build smoke test, both kept every session on one worker on three session workloads with 20, 60 and 10 sessions, where the seeded default split 14, 54 and 1 of them across workers. The `bounded` rule is a threshold on relative load, which a linear utility can only approximate.

## 5.9 Model ladder

Each rung adds capacity only where the previous one leaves a measured gap, and a rung counts only if it improves held-out results. Table 11 lists the arms that were trained. Phase 2 scheduled them in tiers: tier A held M1 and every tuned baseline, tier B the arms that start from the selected M1 or were added by later amendments, and tier C the arms that ran only if an earlier result called for them. Every arm received three restarts of the same budget (Section 6.7). Training and initialization are in Section 6; the rule that picks the headline learned model among the router-observable arms, the same rule that picks the best baseline, is in Section 6.6.

The later router-observable arms test specific questions. M1-noaff pins the session coefficient at zero: the operator’s hypothesis was that session affinity carries little beyond cache overlap in replay, where the router’s cache index is perfectly fresh, but may matter under router-state lag or live, so the arm also ran in the lag robustness check. M1-v2 targets the pilot’s weakest case, the agent workloads on which M1 trailed cache-aware heuristics, with `v2`’s log, set-relative and

prefill-attention terms; one of its starts is LMetric’s representable point. Because  $\theta_0$  stays pinned at  $-1$ , that point is approached as the log-term coefficients grow relative to the anchor rather than reached exactly (our inference from Table 9).

Two ablations were planned and neither ships. Ablation (a) tuned the router’s dispatch-or-hold threshold `router_queue_threshold` jointly with the selected learned model and with the best baseline; it ran, and found nothing measurable (Section 9.7). Ablation (b) would have added simulator-only signals, such as engine waiting counts and KV usage, to measure what the router cannot see. It was first in the pre-registered order of cuts and was skipped, because it needed new plugin inputs, replay plumbing and a separate build, an implementation stage for an arm that never ships; free compute existed later, so the reason was implementation cost, not compute. The AIS league measures part of the same question (Section 9.10).

## 6 Training method

Training searches a policy’s parameters directly against the replayed objective. No part of the procedure scores a single routing decision: a decision changes the queues and caches that later requests see, so its effect is visible only in aggregate over a replay. Each policy configuration is therefore a black box that maps to one number, the objective below, and the search is derivative-free. The spaces are small (7 free coefficients for M1 and 22 for M1-v2, two to six parameters per baseline), and one tool, `lr-train`, tunes every policy, learned or heuristic, with the same settings (Algorithm 2). This section describes phase 2; the pilot used the same method on a 12-cell train subset, with 480 evaluations and two restarts per policy (Section B, in the appendix).

### 6.1 Objective

A configuration  $\pi$  is scored by replaying it on every train cell and averaging the clipped log-ratio of windowed goodput (Equation (6)) against the default router on the same workload replicates:

$$J(\pi) = \frac{1}{|\mathcal{C}_{\text{train}}|} \sum_{c \in \mathcal{C}_{\text{train}}} \frac{1}{K} \sum_{k \in \mathcal{K}_t} \Delta_{c,k}(\pi, \pi_0), \quad \mathcal{K}_t = \{(tK + i) \bmod K_{\text{pool}} : 0 \leq i < K\}, \quad (10)$$

with  $|\mathcal{C}_{\text{train}}| = 34$ ,  $K = 2$  and  $K_{\text{pool}} = 8$ . Each term is a paired comparison: the candidate and  $\pi_0$  replay the same cell on the same common-random-number (CRN) replicate with the same policy seed  $k + 1$  (Section 4.1). Before scoring, `lr-train` checks that every returned record carries its task’s cell, replicate and policy hash, and that candidate and reference cover the same (cell, replicate) set; a mismatch stops the run rather than producing an unpaired score. After its first replay,  $\pi_0$ ’s record comes from the result cache.

The objective is episodic: one value per configuration and generation, aggregated over minutes of simulated traffic per cell, with no per-decision credit assignment. Every cell weighs the same whatever its absolute goodput. The clipped log-ratio was fixed before training and is the same per-cell statistic the headline test uses (Section 8.4); the contract’s original form, the arithmetic mean of ratios, is reported but not optimized, because a few cells in which  $\pi_0$  nearly collapses dominate it (Section 3.3).

A candidate whose replay fails on any cell receives the generation’s lowest finite objective minus one, so CMA-ES ranks it last; a failed reference replay stops the generation, which is retried on the next call because errors are not cached. In the pilot no candidate failed in 4 runs of 480 evaluations (*preliminary (pilot, train subset)*). One generation replays 16 candidates and  $\pi_0$  on 34 cells  $\times$  2 replicates, 1,156 replays, plus 204 for the re-evaluation diagnostic below; a single candidate evaluation on the full train split costs about 700.9 replay-seconds, as measured in the pilot.

**Table 12:** M1’s search space and its three initial points. Within-set sd: geometric mean over the pilot’s 12 train cells of the mean within-candidate-set standard deviation, on reconstructed tables. Ratio:  $\text{sd}(x_{:,0})/\text{sd}(x_{:,f})$ . The search box is  $|\theta_f| \leq \text{bound}$ . Start s1 is  $\theta = -\mathbf{e}_0$  (all free coefficients 0); s2 and s3 are described in Section 6.4.

| $f$ | Feature                 | Within-set sd | Ratio | Bound  | s2     | s3   |
|-----|-------------------------|---------------|-------|--------|--------|------|
| 0   | default_logit_scaled    | 6.607         | 1     | pinned | −1     | −1   |
| 1   | overlap_frac            | 0.2585        | 25.6  | 100    | 63.9   | 76.8 |
| 2   | new_prefill_tokens_k    | 0.3742        | 17.7  | 70     | 4.18   | 0    |
| 3   | active_prefill_tokens_k | 1.239         | 5.33  | 21     | 1.28   | 0    |
| 4   | kv_load_frac            | 0.08173       | 80.8  | 320    | 12.5   | 0    |
| 5   | active_requests_s       | 0.06031       | 110   | 440    | −138   | 0    |
| 6   | session_affinity        | 0.3111        | 21.2  | 85     | −0.324 | 42.4 |
| 7   | isl_x_prefill_load      | 2.053         | 3.22  | 13     | −0.196 | 0    |

## 6.2 Search spaces

**Internal coordinates.** Each policy’s `space.yaml` lists its free parameters with bounds and a linear or log scale, its pinned parameters, and CMA-ES options. CMA-ES searches internal coordinates  $\phi$ : each bounded parameter maps to  $[0, 1]$ , linearly or in log space, and integer parameters are rounded after decoding. `pycma`’s boundary transform keeps every sample inside the box, so every candidate decodes to a valid policy file. Every phase-2 parameter is bounded.

**M1.** At  $\tau = 0$  the decision (Equation (8)) is unchanged when  $\theta$  is multiplied by a positive constant, a direction along which the objective is exactly flat and along which CMA-ES’s step size tends to grow [12]. M1 therefore pins  $\theta_0 = -1$ , which fixes the scale and the default router’s sign and makes the start  $\theta = -\mathbf{e}_0$  the default router (Section 5.6). The temperature is fixed at  $\tau = 0$  rather than searched: with the scale pinned,  $\tau > 0$  is a different, stochastic policy and is a separate variant (Section 8.7). The tie-break mode is `one_draw`. Seven coefficients remain free.

Raw coefficients have no common scale, because the features’ spreads within a candidate set differ by two orders of magnitude (Table 12). We bound each free coefficient at about four times its *scale ratio*  $\text{sd}(x_{:,0})/\text{sd}(x_{:,f})$ , the coefficient at which feature  $f$  varies across a candidate set as much as the anchor does; at its bound, a feature’s term spreads four times as widely as the anchor’s. The standard deviations are measured within candidate sets, as the geometric mean over the pilot’s 12 train cells of each cell’s mean, on candidate tables reconstructed from  $\pi_0$ ’s per-request rows, because replay does not log candidate tables. The reconstruction is approximate (router load is rebuilt from arrivals, first tokens and completions, and overlap is exact only for the chosen worker and for earlier turns of the same session), so the ratios set scales, not policy inputs. The build stage’s placeholder bounds of  $\pm 8$  could not have let `kv_load_frac` or `active_requests_s` matter.

Because each bound is proportional to the feature’s scale ratio, one internal unit moves every coefficient by the same multiple of its ratio, so the initial search distribution is isotropic in units of decision-relevant spread. This is the per-variable scaling Hansen [12] recommends when initial ranges differ by orders of magnitude, and the analogue of the state normalization without which Mania et al. [27] could not train a linear policy on their hardest task.

**Baselines.** Each tuned baseline searches the parameters its catalog entry exposes (Table 13), starting from its shipped defaults. Three rules shape these spaces.

- **Gauge pins.** A parameter that only rescales another is fixed. These are the affinity block

size of `ramjet` (512 tokens), the prefix weight of `llm-d-precise-prefix` (2), and the peak prefill rate of `llm-d-optimized-baseline` (15,928 tokens/s), whose product with `max_ttft_penalty_ms` is the only quantity that policy identifies.

- **Structural parameters** keep their shipped values: `lmetric`’s hot-spot detection on, `ramjet`’s relative basis, `sticky-session`’s map of 65,536 sessions, `llm-d-precise-prefix`’s empty class weights, and `llm-d-optimized-baseline`’s throughput-based TTFT source.
- **Router knobs.** Of the baselines, only `sticky-session` reads the default cost’s router-level knobs (for first turns and fallbacks), so both sticky variants also search the five M0 knobs over M0’s ranges. Sticky-session and M0 at the same knob values gave identical results on a session-free Mooncake cell, both at the start point and at the upper corner of the box. `router_queue_threshold` keeps its default for every policy (Section 5.9).

M0’s space is the pilot’s, with the prefill-load scale  $\beta_{\text{pf}}$  widened to  $[0.05, 500]$  on a log scale because the pilot’s tuned values (17 to 25) sat near the top of the old range  $[0.05, 50]$ . The decode request weight  $\beta_{\text{req}}$  starts at 0.5, the bottom of its log range  $[0.5, 4096]$ , which stands in for the shipped value 0.

**Initial step size.** Parameter units are not comparable across policies, so  $\sigma_0$  is set in decision space. For a step size  $\sigma$ , we draw internal points from  $\mathcal{N}(\phi_0, \sigma^2 I)$  clipped to the box, decode them, and measure the share of routing decisions that differ from the policy’s own decisions at its start  $\phi_0$ , as the total-variation distance between the two decision distributions. The decisions are one-step counterfactuals on the reconstructed candidate tables of the pilot’s 12 train cells, with 8 draws per  $\sigma$  and cell; stateful policies run their own state machine over each cell in arrival order.  $\sigma_0$  is the  $\sigma$ , interpolated on a grid from 0.03 to 0.35, at which this flip rate reaches 17.5%, so every policy starts with a comparable amount of exploration measured in decisions. The literature review’s suggested band of 10–30% is itself a hypothesis (LR-05). For M1,  $\sigma_0 = 0.004$  internal units, a standard deviation of 0.032 scale ratios per coefficient, flips about 20% of decisions relative to  $\theta = -\mathbf{e}_0$ . Five baselines cannot reach the target by  $\sigma = 0.3$ , the step-size cap below, and start at 0.3. Around their shipped defaults their decisions barely respond to their parameters: `lmetric`’s flip rate at  $\sigma = 0.3$  is 0.06%. For `chwb1`’s and `dualmap`’s hashed prefix depths, any change re-draws the whole prefix-to-worker map (flip rates of 0.69 and 0.18 already at  $\sigma = 0.03$ ), so those two coordinates are held at their defaults during the measurement and share their space’s  $\sigma_0$ .

**Step-size cap.** Every baseline space caps `pycma`’s per-coordinate standard deviation (`maxstd`) at 0.3 internal units. The pilot recommended the cap after M0’s first run, whose global step size grew from 0.18 to 10.8 internal units after its validation score had plateaued, along nearly flat directions in the decode request weight and the temperature (*preliminary (pilot, train subset)*). M1’s space, whose step size stayed small in the pilot, sets no explicit cap. Two details qualify the measure. `pycma` 4.5.0 already caps each bounded coordinate at one third of its range by default, so on these  $[0, 1]$  coordinates the explicit cap tightens the default from  $1/3$  to 0.3, and the default applies to M1. The cap also limits the per-coordinate spread of the samples, not the global step size  $\sigma$ , which can still drift.

### 6.3 CMA-ES and noise handling

We use CMA-ES [12] from the `cma` package, version 4.5.0, with population size  $\lambda = 16$ , the package’s default of 8 parents, and a budget of  $B = 400$  objective evaluations per run, which is 25 generations. For the 2 to 7 free coordinates of these spaces the package’s default population would be 6 to 9. Its

**Table 13:** Search spaces of the tuned policies. Dim.: number of free coordinates. Flips: share of one-step decisions that change at  $\sigma_0$  relative to the start, on reconstructed candidate tables; the calibration target is 0.175. <sup>†</sup>Hashed prefix depth held at its default during the measurement. Every space starts at the policy’s shipped defaults, except M1, which has three starts (Table 12).

| Policy                      | Free parameters (pinned)                                                                                                       | Dim. | $\sigma_0$ | Flips              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|------|------------|--------------------|
| M1 (learned-choice, v1)     | $\theta_1, \dots, \theta_7$ ( $\theta_0 = -1$ , $\tau = 0$ )                                                                   | 7    | 0.004      | $\approx 0.20$     |
| M0 (dynamo-default-cost-fn) | overlap_score_credit, overlap_score_credit_decay, prefill_load_scale, decode_active_request_weight, router_temperature         | 5    | 0.11       | 0.169              |
| sticky-session (hard)       | the five M0 knobs                                                                                                              | 5    | 0.17       | 0.179              |
| sticky-session (bounded)    | load_factor and the five M0 knobs                                                                                              | 6    | 0.067      | 0.175              |
| llm-d-precise-prefix        | weights.queue, weights.kv_cache_utilization, prefix_match_length_weight, prefix_match_length_scale_tokens (weights.prefix = 2) | 4    | 0.16       | 0.177              |
| chwbl                       | prefix_tokens, load_factor                                                                                                     | 2    | 0.16       | 0.173 <sup>†</sup> |
| ramjet                      | alpha, max_affinity_blocks, load_unit_tokens (affinity_block_tokens = 512)                                                     | 3    | 0.30       | 0.133              |
| dynamo-two-tier-cost-fn     | cache_threshold, balance_abs_threshold, balance_rel_threshold                                                                  | 3    | 0.30       | 0.083              |
| llm-d-optimized-baseline    | affinity_threshold, max_ttft_penalty_ms, queue_threshold_tokens (peak prefill rate = 15,928)                                   | 3    | 0.30       | 0.034              |
| dualmap                     | hash_prefix_blocks, pending_prefill_token_budget, window_requests                                                              | 3    | 0.30       | 0.018 <sup>†</sup> |
| lmetric                     | class_prefix_blocks, window_requests                                                                                           | 2    | 0.30       | 0.0006             |

function-value stopping rules are switched off, so a run normally ends when the budget is spent; the equal-budget check (Section 6.7) flags any finished run that did not spend exactly  $B$ .

The objective is noisy in a specific way. Replay is deterministic, so a configuration scored on a given set of cells and replicates always gets the same value. What varies is which replicates a generation draws (Section 4.1) and, behind that, which workloads the train split happens to contain. Four measures keep this noise from steering the search.

1. **Common random numbers within a generation.** All 16 candidates of generation  $t$  and the reference  $\pi_0$  replay the same cells and replicates  $\mathcal{K}_t$  with the same policy seeds, so CMA-ES ranks paired differences rather than independent draws. Scoring every candidate on the same fixed scenarios is the device of PEGASUS [34]. In a two-server load-balancing example, conditioning on the exogenous input sequence lowered policy-gradient variance 50-fold [31]; for rank-based CMA-ES the analogue is paired scoring on shared replicates. The `one_draw` tie-break keeps the random stream aligned across candidates whose decisions differ (Section 5.6).
2. **Rotating replicates.**  $\mathcal{K}_t = \{2t, 2t+1\} \bmod 8$ : consecutive generations see different replicates and each pair recurs every fourth generation, so the CMA-ES distribution, which accumulates information over generations, is not fit to one draw.
3. **Population size.**  $\lambda = 16$  is about twice the default. In the analysis behind UH-CMA-ES, raising the population size together with the number of parents, as the package does, is preferable to re-evaluating each candidate several times when step sizes adapt properly [13].
4. **Rank-stability diagnostic.** After each update, the two best candidates are re-scored on the same cells with the next generation’s replicates  $\mathcal{K}_{t+1}$ , and the change in their order is logged. The re-scored share,  $2/\lambda = 0.125$ , equals UH-CMA-ES’s default rate  $\max(0.1, 2/\lambda)$

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**Algorithm 2** Tuning one policy (**lr-train**, one restart). Every policy, learned or baseline, runs this with the same  $B$ ,  $\lambda$ ,  $K$ , cells, validation schedule and selection rule.

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**Require:** space (bounds, pins,  $\sigma_0$ , cap), start  $\phi_0$ , seed, cells  $\mathcal{C}_{\text{train}}$  and  $\mathcal{C}_{\text{val}}$

- 1: initialize CMA-ES at  $\phi_0$  with step size  $\sigma_0$  and population  $\lambda$
- 2:  $best \leftarrow \perp$ ;  $selected \leftarrow \perp$
- 3: **for**  $t = 0, 1, \dots, B/\lambda - 1$  **do**
- 4:    $\mathcal{K}_t \leftarrow \{(tK + i) \bmod K_{\text{pool}} : i < K\}$
- 5:   sample candidates  $\phi_1, \dots, \phi_\lambda$ ; checkpoint ▷ ask
- 6:   replay every candidate and  $\pi_0$  on  $\mathcal{C}_{\text{train}} \times \mathcal{K}_t$ ; check CRN coverage
- 7:   compute  $J$  per candidate (Equation (10)); a failed one gets the worst finite  $J - 1$
- 8:   update CMA-ES with fitness  $-J$  ▷ tell
- 9:    $best \leftarrow$  the candidate with the highest  $J$  seen so far
- 10:   re-score the top 2 on  $\mathcal{C}_{\text{train}} \times \mathcal{K}_{t+1}$ ; log rank changes ▷ diagnostic only
- 11:   **if**  $(t + 1) \bmod 5 = 0$  **then**
- 12:     score  $best$  and the CMA-ES mean on  $\mathcal{C}_{\text{val}} \times \{0, 1, 2\}$
- 13:      $selected \leftarrow$  the validated candidate with the highest score so far
- 14:   **end if**
- 15:   checkpoint (CMA-ES state and random state)
- 16: **end for**
- 17: **return**  $selected$ , and  $best$  for reference

---

[13], but unlike UH-CMA-ES nothing reacts to the measurement: neither the step size nor the number of evaluations per candidate changes. It runs identically for every policy, does not count against  $B$ , and is skipped in a generation whose wall-clock chunk ends during it. In the pilot, the top two candidates swapped in 7 to 11 of 30 generations per run (*preliminary (pilot, train subset)*).

Every 5 generations, **lr-train** scores two candidates on all 14 validation cells with replicates  $k = 0, 1, 2$  ( $K_{\text{val}} = 3$ ): the best train sample so far and the mean of the CMA-ES distribution. The mean is included because it is not itself chosen by a noisy train score. The run keeps the higher-scoring validated candidate as its selection (Section 6.6). Algorithm 2 summarizes one run.

## 6.4 Restarts and initial points

Each tuned policy runs  $R = 3$  restarts with CMA-ES seeds 1, 2 and 3. Restart 1 of every policy runs in the first wave, restart 2 in the second and restart 3 in the third, so a compute cut at any point removes the same restart from every policy. Baselines start every restart at their shipped defaults; only the seed differs. M1’s three restarts start from three different points, all computed from train-split data only (Amendment A11.1; Table 12).

**s1**  $\theta = -\mathbf{e}_0$ , the default router.

**s2** A behavior clone of **llm-d-precise-prefix**, the best of the untuned heuristics evaluated on validation in the pilot. It is the conditional-logit maximum-likelihood fit to that heuristic’s choices: with  $\Pr[i \mid S] \propto \exp(\theta^\top x_i)$  over reconstructed candidate tables, it maximizes the mean over cells of the mean log-probability of the worker the heuristic chose, with  $\theta_0$  pinned at  $-1$  and  $\theta_1, \dots, \theta_7$  held inside M1’s box. The problem is convex and bounded and is solved by L-BFGS-B. The data are the heuristic’s own replays at shipped defaults on all 34 train cells, replicate 0: 144,048 decisions in a stride sample of each cell. The fit chooses the heuristic’s

worker on 0.875 of these decisions (0.866 and 0.883 in a two-fold split by cell), against 0.512 for the default rule  $\theta = -\mathbf{e}_0$ , and no coefficient sits at its bound.

- s3** A cache-heavy point:  $\theta_1 = 3 \times 25.6 = 76.8$  and  $\theta_6 = 2 \times 21.2 = 42.4$ , so that across a candidate set the overlap term spreads three times and the session-affinity term twice as widely as the anchor, with all other free coefficients 0, so load enters only through the anchor. It targets the agent workloads on which the pilot’s M1 trailed the cache-aware heuristic (by  $-0.019$  in mean clipped log-ratio on AgentX, *preliminary (pilot, validation split)*). It was preferred to a fit to the second-best heuristic family because `lmetric`’s product of uncached tokens and load is not representable with `v1` features, and a fit to `ramjet` would be another linear cache-minus-load score close to `s2`.

On one-step decisions, `s2` and `s3` differ from the default rule on 46% and 42% of requests and from each other on 11%. The clone’s agreement should be read with the reconstruction’s bias in mind: overlap is exact only for the chosen worker, which favors overlap-heavy rules. With a hash-aware reconstruction the fit agrees on 0.785 of decisions and its overlap coefficient is 40 instead of 64, and the hand-set point `s3` agrees with the heuristic on 0.872, almost as often as the fit. Agreement is lowest on AgentX (0.797, against 0.914 on Mooncake and 0.865 on sessions).

The initial points are not objective evaluations. The clone uses one replicate of the heuristic’s own train replays (22 fresh and 12 cached), not evaluations of M1, so every M1 restart still spends exactly  $B$ . Bootstrapping a policy search from existing policies is a known lever [29], as is initializing a richer choice model at a simpler model’s maximum-likelihood fit [47]. The starts still give M1 information that a baseline’s restarts do not have, namely another policy’s decisions. The operator approved them with disclosure (Amendment A12): the test also reports *M1-default-init*, M1 selected over restart `s1` alone (Section 8.6).

**Other learned arms.** Table 11 lists the other learned arms. Each is trained like M1, with  $B = 400$ , three restarts, the same cells and the same selection rule; they differ in their spaces and starting points, and all of them ran to completion.

- **M2 rank 2 and M1 continued.** M2 starts at the selected M1 coefficients with  $p = 0$  and context step sizes 0.3 times  $\theta$ ’s; its two named sources, `isl_k` and `mean_active_prefill_tokens_k`, were fixed before any M2 evaluation, and  $\theta_7$  is pinned at 0 because the (`isl_k`, feature 3) context entry duplicates it. It is compared with M1 continued from its selected point for the same added 400 evaluations (LR-15).
- **M1-noaff.** M1’s three starts with  $\theta_6$  pinned at 0; a further check, without retraining, zeroes  $\theta_6$  in the selected M1 (Amendment A15).
- **M1-v2.** Coefficients 0 to 7 keep M1’s bounds, so M1-v2 nests M1; coefficients 8 to 22 are bounded at four times their scale ratio, measured on candidate tables reconstructed from the default router’s train replays.  $\sigma_0 = 0.0028$  flips 19.6% of decisions at the `s1` start, the rate M1’s space has at its own  $\sigma_0$ . The starts are `s1`,  $\theta = -\mathbf{e}_0$ ; `s2`, the behavior clone of `llm-d-precise-prefix` refit on `v2` features (with `hash_home` pinned at 0 in the fit, because its offline reconstruction is approximate); and `s3`, `lmetric`’s representable point  $\theta = -\mathbf{e}_0 - 20.7(\mathbf{e}_{\log\_ptok} + \mathbf{e}_{\log\_bs})$ , 90% of the bound on those coordinates, which agrees with the pure LMetric rule on 0.924 of reconstructed decisions (Amendment A17.1).
- **M1 with a queue threshold** (ablation a). M1 continued’s space plus `router_queue_threshold` on a log scale from 0.25 to 256, starting at 32, a value that replays identically to

the unset knob on every train cell; its baseline counterpart tunes the same knob on top of the best baseline’s selected configuration.

- **AIS league.** M1-ais and M2-ais on feature set v3, M0 with the AIS prefill-load model, and llm-d-optimized-baseline with modeled TTFT are trained the same way on their own build and compete only in a secondary leaderboard (Amendment A16, Section 9.10).

## 6.5 Sign constraints on load coefficients

The midpoint audit found that the first tuned M1 gamed the objective by attracting long requests to a busy worker (Section 10), which triggered a remedy registered in advance (Amendment A11.2): re-run the affected restarts with load coefficients constrained so that load cannot attract, at the same budget. With  $\theta_0$  pinned at  $-1$ , the minimal such set for v1 is  $\theta_3, \theta_4, \theta_5, \theta_7 \leq 0$  (active prefill, KV load, active requests and the prompt-length-by-prefill interaction), so that no load signal adds utility for any prompt length.

- **A clamp, not a box bound.** The restarts s1 and s3 start with those coefficients at 0, which on a box bound of 0 would sit on the edge of pycma’s boundary transform, where its slope is zero and the first generation’s steps on those coordinates shrink by about two orders of magnitude. The space therefore keeps the original box and clamps the decoded value,  $x = \min(x, 0)$ . CMA-ES searches the identical internal space, start, step size and seed as the unconstrained run, and generation 0 asked the same internal points; only the decoded policies differ.
- **Starts.** s1 and s3 already satisfy the constraint. The behavior clone s2 did not (its  $\theta_3$  and  $\theta_4$  were positive, so decode load attracted above about 24K tokens), so it was projected onto the constraint; its agreement with the cloned heuristic stayed at 0.879.
- **Later arms.** Every later learned arm uses the same rule: M1-v2 clamps every load-ordered feature (3, 4, 5, 7, 8, 10 and 12 to 21), M2 also clamps the load columns of its context directions so that every effective load weight stays non-positive, and the AIS league clamps its load and backlog features.
- **Equal footing.** The re-runs used the same seeds, budget, cells, replicates, validation schedule and objective; the unconstrained runs are kept as a disclosed, gaming-flagged secondary row.

## 6.6 Selection

Selection happens at two levels, and neither uses the best train sample, whose score is biased upward by the search itself by an amount comparable to the differences being measured [7].

1. **Within a policy.** A policy’s configuration is the validated candidate with the highest validation objective ( $k = 0, 1, 2$ ) over its three restarts: two candidates at each of five checkpoints per restart, up to 30 per policy. `best.json` keeps the best train sample beside the selection, for reference only.
2. **Across policies.** The best baseline  $\pi_{\text{base}}^*$  is the tuned baseline whose selected configuration has the highest mean clipped log-ratio against  $\pi_0$  on *fresh* validation replicates  $k = 3, \dots, 10$ . The headline learned policy  $\pi_\theta$  is chosen by the same rule among the router-observable learned arms (M1, M2, M1-noaff, M1-v2 and any later learned arm; Amendment A17.2). No within-policy selection reads replicates 3 to 10, so the cross-policy choice is not made on the scores that picked each configuration.

Every learned variant scored on validation is counted, and the number of learned arms compared is reported with the headline (LR-11).

## 6.7 Equal budget

Every tuned policy, learned or baseline, gets the same resources:  $B = 400$  objective evaluations per restart on all 34 train cells with  $K = 2$  replicates,  $\lambda = 16$ , three restarts, five validation checkpoints of two candidates on 14 cells with  $K_{\text{val}} = 3$ , the same reference and objective, and the same selection rule. The orchestrator verifies that every finished run spent exactly 400 evaluations and five validation checkpoints; the re-evaluation diagnostic is outside this check because a wall-clock chunk can skip it. No run starts from the pilot’s tuned configurations, which would give M0 and M1 the pilot’s 480 evaluations on top of  $B$ .

Equal budget fixes what each policy may spend, not how far it can get. The spaces differ in dimension (2 to 22) and in how strongly decisions respond to the parameters (Table 13), and only the learned arms start from points derived from another policy (Section 6.4). Which method wins can depend on the budget [11], so two checks ask whether any policy was still improving at  $B$ . No tuned baseline of tiers B, C or the AIS league was selected at its last validated generation; the learned arms selected there had gained 0.0001 to 0.0033 over the last validation interval, at most 0.09 of the MDE. And an auditor’s validation-oracle probes, which deliberately favor the baselines, found no headroom: 38 **ramjet** configurations around all three restarts’ bases reached at most 0.1421 against the tuned 0.1420, and widened probes of M0 and **llm-d-precise-prefix** stayed below their tuned values (0.1299 against 0.1334 and 0.1202 against 0.1414), all far below M1-v2’s 0.1767.

## 6.8 Drift check before M2

A context term can help only if the best coefficients change with the regime (LR-15). Before M2, M1 is tuned at a quarter of the budget on each of four train subsets: L1 cells only (6 cells), L3 only (8),  $N = 4$  only (18) and  $N = 8$  only (16). The quarter budget is 96 evaluations, six whole generations, because 100 is not a multiple of 16. Each run validates every 3 generations on the matching validation cells and selects on  $k = 0, 1, 2$ . Each selected model is then scored on every regime’s validation cells with fresh replicates  $k = 3, \dots, 10$ . The cross-play loss of the model tuned on regime  $a$ , evaluated on regime  $b$ , is the mean clipped log-ratio against  $\pi_0$  of  $b$ ’s own model on  $b$ ’s validation cells minus that of  $a$ ’s model; the pairs are L1 and L3, and  $N = 4$  and  $N = 8$ . The gate planned to deprioritize M2 if all four losses lay within the MDE of 0.038. Amendment A17 later made M2 at rank 2 mandatory, so the check no longer decides whether M2 trains; it is reported as evidence for or against context terms. It came out inconclusive, with no evidence of drift (Section 5.4).

## 6.9 Compute and numerics

Replays run on CPUs: locally through a pool of 20 slots, and on a pool of shared CPU nodes, one tuning run per node; the first wave started on 10 remote nodes and the workstation. A generation took 225 to 333 s early in phase 2, and about 300 s later, on a 96-core EPYC 9654P node, and 610 to 690 s on a 64-core EPYC 7702P node; the three sign-constrained M1 re-runs took 4 h 24 min to 4 h 31 min each on the latter. Phase 2 ran 66 restarts of 400 evaluations in total (tiers A and B, the ablation and the AIS league), and each run whose ingest the stage log records stored between 30,199 and 30,792 result records. The single test pass added 5,220 records on the tuning build and

900 on the AIS build, and the robustness pass 23,940 (Section 9.9). Replay-seconds for phase 2 were not totaled.

Runs move between machines, so a run must continue bit-exactly on any host. Two things depend on the host. Replay results depend on the operating system’s math library, so every distinct node image was checked against the workstation before use: across 6,312 remote records, every field matched, and 48,901,384 per-request rows matched byte for byte. CMA-ES’s own linear algebra depends on numpy’s and OpenBLAS’s CPU dispatch. In a 60-generation `lr-train` run on a synthetic objective without replays, an EPYC 9654P node with default settings diverged from the workstation in the last bits from the first generation, and matched it in all 60 generations once three environment variables pinned OpenBLAS to a single thread and core type and disabled numpy’s AVX-512 code paths; the pinned values equal the workstation’s defaults. With the pins, two short real `lr-train` runs on that node reproduced the workstation’s history (apart from wall-clock fields), `best.json` and `best_policy.yaml` exactly.

`lr-train` writes its checkpoint twice per generation, after sampling and after the update, including the CMA-ES object and numpy’s random state, so a resumed run continues exactly as an uninterrupted one would; an audit killed a run three times and found its history, selection, CMA-ES state and continuation identical. A run refuses to resume if its space, cells, reference, objective or seeds have changed.

If compute ran short, the last resort in the gate’s cut order is to score each generation on a rotating 12-cell subset of the train cells, for every policy of a wave alike. On the 16 candidates of one pilot generation, subset and full-train objectives had a Spearman correlation of 0.947 (*preliminary (pilot, train subset)*).

## 7 Baselines

The headline compares the learned policy with the best heuristic an operator could deploy after tuning it on the same data. That heuristic is chosen from eleven tuned configurations, each tuned exactly like M1 (Algorithm 2 and Section 6.7): three restarts of  $B = 400$  evaluations, the same objective, train cells, replicates and validation rule. Ten are the catalog policies of Table 14; the eleventh is `ramjet` with a jointly tuned router queue threshold, the baseline arm of ablation (a). Fresh validation replicates then pick  $\pi_{\text{base}}^*$  among the eleven selected configurations (Section 6.6); it was tuned `ramjet`. Every heuristic is also evaluated untuned, at its shipped defaults, and round-robin is evaluated as a floor; these rows are reported for context.

All of them are catalog policies, selected by the same kind of policy YAML as `learned-choice` (Figure 2), and in the headline league every one reads only inputs the live router has. They are evaluated as implemented on the campaign branch: the default cost function and the two-tier policy as on Dynamo’s main branch, the six ported policies as in the pull-request stack #15450 and #15453, and `sticky-session` as written for this campaign. Table 14 summarizes their rules at shipped defaults, which Section 2.4 describes in the API’s terms; Section 7.1 gives what is tuned and Section 7.2 how the ports differ from the methods they are named after.

### 7.1 What is tuned

Each baseline searches its shipped numeric parameters over ranges set in the parameter’s own units, log-scaled where a parameter spans orders of magnitude, starting from the shipped defaults (Table 15); restarts 1, 2 and 3 use CMA-ES seeds 1, 2 and 3. Four kinds of parameter are pinned rather than searched.

**Table 14:** Baselines at their shipped defaults, as implemented on the campaign branch. Every policy except round-robin is also tuned (Table 15).  $o_i$ : overlap blocks of worker  $i$ ;  $L_i^{\text{req}}$ : its active requests; one KV block is 16 tokens. “Tie rotation” is a counter that rotates among tied workers.

| Policy                   | Rule at shipped defaults                                                                                                                                                                                              | State kept by the policy                  |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| round_robin              | requests go to the workers in turn; reads neither cache nor load                                                                                                                                                      | position in the cycle                     |
| dynamo-default-cost-fn   | $\arg \min_i c_i^{\text{def}}$ (Equation (1)): prefill blocks left after the overlap credit, plus decode blocks                                                                                                       | none; random tie-breaks                   |
| dynamo-two-tier-cost-fn  | fewest active requests; but while load is not badly imbalanced (spread $\leq 32$ or largest $\leq 1.1 \times$ smallest) and some worker caches more than half the prompt, the least loaded of the best-cached workers | none                                      |
| lmetric                  | $\arg \min_i \max(\text{ISL}_j - 16 o_i, 1) (L_i^{\text{req}} + 1)$ , with a hot-spot filter for popular prefixes                                                                                                     | class shares over the last 1000 requests  |
| ramjet                   | $\arg \max_i$ of capped relative prefix affinity minus 4 times the worker’s load units                                                                                                                                | tie rotation                              |
| dualmap                  | two hashed candidates per prefix; the warmer one unless its prefill backlog would exceed 65,536 tokens                                                                                                                | prefix shares over the last 1000 requests |
| chwbl                    | hashed owner of the prompt’s first 256 tokens while its active requests stay within 1.25 times the mean (counting this request); reads no cache                                                                       | none                                      |
| llm-d-precise-prefix     | $\arg \max_i$ of prefix, queue and KV-utilization scores weighted 2:1:1                                                                                                                                               | tie rotation                              |
| llm-d-optimized-baseline | lowest prefill token load among workers caching at least 80% of the prompt, unless their TTFT estimate trails by more than 18s                                                                                        | tie rotation                              |
| sticky-session, hard     | the session’s previous worker when eligible, otherwise the default cost’s choice                                                                                                                                      | session map (65,536)                      |
| sticky-session, bounded  | as hard, but leaves that worker when its active requests exceed 1.25 times the candidate mean                                                                                                                         | session map (65,536)                      |

- **Rescalings.** A parameter that only rescales others adds a direction along which decisions do not change, and CMA-ES’s step size drifts along such directions [12]. Three are pinned. Multiplying `ramjet`’s `affinity_block_tokens` by some factor while dividing `max_affinity_blocks` and `alpha` by the same factor leaves its decisions unchanged up to integer rounding; scaling all three `llm-d-precise-prefix` weights never changes its choice; and with the throughput source, `llm-d-optimized-baseline`’s gate compares a token difference divided by `peak_prefill_tokens_per_second` with `max_ttft_penalty_ms`, so only their product matters.
- **Categorical choices.** CMA-ES searches continuous coordinates, so `lmetric`’s hot-spot detector stays on, `ramjet`’s affinity basis stays **relative** and the optimized baseline’s TTFT source stays **throughput**.
- **Capacity.** The sticky-session map keeps its 65,536-session bound.
- **Inputs replay does not supply.** `llm-d-precise-prefix`’s per-class weights stay empty, because replay sets no policy class.

Both sticky variants also search the five default-cost knobs, with M0’s ranges, because their first-turn and fallback choice is the default cost function. On a session-free Mooncake cell, the sticky policy

matched M0 both at the starting point and at the upper corner of the space. No other baseline reads a tunable router-configuration knob, and the router’s queue threshold is left to a separate ablation (Section 5.9).

The initial step size  $\sigma_0$  follows the rule in Section 6.2: it is the step size at which one CMA-ES step changes 17.5% of decisions on candidate tables reconstructed from the pilot’s default-router replays. Five spaces do not reach that rate even at 0.3, the cap on the per-coordinate standard deviation, and start at 0.3: two-tier (0.083 of decisions change), **lmetric** (0.0006), **ramjet** (0.133), **dualmap** (0.018) and the optimized baseline (0.034). For **chwbl** and **dualmap**, the hashed key depth was held at its default for this measurement. Any change of that depth redraws the prefix-to-worker map, so the objective is flat between block boundaries and jumps across them: 0.69 and 0.18 of decisions change already at a step of 0.03. Table 40 in the appendix gives each tuned baseline’s selected restart and checkpoint with its validation and test scores; the selected **ramjet**, for instance, came from restart 3 at generation 25, with **alpha** 0.883, **max\_affinity\_blocks** 68 and **load\_unit\_tokens** 4552 (Figure 2).

## 7.2 Ports, not reproductions

By operator decision, the ports are evaluated as implemented: we built no paper-faithful variants and did not try to reproduce published numbers. Implementations of the same baseline can differ materially [15], so beating **lmetric** or **dualmap** here means beating these ports in this simulator, at tuned and at shipped parameters, not the published systems.

The decision called for only a brief sanity check, that each port’s decisions respond to its inputs in the documented direction. A setup audit confirmed that the parameters of **chwbl**, **dualmap**, **ramjet** and both llm-d ports change their routing deterministically, and that **lmetric** and two-tier replay identically across runs. In the pilot’s screen of the nine untuned heuristics on 12 train cells with replicates 0 and 1, every port ran without error, none routed against its documented direction in aggregate, and seven beat  $\pi_0$ . **dualmap** and **chwbl** lost to it, with mean clipped log-ratios of  $-0.150$  and  $-0.214$ , which the pilot attributed to hash-pinned placement on traffic with few hot prefixes; round-robin scored  $-0.481$ . These screen numbers are preliminary (pilot, train subset).

The known differences between the ports and the published methods are these.

- **lmetric** omits the prefill already queued on a worker from its prefill factor. The paper credits that term for its advantage over a cache-hit-ratio indicator [63]. The port also counts its hot-spot window in decisions rather than time.
- **dualmap**’s budget is a fixed default; the paper derives it as the prefill backlog a GPU can clear within the TTFT SLO [61], and our SLO has no TTFT term. The port uses rendezvous hashing in place of the paper’s consistent-hash rings, and it omits the paper’s migration of queued requests between the two candidates, because Dynamo places each request once.
- **ramjet** approximates Ramjet’s per-request load units from per-worker load, reads Dynamo’s KV-event index instead of Ramjet’s own approximate prefix index, and omits Ramjet’s lane that confines very long prompts to designated replicas.
- **llm-d-precise-prefix**’s queue score counts running as well as waiting requests, where llm-d reads the engine’s waiting queue.
- **llm-d-optimized-baseline**’s prefill rate is llm-d’s hardware calibration for this model, GPU and parallelism, while replay times prefill with AIS; the port also omits llm-d’s optional exploration probability.

**Table 15:** Tuning spaces. Each searched parameter is listed with its range, “log” if it is searched on a log scale, and its starting value, which is the shipped default except where marked.  $d$ : free parameters. Flips: the share of decisions that one CMA-ES step of size  $\sigma_0$  changes, measured from the policy’s defaults on candidate tables reconstructed from the pilot’s default-router replays of 12 train cells (target 0.175). Every space caps the per-coordinate standard deviation at 0.3. M1’s row is for comparison.

| Policy                   | Searched: range (scale), start                                                                                                                                                                             | Pinned                                                                            | $d$ | $\sigma_0$ | Flips                   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|------------|-------------------------|
| default cost (M0)        | $\beta_{\text{ov}}$ [0, 4], 1;<br>$\beta_{\text{decay}}$ [0, 4], 0;<br>$\beta_{\text{pf}}$ [0.05, 500] log, 1;<br>$\beta_{\text{req}}$ [0.5, 4096] log, 0.5 <sup>a</sup> ;<br>router_temperature [0, 1], 0 | —                                                                                 | 5   | 0.11       | 0.169                   |
| two-tier                 | cache_threshold [0, 1], 0.5;<br>balance_abs_threshold [1, 1024] log, 32;<br>balance_rel_threshold [1, 8] log, 1.1                                                                                          | —                                                                                 | 3   | 0.3        | 0.083                   |
| LMetric                  | class_prefix_blocks [1, 64] log, 4;<br>window_requests [50, 50,000] log, 1000                                                                                                                              | hot-spot detector on                                                              | 2   | 0.3        | 0.0006                  |
| Ramjet                   | alpha [0.1, 160] log, 4;<br>max_affinity_blocks [1, 1024] log, 32;<br>load_unit_tokens [256, 262,144] log, 8192                                                                                            | affinity block 512 tokens <sup>b</sup> ;<br>basis <b>relative</b>                 | 3   | 0.3        | 0.133                   |
| DualMap                  | hash_prefix_blocks [1, 64] log, 4 <sup>c</sup> ;<br>pending_prefill_token_budget [2048, 2,097,152] log, 65,536;<br>window_requests [50, 50,000] log, 1000                                                  | —                                                                                 | 3   | 0.3        | 0.018 <sup>c</sup>      |
| CHWBL                    | prefix_tokens [16, 16,384] log, 256 <sup>c</sup> ;<br>load_factor [1, 8] log, 1.25                                                                                                                         | —                                                                                 | 2   | 0.16       | 0.173 <sup>c</sup>      |
| llm-d precise prefix     | queue weight [0.01, 100] log, 1;<br>KV-utilization weight [0.01, 100] log, 1;<br>prefix_match_length_weight [0, 1], 0;<br>prefix_match_length_scale_tokens [512, 131,072] log, 8192                        | prefix weight 2 <sup>b</sup> ;<br>per-class weights empty                         | 4   | 0.16       | 0.177                   |
| llm-d optimized baseline | affinity_threshold [0, 1], 0.8;<br>max_ttft_penalty_ms [100, 3,240,000] log, 18,000;<br>queue_threshold_tokens [16,384, 1,073,741,824] log, 4,194,304                                                      | peak prefill rate 15,928 tokens/s <sup>b</sup> ;<br>TTFT source <b>throughput</b> | 3   | 0.3        | 0.034                   |
| sticky, hard             | the five default-cost knobs, with M0’s ranges and starts                                                                                                                                                   | mode <b>hard</b> ;<br>65,536 sessions                                             | 5   | 0.17       | 0.179                   |
| sticky, bounded          | load_factor [1, 8] log, 1.25;<br>the five default-cost knobs, as M0                                                                                                                                        | mode <b>bounded</b> ;<br>65,536 sessions                                          | 6   | 0.067      | 0.175                   |
| learned (M1)             | $\theta_1, \dots, \theta_7$ , bounds as in Section 6.2                                                                                                                                                     | $\theta_0 = -1$ , $\tau = 0$                                                      | 7   | 0.004      | about 0.20 <sup>d</sup> |

<sup>a</sup> The shipped value of  $\beta_{\text{req}}$  is 0, which a log scale cannot reach; 0.5, at most half a block per active request, stands in for it. <sup>b</sup> Pinned because it only rescales the other parameters. <sup>c</sup> Hashed key depth, held at its default when  $\sigma_0$  was set: any change of it redraws the prefix-to-worker map (0.69 of decisions change for **chwb1** and 0.18 for **dualmap** at a step of 0.03). <sup>d</sup> Measured in the pilot, from  $\theta = -\mathbf{e}_0$ .

- Two-tier reads Dynamo’s KV-event overlap instead of the SGLang router’s approximate cache history, and breaks ties by worker ID instead of the router’s worker-list order.
- `chwb1` replaces the hash ring with rendezvous hashing.

Tuning absorbs two of these: `dualmap`’s budget is searched directly, and a miscalibrated prefill rate in the optimized baseline is equivalent to a different TTFT penalty, which is searched. The missing queued-prefill term in `lmetric` is structural, and no parameter setting adds it. The learned family is not limited the same way. Feature set `v2` can express the paper’s product with the queued term:  $\theta = -(\mathbf{e}_8 + \mathbf{e}_{10})$ , on `log_ptok` and `log_bs` (Table 9), takes its argmin. The arm M1-v2, one of the candidates for the headline learned model, uses `v2`, and one of its restarts starts near that point (Section 5.9). M1-v2 was selected as the headline learned arm, so the headline policy can represent a variant of LMetric that the `lmetric` baseline cannot. This asymmetry bounds the headline’s scope (Section 9.11).

Only `dynamo-default-cost-fn` and `sticky-session` draw random numbers, for ties and softmax samples; the harness seeds them with  $k + 1$  on replicate  $k$ , as it seeds `learned-choice`. The other policies break ties by worker ID or by a rotating counter. Every baseline therefore replays identically on a given replicate.

### 7.3 Excluded and secondary baselines

ThunderAgent’s worker selection [19] follows the preference of a paired request classifier, and replay runs worker-selection policies but not classifiers, so it is not evaluated and lies outside every claim. SMetric [54] is not in the router catalog, and the operator decision rules out building new implementations; its published advantage also assumes a global KV tier, which replay does not have.

A later amendment allows new arms to read signals from AIS, the performance model that times replay, in a separate league with its own leaderboard. Its baselines are M0 run with the router’s AIS prefill-load model (`router_prefill_load_model: ais`) and the optimized baseline with `ttft_source: modeled`, which estimates each worker’s TTFT from that model’s prefill backlog and counts a worker without an estimate as slowest. Both are tuned with the same budget and normalized by the same  $\pi_0$  without AIS. Because these policies read estimates from the model that also times the simulation, the league is a secondary comparison and never enters the headline. Running the default cost with the AIS load model is not neutral: it changed the default router’s own routing on 12 of 12 probe pairs, so M0-ais shows that effect explicitly. All four arms of the league ran with three restarts at the same budget; results are in Section 9.10.

**Fairness.** Equal budget does not mean equal search difficulty. The baseline spaces have 2 to 6 free parameters against M1’s 7 and M1-v2’s 22. In five of them a CMA-ES step changes few decisions even at the largest allowed step. `lmetric`’s two parameters act only through its hot-spot detector, which on these workloads rarely alarms; the space’s notes attribute this to every worker holding the few hot class prefixes, so tuned `lmetric` should stay close to its defaults. The hashed key depths of `chwb1` and `dualmap` make their objectives jump. The equal-budget rule fixes what each policy may spend, not how far each can get. In the other direction, the learned arms’ restarts start from the default router, a behavior clone of the strongest untuned heuristic in the pilot (`llm-d-precise-prefix`) and a third informed point, for M1-v2 the faithful LMetric point (Section 6.4), while every baseline restarts from its shipped defaults. These starting points cost no objective evaluations but carry information from heuristic decisions, so the test also reports M1 restricted to its default start (Amendment A12), and Section 9.11 shows how much of the learned margin the faithful LMetric point carries by itself. No baseline was still improving at the end of its budget (Section 6.7).

**Table 16:** Splits. A segment is a disjoint piece of a source trace (Table 6); all cells of one segment, whatever their  $N$ , load or transform, are one workload for statistics. <sup>†</sup>One train cell pools the plays of AgentX segments A1 and A2.

| Split      | Cells | Segments        | Worker counts $N$  | Open / closed / lanes |
|------------|-------|-----------------|--------------------|-----------------------|
| Train      | 34    | 10 <sup>†</sup> | 4, 8               | 18 / 8 / 8            |
| Validation | 14    | 6               | 4, 6, 8            | 6 / 4 / 4             |
| Test       | 60    | 12              | 2, 4, 6, 8, 16, 32 | 31 / 15 / 14          |

**Table 17:** Test cells by worker count and primary hold-out axis, and the number of test segments at each worker count. Window: held-out Mooncake time window; Family: FAST25; Plays: held-out AgentX plays; Seeds: held-out session seeds. Every cell at  $N \in \{2, 6, 16, 32\}$  is also on a held-out segment.

| $N$ | Worker count | Transform | Window | Family | Plays | Seeds | Cells | Segments |
|-----|--------------|-----------|--------|--------|-------|-------|-------|----------|
| 2   | 8            |           |        |        |       |       | 8     | 6        |
| 4   |              | 4         | 3      | 2      | 2     | 2     | 13    | 7        |
| 6   | 7            |           |        |        |       |       | 7     | 6        |
| 8   |              | 6         | 3      | 4      | 2     | 2     | 17    | 8        |
| 16  | 7            |           |        |        |       |       | 7     | 6        |
| 32  | 8            |           |        |        |       |       | 8     | 6        |
| All | 30           | 10        | 6      | 6      | 4     | 4     | 60    | 12       |

## 8 Evaluation protocol

The protocol that decides every comparison in this paper was fixed before any policy was tuned. The primary test was written at calibration on 2026-10-02, before any training run, in `facts/HEAD LINE_TEST.json`. Later amendments added secondary comparisons and fixed which learned arm is the headline model (Amendments A12 and A14 to A17), all before any test cell was evaluated; none of them changed the primary test.

### 8.1 Splits and hold-out axes

Each test cell is held out from training along one primary axis (Table 17). The 30 cells at unseen worker counts also lie on held-out segments: 14 on held-out Mooncake windows, 8 on held-out session seeds and 8 on held-out AgentX plays. The 10 transform cells use values outside the train ranges: prompt stretches of 2.5 (three cells), output-length multipliers of 0.25, 2.5 and 4, think-time multipliers of 0.25 (one cell) and 4 (two cells), and four times as many prefix roots. Training sees only  $N \in \{4, 8\}$ , so  $N = 2, 16$  and  $32$  are extrapolation and  $N = 6$  is interpolation. Validation also has 4 cells at  $N = 6$ , so every configuration was selected with that worker count in view, and the 7 test cells at  $N = 6$  are labeled *selection-exposed* (Amendment A11.3; 7).

**Unit of independence.** Cells are not independent: all cells cut from one segment share its arrival and length structure, whatever their worker count, load level or transform. Every statistic below is therefore computed over segments. The test has 12, holding between 1 and 15 cells each. The FAST25 conversation cells share the arrival and length skeleton of Mooncake windows w3 and w5 and are averaged into those two segments. CRN replicates and duplicated trace copies (Amendment A4) re-draw or lengthen a segment but add no independent units. One train cell draws from the plays of two AgentX train segments (Table 16); train segments enter no test statistic.

## 8.2 Noise floor

What counts as a real difference depends on three sources of variation, which the campaign measured separately.

- **Repeated runs: none.** Replay is deterministic for a given policy, workload and seed, so a repeated run reproduces every per-request row and measures nothing (Amendment A1). Changing only the default router’s tie-break seed moved its goodput by 1.8% on an 8-worker setup cell. Every policy therefore runs seeded, and noise is measured over workload replicates, each of which re-draws the arbitrary parts of the workload and the policy seed (Section 4.1).
- **Replicate noise.** Over 8 replicates per cell at calibrated loads,  $\pi_0$ ’s windowed goodput had a coefficient of variation between 0.37% and 6.5% across the 48 train and validation cells (median 1.4%). The paired ratio of round-robin to  $\pi_0$  on the same replicate had a pooled standard deviation of 0.027 on train and 0.026 on validation.
- **Segment heterogeneity.** Policies differ by segment far more than replicates differ. In the pilot’s comparison of M1 with `llm-d-precise-prefix`, the pooled replicate standard deviation of the paired ratio was 0.015, while the standard error of the mean over 6 validation segments was 0.019 and was, by the pilot’s own reading, dominated by differences between segments. The mean lead of +0.030 came almost entirely from one Mooncake segment (+0.102), and M1 trailed on all three AgentX segments (−0.012 to −0.025); these numbers are *preliminary (pilot, validation split)*.

Segment heterogeneity, not replicate noise, limits what the campaign can resolve; hence the segment-level test and minimum detectable effect below. The pilot gate used Amendment A1’s two-clause rule: a policy differs from  $\pi_0$  if on some validation cell its mean paired ratio differs from 1 by more than three pooled replicate standard deviations, *and* the mean of its per-segment deltas exceeds two standard errors over at least three segments. The headline test below replaces this rule for the final comparison.

## 8.3 Selection hygiene

The data are split into four disjoint uses, so that selection cannot leak into the result.

1. Train cells, replicates  $k = 0, \dots, 7$ : the CMA-ES objective (Section 6.1).
2. Validation cells,  $k = 0, 1, 2$ : selection within a policy (Section 6.6).
3. Validation cells,  $k = 3, \dots, 10$ : selection across policies, of  $\pi_{\text{base}}^*$  and of the headline learned arm  $\pi_\theta$ ; the pilot gate also used them. No within-policy selection reads them.
4. Test cells,  $k = 0, 1, 2$ : one pass, for a fixed list of finalists.

The test split was frozen at calibration. `facts/test_freeze.json` records the SHA-256 of the test cell file, of each of the 60 cells’ contents, of the 39 test traces and their 7 sources, of the engine configuration and of the split manifest, together with the bindings build and harness version; the test stage verified all of them (149 of 149) before its single pass. A completeness audit later found that the test cells were touched once: the 30,060 test-cell records in the cache are exactly the expected set, all written after the finalists file was frozen. The one exception predates calibration: five default-router-only validation replays of the AgentX trace conversion ran on test play pools, at pre-calibration loads, before the test set was frozen; they compared no policies. Every learned variant scored on validation (rung, rank, feature set, restart) is counted and reported: 314 variants, of which 5 arms competed for the headline.

## 8.4 Headline test (pre-registered)

The headline compares the selected learned policy  $\pi_\theta$  with the best baseline  $\pi_{\text{base}}^*$  (Section 6.6) on the test set. For each test segment  $\omega$ ,

$$D_\omega = \frac{1}{|\omega|} \sum_{c \in \omega} \frac{1}{3} \sum_{k=0}^2 \Delta_{c,k}(\pi_\theta, \pi_{\text{base}}^*), \quad (11)$$

where  $|\omega|$  is the number of test cells in the segment, so each segment weighs the same however many cells it holds. The primary test is a one-sided exact Wilcoxon signed-rank test over the 12 segment statistics, against the alternative that their median is positive, at level 0.05 [10].

It compares one learned policy with one baseline, each chosen on validation before the test, so no multiplicity correction applies. It does not test the stronger claim that the learned policy beats *every* baseline. The literature review proposed that claim as an intersection of per-baseline tests (LR-11); Amendment A2.1 replaced it with the comparison against the validation-selected best baseline.

**What 12 segments allow.** With 12 segments the smallest attainable one-sided  $p$  is  $1/4096$ . The test rejects at 0.05 exactly when the segments on which the learned policy loses have ranks, by  $|D_\omega|$ , summing to at most 17; it can therefore tolerate a few losses if they are small. With 4 segments the exact test cannot reach 0.05 at all (its minimum  $p$  is  $1/16$ ), and with 5 it needs every segment to favor the same policy. Per-family claims, which rest on 2 to 4 segments, are therefore descriptive only.

**Effect size.** We report  $\Pr(\pi_\theta \succ \pi_{\text{base}}^*)$ , the share of segments with  $D_\omega > 0$ , with a 95% bootstrap interval that resamples whole segments together with all their cells. The effect is called significant if the interval’s lower end exceeds 0.5 and meaningful if its upper end exceeds 0.75, following Bouthillier et al. [6]. Next to it we report the interquartile mean and the geometric mean of segment ratios with intervals [2], the arithmetic mean of per-cell ratios (the contract’s original objective), a per-cell map of which policy wins, and the gap to the per-cell best tuned baseline.

**Minimum detectable effect.** The MDE is  $\max(1\%, 2\text{SE})$ , where SE is the standard error of the paired segment mean measured on validation at the pilot. Against `llm-d-precise-prefix` at shipped defaults, the best untuned heuristic on validation then, it is 0.038 in clipped log-ratio (SE 0.019 over 6 segments); against  $\pi_0$  the same rule gives 0.087. Both are *preliminary (pilot, validation split)*. The MDE is not a threshold of the headline test. It is used where a decision needs a size: a difference within it counts as parity at  $N = 2$  (Section 8.5), it set the drift-check rule (Section 6.8), and M2 at ranks 3 and 4 trains only if rank 2 beats M1 on validation by at least the MDE.

## 8.5 Strata and parity

**Strata.** Every table and the bootstrap are also given per load mode (8 open-loop, 6 closed-loop and 4 agentic-lane segments), per worker count, for the 10 transform-extrapolation cells (7 segments) as their own stratum, and per family, descriptively. A significance claim for an individual worker count uses Holm’s correction across the six counts.

Assuming each per-count test is the same exact test, the correction leaves little room. Holm’s first threshold is  $0.05/6 \approx 0.0083$ , below the smallest  $p$  attainable with 6 segments ( $1/64 \approx 0.016$ ). Each unseen worker count,  $N = 2, 6, 16$  and  $32$ , has 6 test segments, so none of them can produce a

significant per-count result, whatever the data; only  $N = 4$  (7 segments, all favoring one policy) and  $N = 8$  (8 segments), both train worker counts, can. Results at unseen worker counts are therefore descriptive, or take the parity and no-regression form below. This is our own arithmetic, not part of the pre-registration.

**Parity and no regression.** Two kinds of claim say “no worse”. For **learned-choice** at  $\theta = -\mathbf{e}_0$  against  $\pi_0$ , and for any claim of no regression at an unseen worker count, the protocol uses two one-sided tests with a margin of  $\pm 0.5\%$  normalized goodput. At  $N = 2$  a difference within the MDE counts as parity rather than failure, because informed routing tends to tie simpler policies when there are few workers to choose from [52]. The  $\pm 0.5\%$  margin is small next to the pilot’s segment standard error of 0.019 (*preliminary (pilot, validation split)*), so before the test we expected most no-regression claims to remain unestablished. In the event, the check held at every unseen worker count against both  $\pi_0$  and **ramjet**: the lower end of the interval exceeded  $-\ln 1.005$  (Section 9.5).

## 8.6 Secondary comparisons

These were registered before the test pass and are reported next to the headline. Only the headline is a confirmatory test; the rest are descriptive or secondary.

- **Learned-arm variants:** M1-default-init, M1 selected over its default start alone (Amendment A12, Section 6.4), and M1-noaff (Amendment A15), as test finalists.
- **Reference points:** M0, round-robin, and every heuristic at its shipped defaults (Amendment A2.5).
- **Oracles:** the best baseline per family and the best tuned baseline per cell, which no single policy can match (Amendment A2.1). The gap to the per-cell oracle shows how much a per-workload choice of heuristic would add, a comparison Xia et al. [58] suggest for learned network policies.
- **The AIS league** (Amendment A16): arms that read AIS-derived estimates at runtime, compared with the best AIS-informed baseline and with the best baseline overall. Every arm is normalized by the same  $\pi_0$ , which does not use AIS, so the two leagues share one scale and the report can state how much the AIS estimates add. These arms cannot become  $\pi_\theta$ .
- **Every tuned baseline and the runner-up.** The learned arm against each of the 11 tuned baselines, as an intersection-union test (LR-11), and against the best baseline of another type, **llm-d-precise-prefix**, registered because the validation choice between it and **ramjet** was a near tie.
- **Token-weighted goodput** (Amendment A19): the headline pipeline with each good in-window request weighted by its prompt plus output tokens, a check that the request-count objective does not hide a loss in served work.

## 8.7 Robustness checks

These checks are reported, not tested, for the finalists on the frozen test cells and replicates. Each asks whether the headline ranking survives a change that training never saw. Agreement between two rankings of the finalists is summarized by Kendall’s rank correlation  $\tau_K$ . As a reading rule taken from the literature review rather than from the pre-registration, a headline gap smaller than the spread these perturbations induce is reported as fragile (LR-14).

**SLO re-scoring.** Every evaluation keeps its per-request rows, so the good-request indicator (Equation (4)) can be recomputed under another SLO without new replays or retraining. The scale sweep multiplies  $(I_{\max}, s_{\max})$  by 0.5, 0.75, 1, 1.5, 2 and 3 and states where the headline ranking flips. Amendment A14 adds four other definitions of a good request:

- the mean-ITL bound alone;
- the end-to-end slowdown bound alone;
- the mean-ITL bound plus a TTFT bound that grows with prompt length,  $b + a \text{ISL}_j$ , where  $a$  is three times AIS’s uncontended per-token prefill time and  $b$  is  $\pi_0$ ’s median TTFT at L2 on train cells;
- the mean-ITL bound plus an absolute end-to-end bound,  $\pi_0$ ’s 90th-percentile end-to-end latency at L2 on train cells.

Their constants are computed per family from train records and written down before test scoring. For each definition the report gives every finalist’s goodput ratio against  $\pi_0$ , the paired segment delta against  $\pi_{\text{base}}^*$ , and  $\tau_K$  against the ranking under the trained SLO. None of them changes selection or the headline test. The constants were computed from 160 train records of  $\pi_0$  at L2 and written down before any test record was scored; for Mooncake, for example, the TTFT bound’s intercept is  $\pi_0$ ’s median TTFT there, 146.7 ms.

**Router-state lag.** Replay hands the router perfectly fresh state, and load balancing on stale information can behave very differently [33]. A replay patch delays three kinds of worker-originated updates by  $\delta$  milliseconds of simulated time: KV cache events, prefill completions and request completions. A delayed update becomes visible at the first router callback at or after its due time; the router’s own bookkeeping of what it has dispatched stays immediate, as in a live router. The patch lives in a separate worktree and build, so the tuning build and its result cache are untouched. On 8 train cells, 4 policies and 2 replicates (64 runs),  $\delta = 0$  reproduced the tuning build in every non-environment record field and in all 404,704 per-request rows. At  $\delta = 50$  and 200 ms routing changed in all 64 runs, and goodput relative to  $\delta = 0$  ranged from 0.972 to 1.023 and from 0.956 to 1.020; these single-replicate ratios show that the knob is live, not how lag affects the ranking. The test stage re-evaluated 17 finalists on all test cells at  $\delta \in \{10, 50, 200\}$  ms, with a lag-0 run on the lag build as its identity check, and reports whether the headline ranking holds; M1-noaff ran alongside M1 to test the hypothesis that session affinity matters more under lag (Amendment A15). Training never sees lag.

**Timing perturbation.** Two timing scale factors of the mock engines are set to 0.8 and 1.2. The first, `speedup_ratio`, scales every forward pass; the second, `decode_speedup_ratio`, scales decode only, so the effective decode speedup is their product. This tests sensitivity of the ranking to a global error in AIS’s timing, not to errors confined to particular batch sizes or context lengths. A timing perturbation also changes what an uncontended request can achieve, so each perturbed record is scored twice: against the nominal  $E_0$  and against the perturbation-consistent  $E'_0 = \text{prefill}/s + \text{decode}/(s d)$  for speedup  $s$  and decode speedup  $d$ . Under the nominal  $E_0$  at  $s = 0.8$ , no zero-reuse AgentX request can be good, a degenerate scoring that is flagged where it appears. The spread rule and  $\tau_K$  use  $E'_0$ .

**Herding guards.** A deterministic argmax over stale state can herd requests onto one worker. Two variants guard against it: `learned-choice` at  $\tau > 0$ , which samples instead of taking the argmax (at

a fixed temperature, the probability left to non-best workers grows with  $N$ ), and a variant restricted to the request’s hash-home workers (feature `hash_home` of `v2`). Both were registered before M1-v2 won, so they ran as variants of M1 (`v1`), at two temperatures, on the lag build (Section 9.9).

**Live validation.** A check on GPUs of the relative result, with its own pre-registered analysis, is described in Section 11; it validates the ranking only and never feeds selection.

## 8.8 Audits

Each campaign stage ends at a checkpoint that independent auditors review before the next stage starts, each auditor through one named lens. A static audit reads code and artifacts; a dynamic audit also re-runs part of the pipeline in a private result root and recomputes results with its own code, without importing the campaign harness. Findings are graded blocker, major or minor. A fixer answers each major finding, either fixing it or rebutting it with evidence, and the lens is audited again.

Table 18 lists the rounds. Through the pilot gate there were 17 audit reports and 6 fix reports. All 10 major findings were judged valid and none was rebutted. Nine were fixed and the fix verified by a later round; the tenth, a second public trace that turned out to be a relabeled copy of the Mooncake trace, was resolved by excluding that family. The pilot gate was reviewed the same way: an independent judge recomputed every pilot number from raw records with its own scripts, reproduced each to four decimals, and approved escalation. Two more checkpoints followed. At the phase-2 midpoint, three dynamic audits looked for overfitting, objective gaming and simulator exploitation; their five major findings were all fixed, including the gaming finding of Section 10. After the single test pass, a completeness critic and three adversarial refuters reviewed the headline before any report was written; their two major findings were fixed by adding the missing secondary metrics and by narrowing the headline’s wording to the scope the evidence supports (Section 9.11).

Every evaluation record carries guard metrics that these audits read: the largest worker’s share of requests against the cap  $\min(2/N, 1/N + 0.25)$ , the largest worker’s prefill tokens over the mean, the share of multi-turn sessions split across workers, the share of requests slower than three times the slowdown bound, and the largest worker’s share within each prompt-length bucket, which exposes length segregation. Black-box search is known to exploit gaps in its objective [22]; an evolution strategy tuning a router on 95th-percentile TTFT alone sent all traffic to one replica [42]; and binary goodput invites sacrificing requests that would miss their SLO anyway [55]. The gaming audit showed that the original guards, computed over all rows, could not see a worker whose in-window requests were all bad, so the midpoint fix added window-basis guards: the share cap over exactly the scored requests, each worker’s good fraction, the good fraction of long prompts against  $\pi_0$ , and the normalized mutual information between worker and prompt-length quartile (Section 10).

**Table 18:** Audit rounds by checkpoint and lens, with each major finding and its resolution.

| Checkpoint, lens                                   | Rounds              | Major findings → resolution                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Setup, AIS and policy in effect                    | r0 pass             | none                                                                                                                                                                                                                                                                                                             |
| Setup, determinism, cost, context                  | r0, r1 fail         | identical re-runs cannot measure noise → CRN workload replicates (Amendment A1); the toolagent trace is a relabeled Mooncake copy → family excluded                                                                                                                                                              |
| Build, goodput and normalization                   | r0–r2 fail, r3 pass | reports paired records across builds and cell contents → pairing on content, build and protocol; closed-loop and lane windows scored the drain → window ends at full occupancy; the slowdown clause was a coin flip at scale 1 → $E_0$ aligned with replay’s decode context, plus a $10^{-6}$ relative tolerance |
| Build, feature leakage                             | r0 fail, r1 pass    | v2’s <code>hash_home</code> keyed on replay-synthesized session identifiers → keyed on the prompt prefix only                                                                                                                                                                                                    |
| Build, harness robustness and splits               | r0 fail, r1 pass    | Mooncake windows overlapped across splits → six disjoint slices; the remote bundle lacked a dependency → shipped and checked at build time                                                                                                                                                                       |
| AgentX lowering, isolation and arrivals            | r0, r1 pass         | none                                                                                                                                                                                                                                                                                                             |
| AgentX lowering, parity and fidelity               | r0 fail, r1 pass    | think-time and output-length transforms could not be lowered → supported                                                                                                                                                                                                                                         |
| Calibration, independent re-run                    | r0 fail, r1 pass    | open-loop session windows opened during the session-population ramp → warm-up of two session-lifetime envelopes, sessions re-calibrated, test re-frozen                                                                                                                                                          |
| Pilot gate, independent recompute                  | escalate            | none; all pilot numbers reproduced                                                                                                                                                                                                                                                                               |
| Phase-2 midpoint, overfitting and selection        | fail, fixed         | the drift check was recorded as decisive although underpowered → recorded as inconclusive, not used for M2’s sources                                                                                                                                                                                             |
| Phase-2 midpoint, gaming and concentration         | fail, fixed         | M1 games the objective (a 0%-good worker of long requests) → sign-constrained re-runs (Section 6.5); the remaining size segregation → re-audited and reported on every learned row                                                                                                                               |
| Phase-2 midpoint, simulator exploitation           | fail, fixed         | the long-prompt sacrifice is stable under perturbation → same re-runs; timing robustness scored against the nominal $E_0$ → consistent $E'_0$ scoring                                                                                                                                                            |
| Final, completeness                                | fail, fixed         | TTFT by prompt length and the planned secondary metrics were missing → computed from the test records, no replay (Section 10)                                                                                                                                                                                    |
| Final, refute the headline (reproduction)          | pass                | none; the headline reproduced bit for bit                                                                                                                                                                                                                                                                        |
| Final, refute the headline (fairness)              | fail, fixed         | faithful LMetric, outside the baseline pool, carries most of the margin → headline worded as beating the branch’s ported heuristics (Section 9.11)                                                                                                                                                               |
| Final, refute the headline (robustness, simulator) | pass                | none                                                                                                                                                                                                                                                                                                             |

## 9 Results

Every number in this section comes from AIS-timed offline replay. The tables marked *generated* are copied character for character from the campaign’s final report, which builds them from the single test pass and the robustness records without any replay. Intervals are 95% percentile intervals from a bootstrap that resamples whole test segments with all their cells; where a stratum has fewer than three segments, cells are resampled within each segment instead, marked †, and those intervals are optimistic because cells of one segment are not independent. Only the pre-registered headline is a test; every other  $p$ -value is descriptive.

### 9.1 Selection: the finalists

Each policy’s configuration was chosen on validation replicates  $k = 0, 1, 2$  over its three restarts and their checkpoints. The cross-policy choices were then made on fresh replicates  $k = 3, \dots, 10$ , by the same rule for baselines and learned arms, and the finalists file was frozen before the first test record was written.

- **Best baseline.** Tuned `ramjet`, 0.1411, ahead of `ramjet` with a tuned queue threshold (0.1405) and `llm-d-precise-prefix` (0.1393). The choice between `ramjet` and `llm-d-precise-prefix` was a near tie: `ramjet` led by 0.0018 in the mean over cells but trailed by 0.0087 in the mean over segments, so the comparison with `llm-d-precise-prefix` was registered as a secondary before the test pass.
- **Headline learned arm.** M1-v2, 0.1783, ahead of M1-noaff (0.1692), M1 with a tuned queue threshold (0.1684), M2 rank 2 (0.1677) and M1 (0.1656). Its configuration is the best-so-far sample at generation 20 of the restart that started from the default router ( $\theta = -\mathbf{e}_0$ ). On the fresh replicates it led `ramjet` by +0.0372 per cell and +0.0290 per segment, on 5 of 6 validation segments.
- **Multiplicity.** 314 learned variants were validated over the campaign (every validation checkpoint of every learned run, the pilot and the post-hoc variants); 5 learned arms competed for the headline.

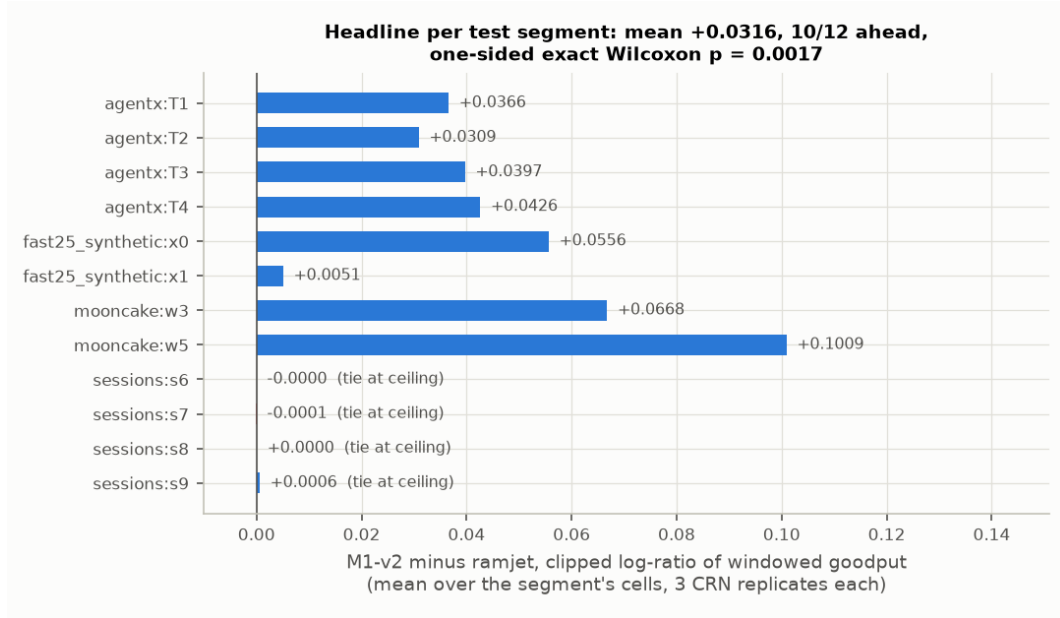
### 9.2 Headline: M1-v2 against the best baseline

**Headline, as the final audit allows.** In AIS-timed simulation, the learned router M1-v2 beats the branch’s ported heuristics, each tuned with an equal budget. Against the validation-selected best baseline, tuned `ramjet`, its windowed goodput on the 60 frozen test cells is higher by a segment-mean clipped log-ratio of +0.0316 (95% segment-bootstrap interval [+0.0150, +0.0501]; geometric-mean ratio 1.032, interval [1.015, 1.051]). It is ahead on 10 of 12 independent test segments and on all 8 informative ones; the 4 session segments are ties at ceiling. The pre-registered one-sided exact Wilcoxon test gives  $p = 0.0017$ , and  $\Pr(\text{M1-v2} \succ \text{ramjet}) = 0.833$  with interval [0.583, 1.0], significant and meaningful by the registered rule.

Table 19 and Figure 5 show the twelve units the test counts. The claim was pre-registered, evaluated once, reproduced bit for bit from the raw per-request rows by an independent auditor, and not refuted by the robustness and simulator-dependence audit. It survives leaving out any one segment (largest  $p$  0.0034), any one family (largest  $p$  0.027), the  $N = 6$  cells and the FAST25 conversation cells; with the latter removed the segment mean is +0.026 and  $p$  is unchanged.

**Table 19:** The headline per test segment:  $D_\omega$  (Equation (11)), the mean over the segment’s cells and replicates  $k = 0, 1, 2$  of the clipped log-ratio of M1-v2’s windowed goodput against tuned **ramjet**’s.

| Segment                                                | Family                      | Cells | Load modes (cells) | $D_\omega$                 |
|--------------------------------------------------------|-----------------------------|-------|--------------------|----------------------------|
| w3                                                     | Mooncake 11, FAST25 conv. 2 | 13    | open 11, closed 2  | +0.0668                    |
| w5                                                     | Mooncake 14, FAST25 conv. 1 | 15    | open 9, closed 6   | +0.1009                    |
| s6                                                     | sessions                    | 6     | open 6             | −0.0000                    |
| s7                                                     | sessions                    | 2     | open 1, closed 1   | −0.0001                    |
| s8                                                     | sessions                    | 2     | open 1, closed 1   | +0.0000                    |
| s9                                                     | sessions                    | 5     | open 1, closed 4   | +0.0006                    |
| T1                                                     | AgentX                      | 4     | lanes 4            | +0.0366                    |
| T2                                                     | AgentX                      | 2     | lanes 2            | +0.0309                    |
| T3                                                     | AgentX                      | 5     | lanes 5            | +0.0397                    |
| T4                                                     | AgentX                      | 3     | lanes 3            | +0.0426                    |
| x0                                                     | FAST25 synthetic            | 2     | open 1, closed 1   | +0.0556                    |
| x1                                                     | FAST25 synthetic            | 1     | open 1             | +0.0051                    |
| Segment mean (SE)                                      |                             |       |                    | +0.0316 (0.0093)           |
| One-sided exact Wilcoxon signed-rank $p$ , 12 segments |                             |       |                    | 0.0017                     |
| Pr(M1-v2 $\succ$ ramjet) over segments, 95% interval   |                             |       |                    | 0.833 [0.583, 1.0]         |
| Geometric mean of segment ratios, 95% interval         |                             |       |                    | 1.032 [1.015, 1.051]       |
| Interquartile mean of segment ratios, 95% interval     |                             |       |                    | 1.026 [1.005, 1.050]       |
| Mean over cells (SE); cells ahead                      |                             |       |                    | +0.0506 (0.0117); 50 of 60 |

**Figure 5:** The headline per test segment (copied from the final report). The four session segments tie at ceiling; every other segment favors M1-v2.

**Limits that travel with the headline.** The final audit requires each of the following next to the claim.

1. **Only the sign is a claim.** The effect is below the pre-registered minimum detectable effect (0.038) and below the spread that timing perturbations induce (0.060). Its magnitude depends on the perturbation (Section 9.9).
2. **It is not “learning beats every heuristic”.** The baseline pool lacks faithful LMetric, which lies inside M1-v2’s class; added to the default cost, it carries most of M1-v2’s validation margin (Section 9.11).
3. **It holds at the calibrated SLO and looser.** At SLO scales 1 to 3 it is significant; at 0.75 and 0.5 the sign holds but significance does not, and at 0.75 all four session segments turn against M1-v2 (Section 9.9).
4. **It is a request-count property.** The token-weighted goodput is mixed; prompts of 32K to 64K tokens wait longer for their first token than under `ramjet`, and prompts of 32K tokens and more lose about 4 points of good fraction on average (Section 10).
5. **The headline is simulated.** The live finalist runs (Section 11; validation only, on six  $N = 4$  test cells chosen after the test pass, not at random, with one run per policy) agree with the simulated policy ranking (pooled Kendall  $\tau_K$ -b 1.00, per cell 0.87 to 1.00) and with the sign of M1-v2 minus `ramjet` on hardware. They test neither the magnitude nor the significance of the headline, and they validate neither absolute goodput (live over simulated 0.72 to 2.18, depending on the policy) nor the size of the gaps between policies: on the sessions cell, the simulated gain of every tuned policy over  $\pi_0$  is not reproduced beyond live noise.

**Secondary statements that hold.**

- **Each tuned baseline.** M1-v2 is Wilcoxon-significant against each of the 11 tuned baselines (intersection-union test, largest  $p$  0.0105). The effect-size leg of LR-11 fails for `lmetric` and sticky-bounded, which M1-v2 trails by up to 0.5% on the at-ceiling session segments. It also beats every heuristic at its shipped defaults (largest  $p$  0.0105).
- **The best baseline on test.** On test the best baseline is `llm-d-precise-prefix` (tuned), not `ramjet`. M1-v2 beats it by +0.0293, ahead on 12 of 12 segments ( $p = 0.00024$ ); this comparison was registered before the test pass.
- **Per cell.** M1-v2 is the single best of the 29 router-observable test policies on 31 of 60 cells, at or above the per-cell best tuned baseline (a virtual best no single policy can match) on 42 of 60 cells with a mean gap of +0.037, and above  $\pi_0$  on all 60.

### 9.3 Every policy

Figure 6 places every test policy against  $\pi_0$ ; Table 40 in the appendix gives the numbers. Four observations frame the headline.

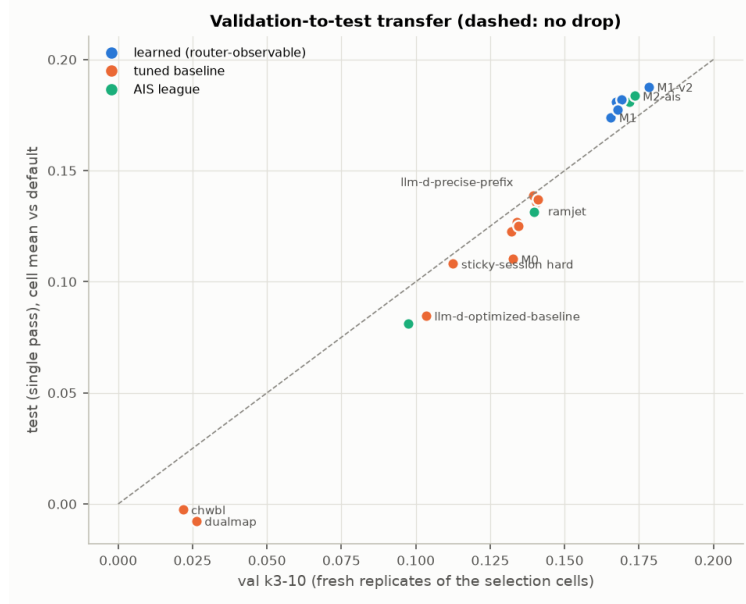
- **Cache-aware routing matters a lot here.** The learned arms and the strong heuristics beat  $\pi_0$  by 0.13 to 0.18 in segment-mean clipped log-ratio, while round-robin, which ignores the cache, loses 0.56.
- **Tuning the default’s knobs is not enough.** M0, the default cost function tuned with the same budget, gains +0.116 over  $\pi_0$ , less than untuned `llm-d-precise-prefix` (+0.144).



**Figure 6:** Every test policy against  $\pi_0$  (60 cells, 12 segments; segment means with 95% segment-bootstrap intervals). Blue: router-observable learned arms; orange: tuned baselines; green: the AIS league; purple: shipped defaults. Copied from the final report.

- **Hash-pinned routing hurts on these workloads.** Tuned chwbl and dualmap are near  $\pi_0$  ( $-0.030$  and  $-0.004$ ), and below it at their shipped defaults.
- **Validation transferred to test.** The ranking on fresh validation predicts the test ranking (Figure 7); test numbers are higher than validation numbers mainly because test holds more cells at  $N = 16$  and  $32$ , where every cache-aware policy gains more.

An independent plain-ratio view of the same records gives M1-v2 a mean normalized goodput of  $1.222$   $[1.154, 1.286]$  against ramjet's  $1.156$   $[1.108, 1.216]$ .



**Figure 7:** Validation (fresh replicates) against test, mean over cells of the clipped log-ratio against  $\pi_0$ . The two compare different cells, so the dashed line is a reference, not an expectation.

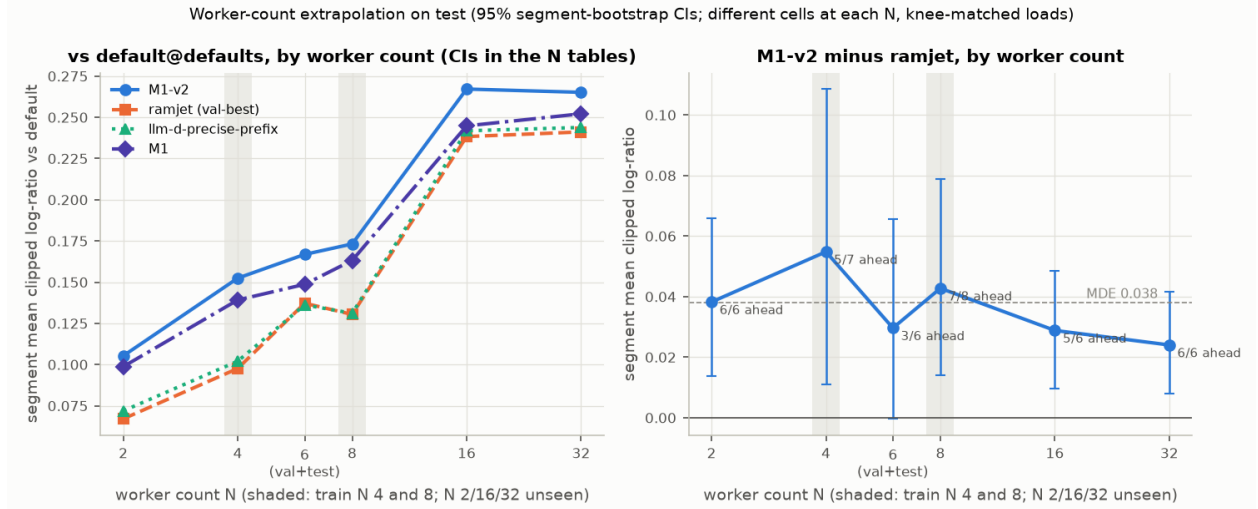
**Table 20:** The headline by stratum, M1-v2 minus tuned **ramjet** (generated). One-sided exact Wilcoxon  $p$  over the stratum’s segments; only the first row is a test. Rows marked  $\dagger$  use the within-segment cell bootstrap.

| Stratum                          | Cells | Seg. | M1-v2 – ramjet [95% CI]              | Ahead | One-sided $p$ |
|----------------------------------|-------|------|--------------------------------------|-------|---------------|
| <b>all (pre-registered test)</b> | 60    | 12   | +0.0316 [+0.0150, +0.0501]           | 10/12 | 0.0017        |
| load mode: open loop             | 31    | 8    | +0.0281 [+0.0032, +0.0608]           | 5/8   | 0.0742        |
| load mode: closed loop           | 15    | 6    | +0.0354 [+0.0106, +0.0633]           | 4/6   | 0.0781        |
| load mode: agentic lanes         | 14    | 4    | +0.0374 [+0.0331, +0.0411]           | 4/4   | 0.0625        |
| $N = 2$ (unseen)                 | 8     | 6    | +0.0382 [+0.0138, +0.0660]           | 6/6   | 0.0156        |
| $N = 4$ (train count)            | 13    | 7    | +0.0548 [+0.0110, +0.1085]           | 5/7   | 0.0391        |
| $N = 6$ (selection-exposed)      | 7     | 6    | +0.0296 [−0.0004, +0.0655]           | 3/6   | 0.2188        |
| $N = 8$ (train count)            | 17    | 8    | +0.0427 [+0.0140, +0.0788]           | 7/8   | 0.0117        |
| $N = 16$ (unseen)                | 7     | 6    | +0.0288 [+0.0096, +0.0484]           | 5/6   | 0.0312        |
| $N = 32$ (unseen)                | 8     | 6    | +0.0240 [+0.0079, +0.0416]           | 6/6   | 0.0156        |
| unseen $N$ (2, 16, 32)           | 23    | 7    | +0.0342 [+0.0159, +0.0515]           | 7/7   | 0.0078        |
| seen $N$ (4, 8)                  | 30    | 12   | +0.0346 [+0.0123, +0.0610]           | 9/12  | 0.0081        |
| transforms beyond train ranges   | 10    | 7    | +0.0250 [+0.0071, +0.0443]           | 4/7   | 0.1094        |
| family: Mooncake                 | 25    | 2    | +0.0505 [+0.0368, +0.0645] $\dagger$ | 2/2   | 0.2500        |
| family: FAST25 conversation      | 3     | 2    | +0.3868 [+0.2857, +0.4879] $\dagger$ | 2/2   | 0.2500        |
| family: FAST25 synthetic         | 3     | 2    | +0.0304 [+0.0164, +0.0443] $\dagger$ | 2/2   | 0.2500        |
| family: sessions                 | 15    | 4    | +0.0001 [−0.0001, +0.0004]           | 2/4   | 0.4375        |
| family: AgentX                   | 14    | 4    | +0.0374 [+0.0331, +0.0411]           | 4/4   | 0.0625        |

#### 9.4 Strata: load mode, worker count, family and transforms

Table 20 breaks the headline down. Only the first row is a test; the rest are descriptive, and per-worker-count claims would need Holm’s correction, under which every per-count  $p$  is at least 0.070.

**Load modes.** M1-v2 leads in all three modes, by +0.028 in open loop, +0.035 in closed loop and +0.037 in agentic lanes, none significant on its own ( $p$  0.06 to 0.08). AgentX ran only in closed-loop lanes.



**Figure 8:** Worker-count extrapolation on test (copied from the final report). Left: against  $\pi_0$  by  $N$ . Right: M1-v2 minus tuned **ramjet** by  $N$ , with segments ahead. Shaded: the train counts 4 and 8. Each  $N$  holds different cells, and loads are matched to  $\pi_0$ 's knee at each  $N$ , not to a fixed per-worker rate.

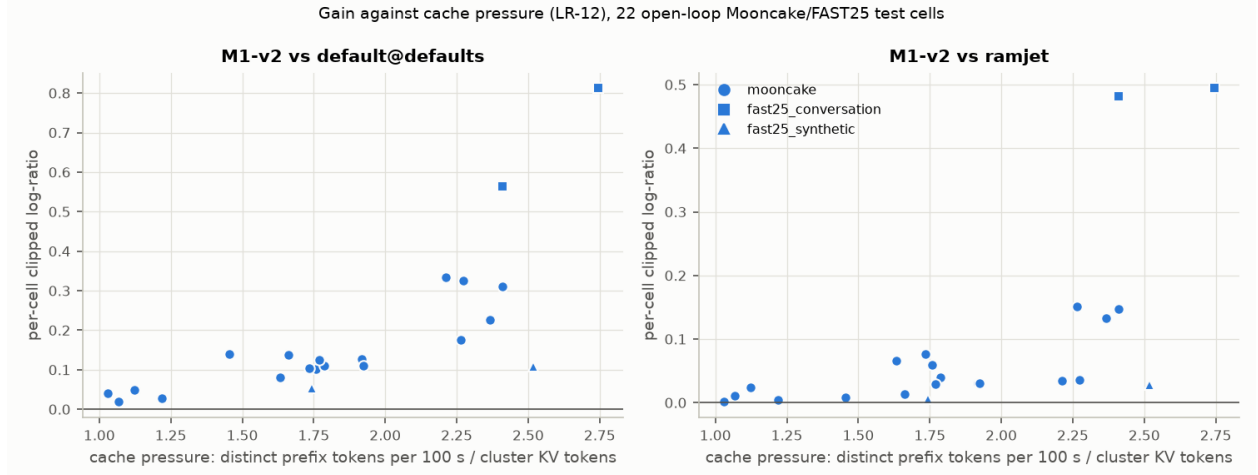
**Table 21:** By worker count, against tuned **ramjet** (generated; selected rows). Segment means with 95% intervals. Cells (segments) per  $N$ : 2: 8 (6); 4: 13 (7); 6: 7 (6), selection-exposed; 8: 17 (8); 16: 7 (6); 32: 8 (6). Table 41 in the appendix gives the same against  $\pi_0$ .

| Policy                            | $N = 2$                 | 4                       | 6                       | 8                       | 16                      | 32                      |
|-----------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.038 [+0.014, +0.066] | +0.055 [+0.011, +0.109] | +0.030 [−0.000, +0.065] | +0.043 [+0.014, +0.079] | +0.029 [+0.010, +0.048] | +0.024 [+0.008, +0.042] |
| M2-ais                            | +0.024 [+0.002, +0.054] | +0.053 [+0.014, +0.101] | +0.029 [+0.004, +0.060] | +0.039 [+0.012, +0.073] | +0.028 [+0.011, +0.045] | +0.026 [+0.009, +0.043] |
| M1 (= A12 M1-default-init)        | +0.032 [+0.007, +0.058] | +0.042 [−0.002, +0.096] | +0.011 [−0.023, +0.047] | +0.033 [+0.006, +0.066] | +0.006 [−0.004, +0.022] | +0.011 [+0.000, +0.023] |
| M1 unconstrained (gaming-flagged) | +0.015 [+0.000, +0.036] | +0.042 [+0.006, +0.089] | +0.014 [−0.005, +0.037] | +0.024 [+0.003, +0.053] | +0.007 [+0.001, +0.012] | +0.013 [+0.003, +0.023] |
| llm-d-precise-prefix              | +0.005 [−0.018, +0.028] | +0.004 [−0.013, +0.021] | −0.001 [−0.018, +0.019] | +0.001 [−0.010, +0.013] | +0.004 [−0.018, +0.026] | +0.003 [−0.013, +0.019] |
| lmetric (port)                    | +0.003 [−0.008, +0.013] | −0.009 [−0.029, +0.010] | −0.014 [−0.039, +0.013] | −0.016 [−0.037, −0.000] | −0.010 [−0.024, +0.004] | −0.012 [−0.023, −0.001] |
| M0 (default cost fn, tuned)       | −0.012 [−0.024, −0.002] | −0.050 [−0.130, −0.003] | −0.030 [−0.067, −0.006] | −0.022 [−0.044, −0.006] | −0.044 [−0.095, −0.010] | −0.029 [−0.052, −0.008] |
| round_robin                       | −0.392 [−0.552, −0.213] | −0.480 [−0.645, −0.332] | −0.786 [−1.043, −0.500] | −0.584 [−0.804, −0.386] | −0.817 [−1.058, −0.530] | −0.813 [−1.080, −0.494] |
| default@defaults (reference)      | −0.067 [−0.141, −0.021] | −0.098 [−0.182, −0.044] | −0.137 [−0.198, −0.077] | −0.130 [−0.191, −0.075] | −0.238 [−0.357, −0.124] | −0.241 [−0.338, −0.137] |

## 9.5 Held-out worker counts

Every learned arm was trained on  $N \in \{4, 8\}$  only and runs unchanged at any  $N$ , because its utility is per worker and its features are set-size invariant (Section 5.5).

- **The margin persists at unseen worker counts.** Over **ramjet** it is positive at every  $N$ : +0.038 at  $N = 2$ , +0.029 at 16 and +0.024 at 32 (Table 21). Over the 7 segments with cells at unseen counts it is ahead on all 7 (+0.034, uncorrected  $p$  0.008); this stratum was not pre-registered as a test.
- **No regression.** The pre-registered no-regression check holds at  $N = 2, 16$  and 32 against both **ramjet** and  $\pi_0$ : the lower end of the interval exceeds  $-\ln 1.005$ .
- **Gains over the default grow with  $N$**  for every cache-aware policy, M1-v2's from +0.105 at  $N = 2$  to +0.27 at 16 and 32, but the learned margin over **ramjet** does not grow at the unseen large counts.
- $N = 6$  is **selection-exposed** and the weakest stratum: 3 of 6 segments ahead,  $p$  0.219.
- **The curve is not a pure  $N$  effect.** Each  $N$  holds different cells and segments, and with loads matched to  $\pi_0$ 's knee per  $N$  the per-worker load varies 1.2 to 1.9 times across  $N$ . A per-worker-matched extrapolation was not run.



**Figure 9:** Per-cell gain against cache pressure on the 22 open-loop Mooncake and FAST25 test cells (copied from the final report). Cache pressure is distinct prefix tokens per 100s over the cluster’s KV capacity. The trend is descriptive: pressure is confounded with load level, and cells of one segment are not independent.

**Table 22:** By family, against tuned **ramjet** (generated; selected rows). Each family has 2 to 4 segments, so these are descriptive; † marks the within-segment cell bootstrap. Table 42 in the appendix gives the same against  $\pi_0$ .

| Policy                            | Mooncake                 | FAST25 conv.             | FAST25 synth.            | Sessions                | AgentX                  |
|-----------------------------------|--------------------------|--------------------------|--------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.050 [+0.037, +0.064]† | +0.387 [+0.286, +0.488]† | +0.030 [+0.016, +0.044]† | +0.000 [−0.000, +0.000] | +0.037 [+0.033, +0.041] |
| M2-ais                            | +0.046 [+0.034, +0.059]† | +0.362 [+0.270, +0.453]† | +0.029 [+0.016, +0.042]† | +0.002 [+0.000, +0.004] | +0.031 [+0.025, +0.034] |
| M1 (= A12 M1-default-init)        | +0.041 [+0.029, +0.053]† | +0.365 [+0.263, +0.467]† | +0.033 [+0.022, +0.043]† | −0.001 [−0.002, −0.000] | +0.004 [−0.019, +0.025] |
| M1 unconstrained (gaming-flagged) | +0.028 [+0.017, +0.040]† | +0.343 [+0.245, +0.441]† | +0.002 [−0.007, +0.012]† | −0.002 [−0.003, −0.000] | +0.015 [+0.010, +0.020] |
| llm-d-precise-prefix              | −0.014 [−0.019, −0.008]† | +0.095 [+0.088, +0.102]† | −0.021 [−0.030, −0.012]† | −0.010 [−0.015, −0.004] | +0.029 [+0.025, +0.034] |
| lmetric (port)                    | −0.027 [−0.031, −0.022]† | −0.097 [−0.098, −0.095]† | −0.034 [−0.046, −0.022]† | +0.003 [+0.001, +0.005] | +0.004 [−0.004, +0.012] |
| M0 (default cost fn, tuned)       | −0.010 [−0.014, −0.005]† | −0.151 [−0.155, −0.147]† | −0.036 [−0.038, −0.034]† | −0.012 [−0.019, −0.005] | −0.044 [−0.079, −0.010] |
| round_robin                       | −0.272 [−0.324, −0.223]† | −0.430 [−0.432, −0.427]† | −0.360 [−0.373, −0.347]† | −0.935 [−1.040, −0.829] | −0.740 [−0.865, −0.562] |
| default@defaults (reference)      | −0.104 [−0.132, −0.077]† | −0.200 [−0.200, −0.200]† | −0.089 [−0.115, −0.064]† | −0.178 [−0.264, −0.088] | −0.155 [−0.219, −0.063] |

**Cache pressure.** On the 22 open-loop Mooncake and FAST25 test cells, M1-v2’s gain rises with cache pressure: Spearman  $\rho$  0.79 against  $\pi_0$  and 0.71 against **ramjet** (Figure 9). Pressure ranges from 1.0 to 2.7, so no test cell has little cache pressure, a design gap calibration recorded.

## 9.6 Families and transforms

Table 22 shows where the margin comes from. It is largest on the FAST25 conversation cells (+0.387), a family no policy was trained on; those three cells carry about 35% of M1-v2’s cell-mean gain, and their request-count gains (+0.48, +0.49, +0.08) contrast with token-weighted deltas of +0.02, −0.15 and +0.02 (Section 10). On Mooncake the lead is +0.050, on AgentX +0.037 with all four segments ahead, and on FAST25 synthetic +0.030. Sessions sit at ceiling at the calibrated SLO: every strong policy serves nearly every in-window session request well, so the four session segments tie. The ten cells with transforms beyond the train ranges show +0.025 with 4 of 7 segments ahead ( $p$  0.109), so generalization to held-out transform values is not established.

## 9.7 The model ladder and the ablations

Tables 23 and 24 and Figure 10 read as follows.

- **M0 → M1 is the one clear step.** Learning over the default cost’s features beats tuning the default cost’s knobs: +0.047 per segment, 11 of 12 segments ahead.

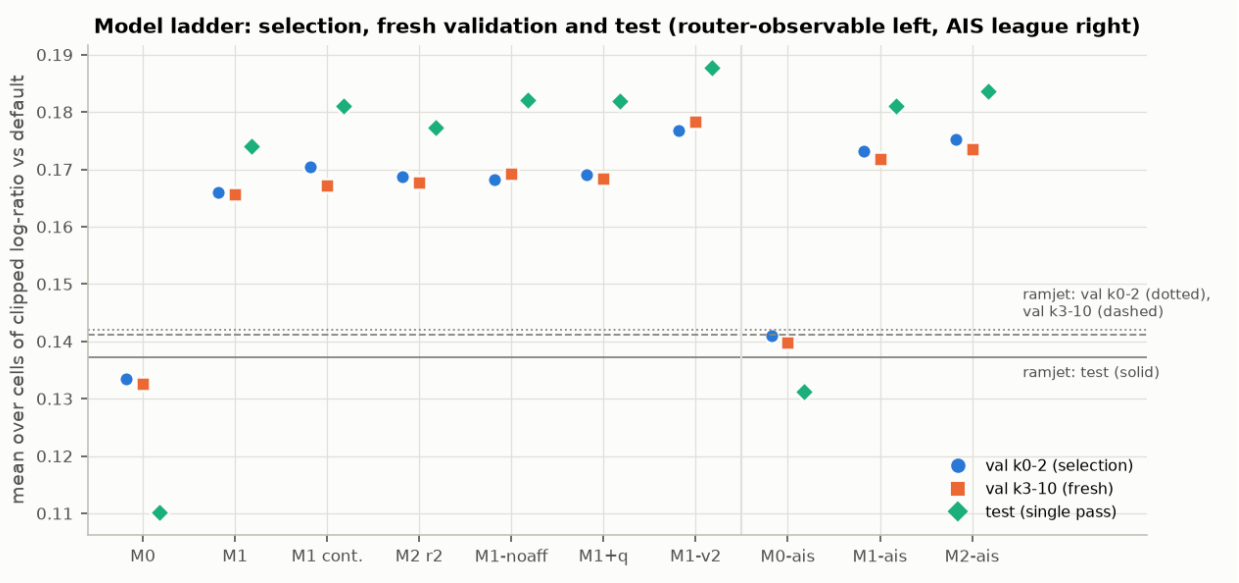
**Table 23:** The model ladder (generated). Selected: restart and generation of the configuration chosen on validation replicates  $k_{0..2}$ ; columns 3 and 4 are mean clipped log-ratios against  $\pi_0$  on the selection and on the fresh replicates; test against  $\pi_0$  is a mean over cells; test against tuned **ramjet** a segment mean with its 95% interval and one-sided  $p$ .

| Rung                                      | Selected (restart, generation) | Val $k_{0..2}$ | Val $k_{3..10}$ | Test vs $\pi_0$ | Test vs ramjet [95% CI]    | $p$    |
|-------------------------------------------|--------------------------------|----------------|-----------------|-----------------|----------------------------|--------|
| M0                                        | m0-s3 gen 20 mean              | 0.1334         | 0.1326          | +0.1102         | −0.0281 [−0.0439, −0.0137] | 0.9993 |
| M1                                        | m1c-s1 gen 20 mean             | 0.1660         | 0.1656          | +0.1741         | +0.0186 [+0.0013, +0.0378] | 0.0549 |
| M1 continued (+400 evals)                 | m1cont-s3 gen 25 best_so_far   | 0.1704         | 0.1672          | +0.1811         | +0.0256 [+0.0090, +0.0437] | 0.0212 |
| M2 rank 2                                 | m2r2-s3 gen 25 mean            | 0.1688         | 0.1677          | +0.1774         | +0.0211 [+0.0043, +0.0406] | 0.0549 |
| M1-noaff (A15)                            | m1noaff-s1 gen 25 best_so_far  | 0.1682         | 0.1692          | +0.1821         | +0.0270 [+0.0109, +0.0443] | 0.0134 |
| M1 + queue threshold (ablation a)         | ablaml-s1 gen 25 mean          | 0.1690         | 0.1684          | +0.1819         | +0.0272 [+0.0115, +0.0444] | 0.0171 |
| M1-v2 (A17.1)                             | m1v2-s1 gen 20 best_so_far     | 0.1767         | 0.1783          | +0.1877         | +0.0316 [+0.0150, +0.0501] | 0.0017 |
| M1 unconstrained (tier A, gaming-flagged) | m1-s2 gen 15 mean              | 0.1636         | n/a             | +0.1687         | +0.0147 [+0.0032, +0.0284] | 0.0386 |
| M0-ais                                    | m0ais-s2 gen 20 best_so_far    | 0.1410         | 0.1397          | +0.1313         | −0.0019 [−0.0086, +0.0048] | 0.6890 |
| M1-ais                                    | m1ais-s1 gen 25 best_so_far    | 0.1731         | 0.1718          | +0.1810         | +0.0256 [+0.0113, +0.0423] | 0.0012 |
| M2-ais                                    | m2ais-s3 gen 10 mean           | 0.1751         | 0.1735          | +0.1837         | +0.0286 [+0.0138, +0.0458] | 0.0005 |

**Table 24:** Rung-to-rung comparisons on test (generated; descriptive).

| Comparison ( $A - B$ )                                         | Segment mean [95% CI]      | Ahead | $p$ ( $A > B$ ) | Cell mean |
|----------------------------------------------------------------|----------------------------|-------|-----------------|-----------|
| M1 (= A12 M1-default-init) − M0 (default cost fn, tuned)       | +0.0467 [+0.0248, +0.0700] | 11/12 | 0.0007          | +0.0639   |
| M2 rank 2 − M1 (= A12 M1-default-init)                         | +0.0025 [−0.0008, +0.0070] | 6/12  | 0.2349          | +0.0033   |
| M1 continued (secondary) − M1 (= A12 M1-default-init)          | +0.0071 [−0.0024, +0.0195] | 6/12  | 0.3386          | +0.0070   |
| M2 rank 2 − M1 continued (secondary)                           | −0.0045 [−0.0134, +0.0029] | 8/12  | 0.5452          | −0.0037   |
| M1-v2 (headline learned arm) − M1 (= A12 M1-default-init)      | +0.0130 [+0.0039, +0.0244] | 11/12 | 0.0034          | +0.0137   |
| M1-noaff − M1 (= A12 M1-default-init)                          | +0.0084 [−0.0012, +0.0211] | 7/12  | 0.2119          | +0.0081   |
| M1 + queue threshold (abl. a) − M1 continued (secondary)       | +0.0015 [−0.0002, +0.0035] | 8/12  | 0.0549          | +0.0008   |
| ramjet + queue threshold (abl. a) − ramjet (val-best baseline) | −0.0016 [−0.0050, +0.0010] | 4/12  | 0.8833          | −0.0017   |
| M1 (= A12 M1-default-init) − M1 unconstrained (gaming-flagged) | +0.0039 [−0.0065, +0.0144] | 8/12  | 0.1902          | +0.0053   |
| M1-ais − M0-ais                                                | +0.0275 [+0.0104, +0.0486] | 10/12 | 0.0017          | +0.0497   |
| M2-ais − M1-ais                                                | +0.0030 [+0.0010, +0.0053] | 9/12  | 0.0046          | +0.0027   |
| M2-ais − M1-v2 (headline learned arm)                          | −0.0029 [−0.0052, −0.0006] | 3/12  | 0.9788          | −0.0040   |
| M0-ais − M0 (default cost fn, tuned)                           | +0.0262 [+0.0101, +0.0445] | 10/12 | 0.0017          | +0.0211   |

- **M1 is the sign-constrained re-run.** The unconstrained tier-A M1 gamed the objective (Section 10); it is reported only as a flagged secondary row, and it is weaker on test.
- **M1 → M2 rank 2: no gain.** The context term adds +0.0025 on test (6 of 12 segments) and does not beat M1 continued for the same added 400 evaluations (−0.0045). The pre-registered gate for ranks 3 and 4 failed: M2 rank 2 minus the better of M1 and M1 continued was −0.0016 on validation, against an MDE of 0.038, so ranks 3 and 4 were not run. M3 (nested logit) was never built, since this deployment has no nests.
- **M1 → M1-v2 is the step that passes.** The extended feature set, with log, set-relative, interaction and prefill-attention terms, beats M1 by +0.013 on 11 of 12 segments ( $p$  0.0034). The originally planned M1 and M2 alone do *not* pass against **ramjet** on test ( $p$  0.055 each); the passing arm is a redesign made after the pilot, on validation evidence (Section 9.11).
- **Ablation (a), a jointly tuned queue threshold: nothing measurable.** The learned arm with the threshold against M1 continued is +0.0015 ( $p$  0.055), and **ramjet** with the threshold against **ramjet** is −0.0016. Removing the tuned threshold after the fact changes the validation score by +0.0001 and −0.0006.



**Figure 10:** The model ladder on the selection replicates, the fresh validation replicates and test (copied from the final report). Left: router-observable arms; right: the AIS league.

- **Session affinity (A15).** M1-noaff, retrained without  $\theta_6$ , is not worse than M1 on test (+0.0084,  $p$  0.21), and zeroing  $\theta_6$  in the selected M1 changes its validation score by  $-0.0010$ . Under router-state lag, M1 minus M1-noaff stays between  $-0.008$  and  $-0.005$ , so the hypothesis that affinity gains value as the router’s state goes stale is not supported. For M1-v2, zeroing  $\theta_6$  changes the validation score by  $+0.0005$ .
- **Initialization (A12).** The headline M1 came from the s1 restart, the default router, so the registered M1-default-init row is the same policy as M1. M1-v2 also came from s1; its informed restarts s2 (the behavior clone) and s3 (the faithful LMetric point) reached 0.1609 and 0.1724 on the selection replicates, against s1’s 0.1767.

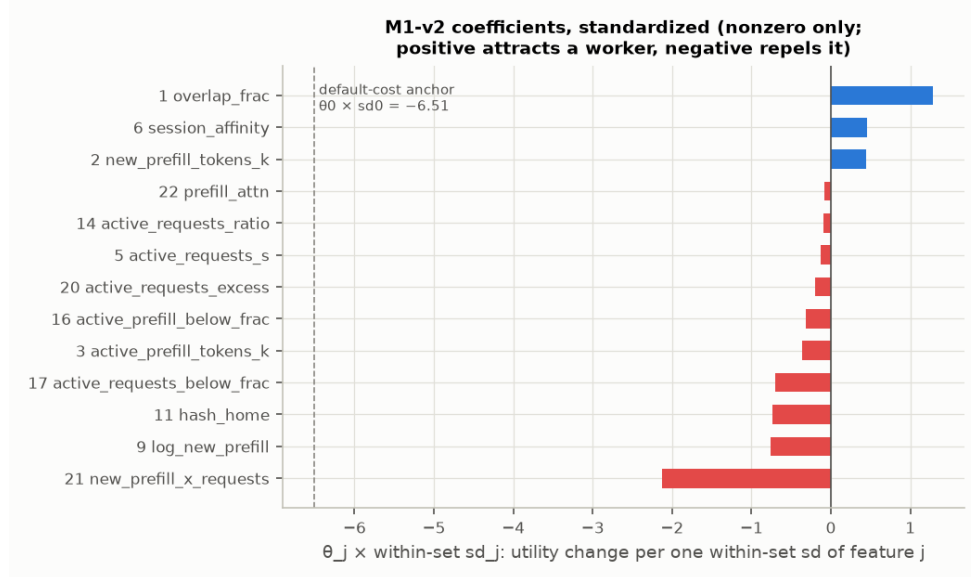
## 9.8 What M1-v2 learned

The shipped artifact is one policy YAML: `learned-choice` with `feature_set: v2`,  $\tau = 0$ , `tie_break: one_draw` and the 23 coefficients of Table 25. It reads no AIS signal; a decision costs  $O(Nd)$  with  $d = 23$ . Following LR-15, we read the coefficients through standardized magnitudes and ablations, not raw values: Table 25 scales each  $\theta_f$  by the feature’s within-candidate-set standard deviation on train, so  $\theta_f \text{sd}_f$  is the utility change for a typical difference between two candidates. The anchor, the default cost per request block, has  $\theta_0 \text{sd}_0 = -6.51$ .

1. **The default router plus corrections.**  $\theta_0 = -1$  dominates; every learned term is a fraction of the anchor’s spread.
2. **Uncached prefill costs more the busier the worker.** The largest learned term is `new_prefill_x_requests` ( $\theta_{21} \text{sd}_{21} = -2.13$ ). With  $\theta_2$ , the linear weight on uncached prefill is  $+1.275 - 0.525 \times (\text{active requests})$  per 8K tokens, negative on any worker with three or more active requests: uncached work is steered to near-idle workers, while cache affinity is kept on lightly loaded ones. Zeroing  $\theta_{21}$  costs 0.018 on validation (13 of 14 cells worse), and the fairness audit attributes M1-v2’s increment over the in-class LMetric heuristic to this term.

**Table 25:** M1-v2’s coefficients (generated).  $sd_f$ : pooled within-candidate-set standard deviation of feature  $f$  on train; constraint: the sign rule of Section 6.5.

| $f$ | Feature                    | $\theta_f$ | $sd_f$  | $\theta_f sd_f$ | Constraint |
|-----|----------------------------|------------|---------|-----------------|------------|
| 0   | default_logit_scaled       | −1         | 6.51    | −6.510          | pinned −1  |
| 1   | overlap_frac               | +5.034     | 0.2561  | +1.289          | free       |
| 2   | new_prefill_tokens_k       | +1.275     | 0.3519  | +0.449          | free       |
| 3   | active_prefill_tokens_k    | −0.2811    | 1.285   | −0.361          | $\leq 0$   |
| 4   | kv_load_frac               | +0         | 0.08438 | +0.000          | $\leq 0$   |
| 5   | active_requests_s          | −2.146     | 0.06    | −0.129          | $\leq 0$   |
| 6   | session_affinity           | +1.391     | 0.3244  | +0.451          | free       |
| 7   | isl_x_prefill_load         | +0         | 2.13    | +0.000          | $\leq 0$   |
| 8   | log_ptok                   | +0         | 1.156   | +0.000          | $\leq 0$   |
| 9   | log_new_prefill            | −0.9276    | 0.8219  | −0.762          | free       |
| 10  | log_bs                     | +0         | 0.2708  | +0.000          | $\leq 0$   |
| 11  | hash_home                  | −1.582     | 0.4668  | −0.738          | free       |
| 12  | kv_load_ratio              | +0         | 0.2377  | +0.000          | $\leq 0$   |
| 13  | active_prefill_ratio       | +0         | 0.9491  | +0.000          | $\leq 0$   |
| 14  | active_requests_ratio      | −0.348     | 0.2576  | −0.090          | $\leq 0$   |
| 15  | kv_load_below_frac         | +0         | 0.2817  | +0.000          | $\leq 0$   |
| 16  | active_prefill_below_frac  | −1.326     | 0.2398  | −0.318          | $\leq 0$   |
| 17  | active_requests_below_frac | −2.617     | 0.267   | −0.699          | $\leq 0$   |
| 18  | kv_load_excess             | +0         | 0.05047 | +0.000          | $\leq 0$   |
| 19  | active_prefill_excess      | +0         | 0.9154  | +0.000          | $\leq 0$   |
| 20  | active_requests_excess     | −5.845     | 0.03463 | −0.202          | $\leq 0$   |
| 21  | new_prefill_x_requests     | −16.79     | 0.1268  | −2.128          | $\leq 0$   |
| 22  | prefill_attn               | −0.2967    | 0.2846  | −0.084          | free       |



**Figure 11:** M1-v2’s nonzero coefficients, standardized (copied from the final report). Positive attracts a worker, negative repels it; the default-cost anchor is at −6.51.

**Table 26:** The headline under perturbations and alternative SLOs (generated). “Nominal  $E_0$ ” scores timing-perturbed replays against the unperturbed  $E_0$ ; “ $E'_0$ ” uses the perturbation-consistent reference (Section 8.7).  $s$ : speedup;  $d$ : decode speedup.  $\tau_K$ -b compares the condition’s ranking with the nominal ranking, over the 17 robustness finalists for lag and timing and over all 32 non-default policies for SLO variants. A19 is the token-weighted secondary metric.

| Condition                   | M1-v2 – ramjet [95% CI]    | Ahead | $p$    | $\tau_K$ -b | Rank M1-v2 / ramjet |
|-----------------------------|----------------------------|-------|--------|-------------|---------------------|
| nominal                     | +0.0316 [+0.0150, +0.0501] | 10/12 | 0.0017 | 1.000       | 1 / 7 of 17         |
| lag 10 ms                   | +0.0312 [+0.0139, +0.0508] | 9/12  | 0.0081 | 0.950       | 1 / 7 of 17         |
| lag 50 ms                   | +0.0288 [+0.0125, +0.0477] | 10/12 | 0.0024 | 0.950       | 1 / 6 of 17         |
| lag 200 ms                  | +0.0304 [+0.0135, +0.0497] | 9/12  | 0.0061 | 0.917       | 1 / 7 of 17         |
| s0.8 nominal $E_0$          | +0.0790 [+0.0042, +0.1666] | 8/12  | 0.1167 | 0.783       | 1 / 10 of 17        |
| s0.8 $E'_0$                 | +0.0814 [+0.0229, +0.1511] | 9/12  | 0.0105 | 0.800       | 1 / 10 of 17        |
| s1.2 nominal $E_0$          | +0.0151 [+0.0064, +0.0247] | 10/12 | 0.0081 | 0.950       | 1 / 6 of 17         |
| s1.2 $E'_0$                 | +0.0219 [+0.0105, +0.0333] | 10/12 | 0.0046 | 0.950       | 1 / 6 of 17         |
| d0.8 nominal $E_0$          | +0.0473 [+0.0131, +0.0872] | 9/12  | 0.0261 | 0.817       | 1 / 7 of 17         |
| d0.8 $E'_0$                 | +0.0391 [+0.0145, +0.0691] | 10/12 | 0.0024 | 0.917       | 1 / 8 of 17         |
| d1.2 nominal $E_0$          | +0.0224 [+0.0099, +0.0360] | 8/12  | 0.0134 | 0.950       | 1 / 6 of 17         |
| d1.2 $E'_0$                 | +0.0304 [+0.0140, +0.0477] | 9/12  | 0.0105 | 0.950       | 1 / 6 of 17         |
| SLO scale 0.5               | +0.0716 [−0.0289, +0.1734] | 10/12 | 0.0647 | 0.544       | 1 / 13 of 32        |
| SLO scale 0.75              | +0.0353 [−0.0015, +0.0776] | 8/12  | 0.1167 | 0.907       | 1 / 10 of 32        |
| SLO scale 1.5               | +0.0156 [+0.0064, +0.0256] | 12/12 | 0.0002 | 0.956       | 1 / 10 of 32        |
| SLO scale 2.0               | +0.0113 [+0.0052, +0.0178] | 12/12 | 0.0002 | 0.931       | 2 / 10 of 32        |
| SLO scale 3.0               | +0.0120 [+0.0048, +0.0207] | 12/12 | 0.0002 | 0.919       | 2 / 10 of 32        |
| A14 itl_only                | +0.0089 [+0.0022, +0.0161] | 10/12 | 0.0134 | 0.879       | 1 / 16 of 32        |
| A14 e2e_only                | +0.0351 [+0.0152, +0.0574] | 11/12 | 0.0012 | 0.988       | 1 / 11 of 32        |
| A14 ttft_len_itl            | +0.0134 [+0.0046, +0.0227] | 10/12 | 0.0105 | 0.847       | 1 / 10 of 32        |
| A14 abs_e2e_itl             | +0.0189 [+0.0082, +0.0303] | 10/12 | 0.0061 | 0.891       | 1 / 15 of 32        |
| A19 good tokens (secondary) | −0.0092 [−0.0645, +0.0244] | 10/12 | 0.0320 | n/a         | n/a                 |

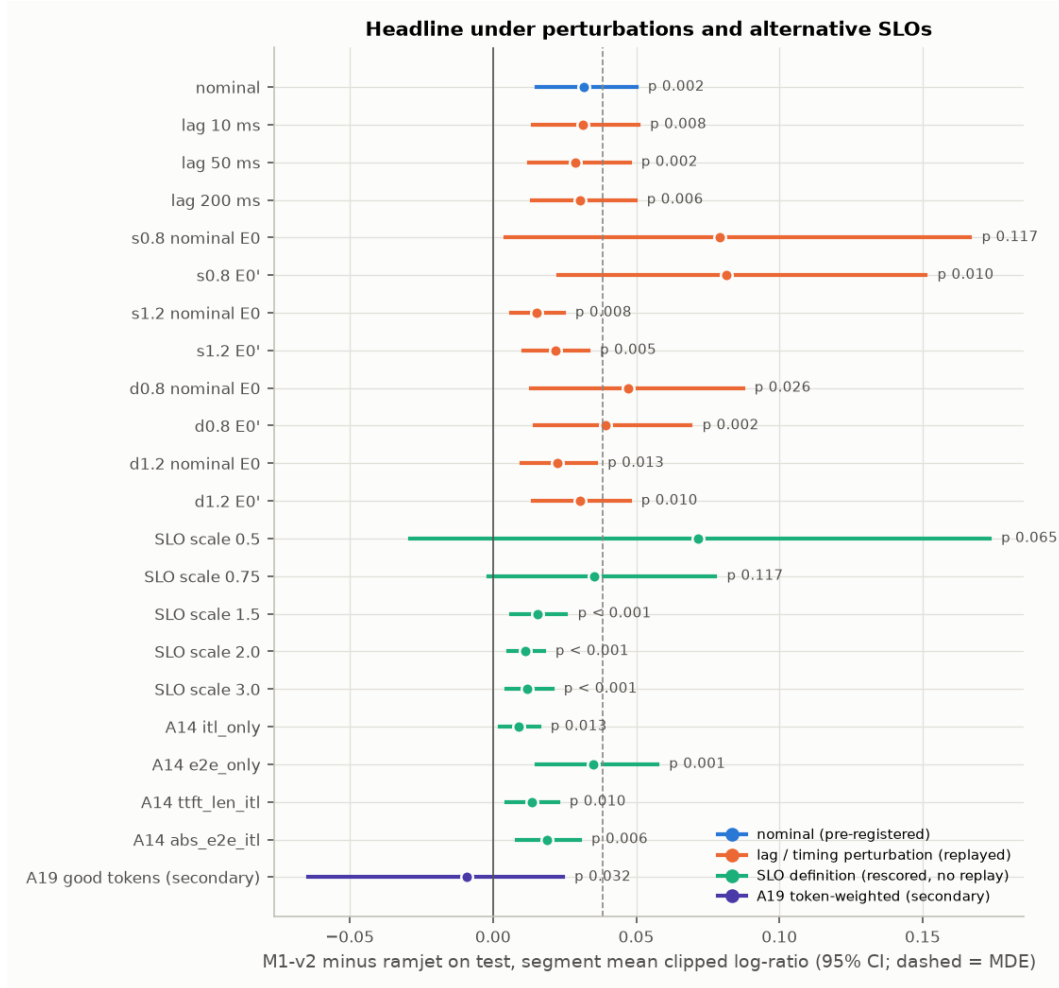
- Cache terms reward hits beyond the default’s credit.** `overlap_frac` (+1.29 per sd) and the log of uncached prefill (−0.76 per sd) both favor the worker holding more of the prompt; the log term rewards near-complete hits most. Zeroing  $\theta_1$  alone costs only 0.002 on validation: overlap is largely redundant with the anchor and the prefill terms.
- Rank-like load terms do the balancing.** The set-relative terms `active_requests_below_frac` (−0.70) and `active_prefill_below_frac` (−0.32), with request excess and ratio, outweigh the absolute load terms (−0.36 and −0.13). LMetric’s own product terms (`log_ptok`, `log_bs`) are 0 in the selected policy.
- Session and prefix-home terms are unneeded.** `session_affinity` (+0.45) and `hash_home` (−0.74, a mild avoidance of the prefix’s rendezvous pair) can each be zeroed with no validation loss (+0.0005 and +0.0010), so the policy depends on neither session IDs nor rendezvous homes.

The v1 arms agree on a simpler structure (Table 45 in the appendix): a strong extra overlap credit ( $\theta_1$  sd<sub>1</sub> of +4.2 to +4.9) and a strong penalty on active requests (−2.1 to −2.9) on top of the default. The unconstrained M1 used overlap and request weights about 3.5 to 3.7 times larger and a positive  $\theta_7$ , the gaming pattern of Section 10.

## 9.9 Robustness

Table 26 and Figure 12 answer whether the ranking survives changes training never saw.

- Router-state lag.** At 10, 50 and 200 ms the lead is +0.031, +0.029 and +0.030 ( $p \leq 0.0081$ ), M1-v2 stays first among the 17 robustness finalists, and the ranking’s  $\tau_K$ -b is 0.92 to 0.95. The lag build reproduces the tuning build at lag 0 per request (180 of 180 test records for  $\pi_0$



**Figure 12:** The headline under perturbations and alternative SLOs (copied from the final report; dashed: the MDE).

and M1, and 448 of 448 validation records for M1-v2, `ramjet`, `llm-d-precise-prefix` and  $\pi_0$ ). Beyond the registered range, on validation only, 1,000 ms gives +0.033 against +0.029 at 0: the baselines lose more to staleness than M1-v2 does.

- **Engine timing.** The sign holds under speedup and decode speedup of 0.8 and 1.2, on both references, and with the ITL bound also rescaled the lead is +0.022 to +0.080 ( $p \leq 0.0081$ ). Speedup 0.8 on the nominal  $E_0$  is not significant ( $p 0.117$ ), a degenerate scoring in which no zero-reuse AgentX request can be good. The spread across conditions (0.060, set by speedup 0.8) exceeds the nominal effect, so by LR-14 the direction is robust and the magnitude is perturbation-dependent.
- **SLO scale.** The sign is positive at every scale; significance holds at 1 to 3 and is lost at 0.75 and 0.5. At 0.75 the four session segments reverse ( $-0.014$  to  $-0.042$ ), as they do at speedup 0.8 on the nominal reference, so “sessions at ceiling” holds only where the SLO does not bind. At 0.5 the scaled slowdown bound falls below 1 for AgentX, so only cache hits can be good there.
- **SLO definition (A14, registered before the test pass).** M1-v2 ranks first under all four

**Table 27:** Herding guards, as variants of M1 on the lag build at lag 0 (generated): segment-mean clipped log-ratio against M1 with 95% intervals, over all test cells and at the unseen worker counts.

| Variant of M1       | vs M1, all cells           | $N = 2$                    | $N = 16$                   | $N = 32$                   |
|---------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| $\tau = 0.747$      | -0.0044 [-0.0103, +0.0008] | -0.0114 [-0.0328, +0.0006] | -0.0016 [-0.0049, +0.0019] | -0.0044 [-0.0077, -0.0014] |
| $\tau = 2.75$       | -0.0277 [-0.0352, -0.0202] | -0.0162 [-0.0227, -0.0092] | -0.0391 [-0.0557, -0.0199] | -0.0383 [-0.0566, -0.0184] |
| hash_home pair only | -0.2626 [-0.4025, -0.1394] | +0.0000 [+0.0000, +0.0000] | -0.5716 [-0.7895, -0.3406] | -0.6844 [-0.9727, -0.3910] |

alternative definitions of a good request, and the ranking’s  $\tau_K$ -b against the trained SLO is 0.85 to 0.99: ITL only +0.0089 ( $p$  0.013), slowdown only +0.035 ( $p$  0.0012), length-scaled TTFT plus ITL +0.013 ( $p$  0.0105), absolute end-to-end plus ITL +0.019 ( $p$  0.0061).

- **Window and warm-up.** Starting the window 25% or 50% later gives +0.0345 and +0.0297; halving or doubling the warm-up, or a common completion window, leaves the result unchanged on the same cells.
- **Simulator model** (validation only, an audit-only AISim build). Against +0.0290 on the unmodified simulator, the lead is +0.0274 when mixed prefill passes are priced attention-equivalent, +0.0291 when priced as a per-request sum, +0.0268 with an extra 0.1 ms of decode cost per running sequence, and +0.0269 when decode inside a prefill pass costs a quarter. These bracket fidelity errors that act on M1-v2’s concentration mechanism; they are not measurements of reality, and FAST25, a test-only family, is not covered.

**Temperature and prefix-home guards.** Two guards against herding were registered before M1-v2 won, so they ran as variants of M1 (Table 27). A positive temperature does not help:  $\tau = 0.747$  is within noise of M1, and  $\tau = 2.75$  costs 0.028, more at  $N = 16$  and 32 than at 2, as expected when the softmax mass on non-best workers grows with  $N$ . Restricting candidates to the prefix’s hash\_home pair is harmful at large  $N$  (−0.57 at 16 and −0.68 at 32); at  $N = 2$  every worker is home, so the variant is identical there. No herding failure of the argmax appeared under the tested lags.

## 9.10 The AIS league

The AIS league may read AIS-modeled prefill and decode estimates at runtime (Sections 5.2 and 7.3); it is a pre-registered secondary comparison.

- **Against its own baseline and the overall best.** M2-ais, the selected AIS-league learned arm, beats M0-ais, the better AIS-informed baseline on validation, by +0.0305 ( $p$  0.0012), and tuned ramjet by +0.0286 ( $p$  0.0005).
- **AIS features add nothing at the learned level.** M2-ais minus M1-v2 is −0.0029, ahead on 3 of 12 segments.
- **The AIS load model helps the default cost function.** M0-ais beats M0 by +0.026 and roughly matches ramjet (−0.002).
- **No robustness evidence.** The league has no lag robustness, because the lag build predates feature set v3, and no timing robustness, which was never registered; under LR-14 even its direction is unverified under perturbation.
- **Not portable as is.** M1-ais and M2-ais read a request’s own modeled prefill time, a research input only offline replay fills; live they would need the closed-form estimate (6.6% mean error).

**Table 28:** The in-class heuristic. *Auditor diagnostic, not pre-registered, never used for selection; train and validation only, the test split was not evaluated.* Mean clipped log-ratio against  $\pi_0$ ; the last column is the margin over **ramjet** on the fresh validation replicates, per cell and per segment.

| Policy (tuning)                                              | Train $k_{0..1}$ | Val $k_{0..2}$ | Val $k_{3..10}$ | vs ramjet on $k_{3..10}$ : cell / segment |
|--------------------------------------------------------------|------------------|----------------|-----------------|-------------------------------------------|
| M1-v2 ( $3 \times 400$ evaluations)                          | 0.1851           | 0.1767         | <b>0.1783</b>   | +0.0372 / +0.0290 (5 of 6)                |
| default + $10 \times$ faithful LMetric (8 train evaluations) | 0.1566           | 0.1609         | <b>0.1609</b>   | +0.0197 / +0.0179 (4 of 6)                |
| M1-v2’s s3 start, $c = 20.7$ (untuned)                       | 0.1566           | 0.1596         | <b>0.1612</b>   | +0.0200 / +0.0190 (6 of 6)                |
| faithful LMetric alone (untuned)                             | 0.1466           | 0.1493         | <b>0.1490</b>   | +0.0079 / +0.0110 (6 of 6)                |
| <b>ramjet</b> (best of 11 tuned baselines, $3 \times 400$ )  | 0.1435           | 0.1420         | <b>0.1411</b>   | —                                         |

### 9.11 Scope: the in-class heuristic and asymmetric iteration

Equal footing was verified from the raw tuning histories by two auditors: every tuned policy, learned or baseline, ran three restarts of 400 evaluations with the same cells, replicates, validation schedule, objective and reference, and the splits share no trace files. The fairness audit then found a gap that equal budgets do not close. By operator decision the baseline pool holds the branch’s ports as implemented, and the **lmetric** port lacks LMetric’s queued-prefill term (Section 7.2). Faithful LMetric is inside M1-v2’s hypothesis class, and M1-v2’s restart s3 started from default plus 20.7 times the faithful LMetric point. Table 28 shows what that heuristic does on its own.

- **Faithful LMetric beats every tuned baseline untuned**, on the selection replicates (0.1493 against at most 0.1423) and on the fresh ones (0.1490 against 0.1411).
- **Most of the margin is the heuristic.** Default plus faithful LMetric reproduces 53 to 66% of M1-v2’s fresh-validation margin over **ramjet**.
- **The learned increment** over it is about +0.011 per segment (standard error 0.0065, 5 of 6 segments; +0.017 per cell), about 0.29 of the MDE, and it *was not measured on test*; it sits mostly on one Mooncake validation segment.
- **Follow-up.** A pre-registered comparison with faithful LMetric and an SMetric-style baseline, each tuned with the equal budget and evaluated on fresh segments or on live runs over cells outside the frozen test set, has not been run and is not part of this paper. (The live runs of Section 11 used frozen test cells and so cannot serve it.)

**Asymmetric iteration.** The learned side was redesigned on validation evidence: multi-start initialization, the sign constraints, and M1-v2 itself, which was written to target the pilot’s weakest family, plus M2, M1-noaff and the queue-threshold arm. The baseline side gained one arm and no new heuristic types. Selection among the five learned arms used the same fresh-validation rule as the baselines, and the test pass was single, so the headline  $p$ -value is valid for the selected arm; but the originally planned M1 and M2 do not pass on test. Two registration choices (excluding M1 continued from the headline candidates and including the queue-threshold **ramjet** in the baseline pool) cannot be shown to predate the fresh-validation results; both were outcome-neutral on those values, and we do not claim strict pre-registration for them.

**Table 29:** Compute by stage. Every row is measured and comes from the stage facts, except where it says otherwise.

| Stage                                       | Value                                                                                                                                                        | Source                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Calibration, first pass                     | 13,532 replays, 122,436 replay-s, about 3.3 h on the workstation                                                                                             | facts/calibration.json  |
| Calibration fix                             | 9,792 replays, 112,561 replay-s, 0 errors                                                                                                                    | facts/STATE.md          |
| Pilot                                       | 56,552 fresh replays, 0 errors                                                                                                                               | facts/pilot.json        |
| One evaluation, all 34 train cells, $K = 2$ | 700.9 replay-s; 37.0s wall on the 20-slot workstation pool (pilot)                                                                                           | facts/pilot.json        |
| Phase 2                                     | 66 restarts of 400 evaluations; about 30,000 result records per run; generations of about 300s (EPYC 9654P) to 690s (EPYC 7702P); replay-seconds not totaled | facts/STATE.md          |
| Test pass                                   | 5,220 + 900 records, 0 errors                                                                                                                                | facts/test_results.json |
| Robustness                                  | 23,940 records, 0 errors                                                                                                                                     | facts/robustness.json   |
| Per-decision policy latency                 | not measured                                                                                                                                                 | —                       |

## 9.12 Cost

Table 29 collects the compute. Replay-seconds sum the wall time of single-threaded replays, so they measure work, not elapsed time.

# 10 Objective gaming, concentration and long prompts

The objective counts good requests. A request is good or not, whatever its size, so a policy can raise the count by serving many short requests well at the expense of a few long ones. This section reports how the search exploited that, how the exploit was removed, and what remains in the headline policy.

## 10.1 The finding: a sacrificial worker

At the phase-2 midpoint an adversarial audit replayed the selected tier-A M1 (unconstrained) and every contender on the 14 validation cells with its own scorer, and examined each request.

- **Concentration.** M1 sent more than the cap  $\min(2/N, 1/N + 0.25)$  of a cell’s requests to one worker on 4 of the 5 Mooncake validation cells (up to 0.454 against a cap of 0.333), while every contender baseline stayed within the cap. The concentration was in requests, not work: the hottest worker’s share of prefill tokens stayed near  $1/N$ . It carried one shared-prefix group, and the extra concentration bought almost no cache: on the violating cells, M1’s prefix reuse was at most 0.0065 above the best within-cap baseline and below it on two.
- **A worker of lost requests.** On the heaviest Mooncake cell, in every replicate, M1 left one worker receiving only 14 to 16 in-window requests, all very long (median prompt 98K to 100K tokens), and none of them good, while the hot workers served 519 to 1,014 mostly short, cached requests at 0.90 to 0.99 good. These long requests share only one 512-token block, so stacking them has no cache rationale, and they were not doomed:  $\pi_0$ , ramjet and M0 served every request of 60K tokens or more well on that cell, M1 only 42.5%.
- **Mechanism: load attracted long requests.** When a request of 60K tokens or more arrived, M1 sent it to the worker with the most outstanding prefill 58% of the time, against a chance rate of 12.5% and 0 to 2% for the baselines. The selected coefficients predict this:

with  $\theta_7 > 0$  (the product of prompt length and active prefill), the net coefficient on a worker’s active prefill turns positive for prompts above about 77K tokens, so a long request is drawn to a worker that is already busy. Setting  $\theta_7 = 0$  alone removed the lost-requests worker on all three restarts and brought the long requests back to 0.94 to 1.00 good.

- **Not starvation in the long-wait sense.** M1’s share of requests slower than three times the slowdown bound was below every baseline’s; the sacrificed requests missed narrowly (median slowdown 1.13 times the bound).

The audit’s verdict was objective gaming, which triggered the remedy registered in advance (Amendment A11.2). Black-box search exploiting a gap in its objective is a known pattern [22], and a binary per-request goodput invites exactly this trade [55]; a router tuned by an evolution strategy on tail TTFT alone sent all traffic to one replica [42]. A companion audit of simulator exploitation found the long-prompt sacrifice stable across 26 perturbations of timing, lag, window and SLO, so it is a property of the policy, not of a fragile corner of the simulator.

## 10.2 The remedy: load must not attract

The remedy re-ran all three M1 restarts with  $\theta_3, \theta_4, \theta_5, \theta_7 \leq 0$ , at the same budget, seeds and starting points (Section 6.5), and every later learned arm used the same rule. A re-audit with the audit’s own scripts, unchanged, then compared the constrained M1 with the unconstrained one.

- **Removed.** No Mooncake validation cell has a lost-requests worker (the unconstrained M1 had one per replicate on two cells); requests of 60K tokens or more are 0.92 to 1.00 good (0.425 to 0.966 before); the share of them sent to the worker with the most outstanding prefill fell to 0.07 to 0.20, near or below chance (0.37 to 0.58 before).
- **Remaining.** The constrained M1 still exceeds the share cap on 4 of 5 Mooncake cells, at higher shares than before (0.56 against a cap of 0.25 on the two cells at  $N = 8$ ), and still segregates by prompt length: the normalized mutual information between worker and prompt-length quartile is 0.21 to 0.23 at  $N = 8$ , against at most 0.095 for the contender baselines. Long prompts still lose: on the heaviest cell 0.826 of prompts of 32K tokens or more are good, against 0.995 under  $\pi_0$  and 0.963 under **ramjet**, and 65 to 69% of the requests that turn bad against  $\pi_0$  fall in the top prompt-length quartile.

The sign constraint removed the pathological mechanism, a worker that collects long requests it cannot serve, but not the underlying trade: short cached requests are packed onto a hot worker and long prompts compete for the rest. By the outcome rule registered with the remedy, the flags are reported on every learned row and are never a reason to retune or reselect; retuning under a size-aware objective is a design decision for the operator. The operator kept the request-count objective, headline and selection unchanged and added one token-weighted secondary metric (Amendment A19, Section 10.5). The same re-audit on the later learned arms (M2, M1-v2, M1-noaff and M1 continued) flagged each on 3 or 4 Mooncake validation cells, with no lost-requests worker for any.

## 10.3 Concentration on test

Table 30 applies the window-basis guards to every test policy. Every learned arm concentrates and segregates by prompt length far more than the strong baselines: M1-v2 exceeds the share cap on 24 of 60 test cells, against 5 for **ramjet** and 0 for  $\pi_0$ , and its worker assignment carries information about prompt length on 19 cells, against 5. A worker with no good request (“dump worker” in the table) appears on 5 of M1-v2’s cells and 3 of **ramjet**’s; the unconstrained M1 has 9, and more long-request flags than M1-v2 (20, 20 and 15 cells against 11, 10 and 10).

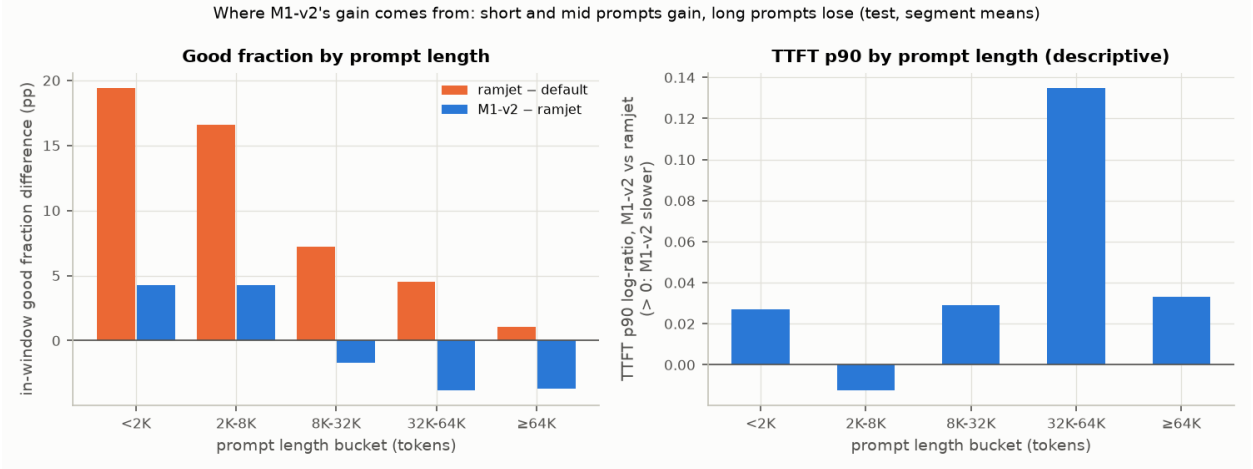
**Table 30:** Concentration and long-request flags on test, counts of the 60 test cells (generated). Share > cap: the window-basis largest worker request share exceeds  $\min(2/N, 1/N + 0.25)$ . NMI: normalized mutual information between worker and prompt-length quartile above 0.10.  $\geq 32K$ ,  $\geq 64K$ , top decile: the good fraction of those prompts is more than 0.05 below  $\pi_0$ ’s. Dump worker: a worker with 0% good in-window requests. Last column: mean over cells of the smallest per-worker good fraction.

| Policy                                | Share > cap | NMI > 0.10 | $\geq 32K$ | $\geq 64K$ | Top decile | Dump worker | Min worker good |
|---------------------------------------|-------------|------------|------------|------------|------------|-------------|-----------------|
| M1-v2 (headline learned arm)          | 24          | 19         | 11         | 10         | 10         | 5           | 0.841           |
| M2-ais                                | 22          | 20         | 13         | 7          | 10         | 3           | 0.844           |
| M1-noaff                              | 23          | 19         | 11         | 6          | 11         | 2           | 0.836           |
| M1 + queue threshold (abl. a)         | 24          | 19         | 12         | 7          | 11         | 3           | 0.840           |
| M1 continued (secondary)              | 23          | 18         | 14         | 8          | 11         | 2           | 0.830           |
| M1-ais                                | 25          | 22         | 11         | 6          | 10         | 5           | 0.837           |
| M2 rank 2                             | 23          | 20         | 11         | 6          | 10         | 3           | 0.832           |
| M1 (= A12 M1-default-init)            | 21          | 20         | 11         | 7          | 10         | 1           | 0.834           |
| M1 unconstrained (gaming-flagged)     | 20          | 20         | 20         | 20         | 15         | 9           | 0.753           |
| llm-d-precise-prefix                  | 5           | 6          | 6          | 4          | 5          | 2           | 0.845           |
| ramjet (val-best baseline)            | 5           | 5          | 1          | 0          | 3          | 3           | 0.849           |
| ramjet + queue threshold (abl. a)     | 5           | 5          | 0          | 0          | 1          | 2           | 0.852           |
| M0-ais                                | 5           | 5          | 3          | 1          | 2          | 2           | 0.850           |
| llm-d-precise-prefix@defaults         | 6           | 8          | 4          | 2          | 4          | 3           | 0.840           |
| sticky-session bounded                | 3           | 3          | 2          | 0          | 4          | 2           | 0.850           |
| lmetric@defaults                      | 2           | 2          | 5          | 3          | 5          | 4           | 0.835           |
| two-tier                              | 3           | 1          | 5          | 1          | 4          | 3           | 0.832           |
| ramjet@defaults                       | 3           | 2          | 2          | 1          | 3          | 2           | 0.854           |
| two-tier@defaults                     | 4           | 5          | 3          | 2          | 3          | 3           | 0.827           |
| lmetric (port)                        | 2           | 0          | 5          | 3          | 5          | 3           | 0.837           |
| M0 (default cost fn, tuned)           | 3           | 3          | 2          | 1          | 3          | 1           | 0.848           |
| sticky-session hard                   | 0           | 0          | 3          | 6          | 11         | 1           | 0.843           |
| llm-d-optimized-baseline (throughput) | 3           | 4          | 2          | 4          | 9          | 3           | 0.799           |
| llm-d-optimized-baseline@throughput   | 5           | 5          | 1          | 3          | 11         | 2           | 0.793           |
| llm-d-optimized-baseline (modeled)    | 5           | 5          | 15         | 8          | 19         | 2           | 0.792           |
| sticky-bounded@defaults               | 0           | 0          | 1          | 0          | 2          | 1           | 0.800           |
| sticky-hard@defaults                  | 0           | 0          | 1          | 6          | 10         | 1           | 0.785           |
| chwhl                                 | 14          | 7          | 14         | 18         | 27         | 0           | 0.724           |
| dualmap                               | 14          | 16         | 11         | 11         | 31         | 3           | 0.646           |
| dualmap@defaults                      | 13          | 22         | 18         | 19         | 33         | 4           | 0.542           |
| chwhl@defaults                        | 14          | 16         | 25         | 22         | 44         | 1           | 0.555           |
| round_robin                           | 0           | 0          | 39         | 30         | 57         | 0           | 0.419           |
| default@defaults (reference)          | 0           | 0          | 0          | 0          | 0          | 0           | 0.755           |

**Table 31:** Good fraction and TTFT by prompt length on test (generated; descriptive, computed after the test pass from its records). Good-fraction differences in percentage points, with the number of segments on which M1-v2 is higher; TTFT p90 as a segment-mean log-ratio.

| Prompt tokens | Good frac. M1-v2 – ramjet (pp) | Seg. higher | Good frac. ramjet – $\pi_0$ (pp) | TTFT p90 log-ratio, M1-v2 vs ramjet | Seg. higher |
|---------------|--------------------------------|-------------|----------------------------------|-------------------------------------|-------------|
| 0-2048        | +4.3                           | 8/12        | +19.4                            | +0.027                              | 6/12        |
| 2048-8192     | +4.3                           | 8/12        | +16.6                            | −0.012                              | 5/12        |
| 8192-32768    | −1.7                           | 7/12        | +7.2                             | +0.029                              | 10/12       |
| 32768-65536   | −3.8                           | 5/10        | +4.5                             | +0.135                              | 11/12       |
| 65536-        | −3.7                           | 4/8         | +1.1                             | +0.033                              | 4/8         |

The engine state stays inside the simulator’s measured range. On Mooncake, M1-v2’s hottest worker carries 1.85 times the per-worker mean of in-flight requests (**ramjet** 1.04,  $\pi_0$  1.18), and 2.21 and 1.67 times on FAST25 conversation and synthetic. Peak in-flight requests are 59 and 65, inside AIS’s measured vLLM decode grid and far below **max\_num\_seqs**, so the concentration does not push the engine into the performance model’s extrapolation. M1-v2 also shows no preference for pairing short prompts with long prefill chunks, the mixed-batch pricing artifact a search could exploit (caveat 13 in Table 2), and has fewer requests that finish below their uncontended latency than **ramjet** or **llm-d-precise-prefix**.



**Figure 13:** Where M1-v2's gain comes from (copied from the final report). Left: in-window good fraction by prompt length, `ramjet` minus  $\pi_0$  (orange) and M1-v2 minus `ramjet` (blue). Right: TTFT p90 log-ratio of M1-v2 against `ramjet`.

## 10.4 Where the gain comes from: prompt length

Table 31 and Figure 13 show the trade the request-count objective rewards.

- **The gain sits in short and mid prompts.** M1-v2 gains 4.3 points of good fraction below 8K tokens and gives up 1.7 to 3.8 points at 8K and above.
- **Long prompts wait longer than under `ramjet`.** Its TTFT p90 for 32K to 64K prompts is higher on 11 of 12 segments (log-ratio +0.135, a factor of 1.145); counting only records with at least 20 such prompts in both arms it is +0.169 on 8 of 8. That is about the default router's own long-prompt TTFT ( $-0.011$  against  $\pi_0$ ), while `ramjet` is  $-0.180$  below  $\pi_0$  there. Overall TTFT is on par with `ramjet`: p50 +0.006, p90  $-0.002$ , p99  $-0.082$ .

## 10.5 Token-weighted goodput (A19)

Amendment A19 added one secondary metric, registered before it was computed: the headline pipeline with each good in-window request weighted by its prompt plus output tokens, on exactly the requests and good rule that produce the headline's goodput. It never selects a policy or decides the headline. Against `ramjet`, M1-v2's token-weighted goodput is mixed: segment mean  $-0.0092$ , interval  $[-0.0645, +0.0244]$ , ahead on 10 of 12 segments ( $p$  0.032). One segment drives the mean: FAST25 synthetic x0, at  $-0.29$ . The three FAST25 conversation cells, which carry about 35% of the headline's cell-mean gain, gain +0.48, +0.49 and +0.08 in request count but +0.02,  $-0.15$  and +0.02 in tokens. The headline's magnitude is therefore partly a request-count effect.

## 10.6 Other secondary metrics

Against `ramjet` (Table 32), M1-v2's end-to-end p90 over the scored in-window requests is better ( $-0.048$ , higher on 3 of 12 segments), its prefix reuse lower ( $-0.024$ , higher on 1 of 12) and its per-request ITL p90 on par (+0.006); AgentX trajectory latency is slightly better ( $-0.012$ ). Against  $\pi_0$  it cuts TTFT p50 by 44% ( $-0.573$  log-ratio) and raises prefix reuse by 27% (+0.238).

**Table 32:** Secondary metrics against tuned **ramjet** on test (generated; selected rows and columns; descriptive). Segment-mean log-ratio, with the number of segments on which the policy’s value is higher; lower is better for latencies, higher for prefix reuse. Table 46 in the appendix has every policy and metric.

| Policy                            | TTFT p50       | TTFT p90       | TTFT p90, 32-64K | E2E p90        | ITL p90        | Prefix reuse  |
|-----------------------------------|----------------|----------------|------------------|----------------|----------------|---------------|
| M1-v2 (headline learned arm)      | +0.006 (9/12)  | −0.002 (6/12)  | +0.169 (8/8)     | −0.048 (3/12)  | +0.006 (3/12)  | −0.024 (1/12) |
| M2-ais                            | +0.018 (7/12)  | −0.018 (3/12)  | +0.111 (6/8)     | −0.054 (0/12)  | +0.020 (1/12)  | −0.013 (6/12) |
| M1 (= A12 M1-default-init)        | +0.020 (9/12)  | +0.046 (7/12)  | +0.105 (5/8)     | −0.032 (5/12)  | +0.023 (6/12)  | −0.011 (5/12) |
| M1 unconstrained (gaming-flagged) | +0.015 (9/12)  | +0.053 (8/12)  | +0.207 (5/8)     | −0.031 (4/12)  | +0.035 (5/12)  | −0.010 (7/12) |
| llm-d-precise-prefix              | +0.125 (12/12) | +0.144 (9/12)  | +0.075 (7/8)     | −0.005 (6/12)  | −0.010 (5/12)  | −0.008 (3/12) |
| lmetric (port)                    | +0.125 (8/12)  | +0.134 (7/12)  | +0.130 (8/8)     | −0.004 (4/12)  | −0.003 (2/12)  | −0.004 (5/12) |
| M0 (default cost fn, tuned)       | +0.090 (10/12) | +0.047 (10/12) | +0.036 (7/8)     | +0.038 (12/12) | +0.066 (11/12) | −0.133 (0/12) |
| default@defaults (reference)      | +0.579 (12/12) | +0.451 (12/12) | +0.180 (6/8)     | +0.116 (12/12) | +0.277 (11/12) | −0.262 (0/12) |

## 11 Live GPU validation

The headline is simulated, and a simulator that matches aggregate measurements can still rank policies differently from real hardware [52, 59]. The live check asks only whether the *relative* result holds on GPUs: whether M1-v2 still beats tuned **ramjet**, and whether the finalists keep their simulated order. It validates; it never feeds selection and never changes a finalist, the headline or any pre-registered number.

### 11.1 The live lane

The lane deploys Dynamo’s frontend with its KV router in front of vLLM 0.24.0 workers serving Qwen3-32B at tensor parallelism 2, four workers per 8×H100 node, with KV events from every worker, no session affinity and no busy thresholds. The router loads the same policy YAML files the replays used, byte for byte. Load comes from AIPerf 0.13.0 [36], driven from inputs generated from the campaign’s cells. A scoring adapter turns AIPerf’s per-request export into the records the replay scorer reads, so live and replay are scored by the same code with the same window and SLO.

**Load-generator equivalence.** Before any comparison, a CPU-only audit checked, request by request, that AIPerf replays each cell as the offline replay does. For every family and load mode the generator supports (Mooncake, FAST25 conversation and synthetic, and generated sessions, each in open and closed loop) it found the same request set and order, bit-identical open-loop arrivals including the replicate re-draws and speedup, exact prompt and output lengths with output length forced, the same prefix-sharing structure (a one-to-one map between the KV blocks the router and vLLM see and those of replay), the same session membership and think delays, the same closed-loop admission rule, and the same warm-up and scoring window. Replaying replay’s own per-request latencies through AIPerf and a stub server moved windowed goodput by at most 0.06%. Two non-equivalences remain. AgentX has no live load generator, because its recycled lanes need dependency, join and recycling semantics that neither AIPerf nor its prior fork implements, so the live check covers Mooncake, FAST25 and sessions only. And live latency is measured at the client, so it includes frontend, transport and streaming overhead that replay and the reference latency  $E_0$  do not; this biases every policy slightly toward “not good” at the slowdown boundary.

**The smoke run.** One GPU smoke ran  $\pi_0$  on one Mooncake train cell, open loop at load L2 with four workers, on one 8×H100 node, and compared it with the cached replay of the same cell and replicate.

- **Fidelity of the workload.** All 4,002 requests arrived with exact prompts, prompt lengths and forced output lengths; AIPerf’s dispatch was within 2.5 ms of the schedule at the 99th percentile, with no drift.
- **What agreed.** Windowed goodput was 3.4454 live against 3.4535 in replay (ratio 0.9976; 2113 against 2118 good of 2376 in-window requests), the prefix-cache hit rate 0.360 against 0.359, and the end-to-end latency distributions differed by a Kolmogorov–Smirnov statistic of 0.022.
- **What differed, in offsetting directions.** Live TTFT was higher (p50 176 against 130 ms) and live decode faster (per-request ITL p50 22.9 against 25.6 ms). Idle single-request calibration on two nodes agreed within 2.2%; in it, decode steps were 7 to 10% faster than AIS’s, and prefill 5 to 10% slower above 4K tokens.

One cell, one policy and one run validate the lane, not any policy ranking.

## 11.2 Pre-registered analysis

The analysis was registered in the campaign’s live plan before any finalist live job was submitted (Amendment A20). It is summarized here; the plan file is the binding version.

- **Deployment.** One 8×H100 SXM node per job, four vLLM workers at tensor parallelism 2, as in the smoke. A second worker count was not planned: the two-node path is scripted but untested and would double the cost of every run.
- **Cells.** Six frozen test cells at  $N = 4$ : Mooncake w3 open loop at L2, Mooncake w5 open loop at L3 and closed loop at L3, FAST25 conversation (w3) open loop at L2, FAST25 synthetic x0 open loop at L2, and sessions s6 open loop at L2. There is no AgentX cell, since AgentX has no live load generator. The cells were chosen at registration, after the frozen test pass, for family and load-mode coverage and run time, so their simulated outcomes were known when they were chosen (Section 11.4).
- **Policies,** each exactly as frozen, with policy files byte-identical to those the test pass replayed:  $\pi_0$ , run twice per cell to estimate live run-to-run noise; tuned **ramjet**; M1-v2; M1; M0; and round-robin. Every run uses replicate  $k = 0$  with policy seed 1, and the order of the seven runs within a cell is randomized by a fixed seed.
- **Scoring.** The same A2 scorer as replay, with the AIS reference latency  $E_0$  as the primary metric; a run counts only if every request matched its intended prompt, prompt length and forced output length and the load generator exited cleanly. The token-weighted goodput, goodput under a live-measured  $E_0$ , and TTFT, ITL, prefix hits and worker shares are secondary.
- **Simulated counterparts** are the frozen test pass’s own records for the same cells, policies and replicate; no test cell is replayed again. In those records M1-v2 minus **ramjet** is +0.069, +0.122, +0.141, +0.470, +0.011 and  $-0.0004$  on the six cells in the order above.
- **Readouts and wording rules.** Paired deltas against  $\pi_0$  (the mean of the default runs in the same job) and against **ramjet**, per cell, without significance claims. The Kendall  $\tau_K$ -b between simulated and live deltas against  $\pi_0$ , per cell and pooled over the six policies; the live ranking “agrees” with simulation if the pooled  $\tau_K$ -b is at least 0.6. The sign of M1-v2 minus **ramjet** per cell; “M1-v2 > **ramjet** holds live” only if it is positive on at least two thirds of the five informative cells and positive on average over them. A cell is informative if its simulated M1-v2 minus **ramjet** exceeds 0.01 in size; the sessions cell is not, because the four tuned

policies tie at ceiling there in simulation. Live noise from the repeated  $\pi_0$  runs is reported next to every delta.

### 11.3 Results

**In brief.** Both registered verdicts are positive. The live ranking of the six policies agrees with simulation (pooled Kendall  $\tau_{K-b}$  1.00), and  $M1-v2 > \text{ramjet}$  holds live:  $M1-v2$  leads in all five informative cells, by +0.219 on average against +0.162 in simulation. That is all the live runs validate. Absolute goodput departs from simulation by factors of 0.72 to 2.18, differently for different policies in the same cell, so the size of a gap between two policies is not validated either; on the sessions cell, the simulated gain of every tuned policy over  $\pi_0$  is not reproduced beyond live noise. On the token-weighted secondary,  $M1-v2 > \text{ramjet}$  does not hold live, as it does not in simulation.

**What ran.** All 42 planned cell runs (six cells, seven runs each) and all four applicable idle calibration runs are valid, so no cell is short. Two cell runs failed the validity rule at their first attempt: after every request had completed, AIPerf exited non-zero in its shutdown phase, because a slow export to a shared file system tripped its heartbeat watchdog. Each was re-run once under the registered rule, and the rerun is used. The first attempts are descriptive only:  $\pi_0$ 's second run on FAST25 conversation scored 0.886 good requests per second (0.893 in the rerun), and round-robin on Mooncake open L2 scored 3.085 (3.087). Jobs submitted after the first failure raised the watchdog's missed-heartbeat threshold from 3 to 60; the setting does not touch the request schedule, prompts, lengths, headers, timing or scoring. Some comparisons span two jobs, and so two nodes: both noise pairs on FAST25 conversation and sessions, round-robin against  $\pi_0$  and against  $\text{ramjet}$  on Mooncake open L2, and M0 and round-robin against  $\text{ramjet}$  on sessions. Every other comparison is within one job. Two jobs were scheduled through a different queue than planned, on the same node type and with the same inputs. Two properties of the workload carry over from replay unchanged: the closed-loop cell admits requests by concurrency, not by an arrival schedule, and in FAST25 synthetic seven groups of two requests share an arrival instant, whose order within the group the live load generator cannot enforce.

**Live noise.** The two runs of  $\pi_0$  in each cell differ by a clipped log-ratio of  $-0.010$ ,  $-0.016$ ,  $+0.023$ ,  $+0.008$ ,  $-0.004$  and  $-0.006$  (cells in the order of Table 36), which gives a per-run noise  $\sigma_{\text{live}} = \sqrt{r^2/2} = 0.0090$ . A live delta against  $\pi_0$  then has a standard deviation of about 0.011 where the reference is the mean of two same-job default runs and 0.013 where it is one (on FAST25 conversation and sessions, only one default ran in the same job as each policy), and a delta against  $\text{ramjet}$  about 0.013. Of the 30 live deltas against  $\pi_0$ , 24 exceed twice their standard deviation; the six that do not are round-robin on FAST25 conversation, M0 on FAST25 synthetic and the four tuned policies on sessions. The estimate rests on six pairs, so it has six degrees of freedom: its 95% interval is about 0.006 to 0.020, and it is assumed to hold for every policy, including the concentrating learned ones whose run-to-run noise was not measured.

**Ranking agreement.** Table 33 gives every paired delta, live and simulated, and Table 34 the Kendall  $\tau_{K-b}$  between them. The registered primary,  $\tau_{K-b}$  between the six policies' mean gains over  $\pi_0$  across cells, is 1.00: the policies have the same order live as in simulation,  $M1-v2$  (+0.273 live, +0.219 simulated),  $M1$  (+0.256, +0.218), tuned  $\text{ramjet}$  (+0.090, +0.084),  $M0$  (+0.059, +0.066),  $\pi_0$  (0) and round-robin ( $-0.295$ ,  $-0.332$ ). By the registered rule ( $\tau_{K-b} \geq 0.6$ ) the live ranking agrees with simulation. The exact one-sided permutation  $p$  over the 720 orderings is 0.0014; it is descriptive,

**Table 33:** Live GPU validation: paired deltas, live (simulated at  $k = 0$ ), as the clipped log-ratio of the headline. Top: against  $\pi_0$ , the mean of its runs in the same job. Bottom: against tuned **ramjet**. One run per cell and policy; live noise  $\sigma_{\text{live}} = 0.0090$  per run.

| Policy              | Mooncake open L2 | Mooncake open L3 | Mooncake closed L3 | FAST25 conv. open L2 | FAST25 synth. open L2 | Sessions open L2 | Mean live [range]       | Mean sim |
|---------------------|------------------|------------------|--------------------|----------------------|-----------------------|------------------|-------------------------|----------|
| tuned <b>ramjet</b> | +0.023 (+0.037)  | +0.086 (+0.102)  | +0.147 (+0.135)    | +0.171 (+0.056)      | +0.096 (+0.084)       | +0.016 (+0.086)  | +0.090 [+0.016, +0.171] | +0.084   |
| M1-v2               | +0.100 (+0.106)  | +0.276 (+0.224)  | +0.280 (+0.276)    | +0.833 (+0.526)      | +0.131 (+0.095)       | +0.016 (+0.086)  | +0.273 [+0.016, +0.833] | +0.219   |
| M1                  | +0.095 (+0.085)  | +0.244 (+0.225)  | +0.267 (+0.244)    | +0.771 (+0.537)      | +0.140 (+0.131)       | +0.016 (+0.086)  | +0.256 [+0.016, +0.771] | +0.218   |
| M0                  | +0.025 (+0.029)  | +0.075 (+0.059)  | +0.130 (+0.140)    | +0.100 (+0.043)      | +0.012 (+0.038)       | +0.011 (+0.084)  | +0.059 [+0.011, +0.130] | +0.066   |
| round-robin         | -0.139 (-0.106)  | -0.383 (-0.297)  | -0.262 (-0.193)    | -0.010 (-0.132)      | -0.206 (-0.165)       | -0.770 (-1.099)  | -0.295 [-0.770, -0.010] | -0.332   |

| Policy      | Mooncake open L2 | Mooncake open L3 | Mooncake closed L3 | FAST25 conv. open L2 | FAST25 synth. open L2 | Sessions open L2  | Mean live [range]        | Mean sim |
|-------------|------------------|------------------|--------------------|----------------------|-----------------------|-------------------|--------------------------|----------|
| M1-v2       | +0.077 (+0.069)  | +0.190 (+0.122)  | +0.133 (+0.141)    | +0.663 (+0.470)      | +0.035 (+0.011)       | +0.0002 (-0.0004) | +0.183 [+0.0002, +0.663] | +0.135   |
| M1          | +0.072 (+0.048)  | +0.159 (+0.123)  | +0.119 (+0.109)    | +0.600 (+0.481)      | +0.043 (+0.046)       | -0.0002 (-0.001)  | +0.166 [-0.0002, +0.600] | +0.134   |
| M0          | +0.001 (-0.008)  | -0.011 (-0.043)  | -0.017 (+0.005)    | -0.070 (-0.014)      | -0.084 (-0.046)       | +0.001 (-0.002)   | -0.030 [-0.084, +0.001]  | -0.018   |
| $\pi_0$     | -0.023 (-0.037)  | -0.086 (-0.102)  | -0.147 (-0.135)    | -0.171 (-0.056)      | -0.096 (-0.084)       | -0.016 (-0.086)   | -0.090 [-0.171, -0.016]  | -0.084   |
| round-robin | -0.162 (-0.143)  | -0.469 (-0.399)  | -0.410 (-0.328)    | -0.181 (-0.188)      | -0.303 (-0.249)       | -0.780 (-1.099)   | -0.384 [-0.780, -0.162]  | -0.401   |

**Table 34:** Kendall  $\tau_K$ -b between simulated and live deltas against  $\pi_0$  per cell, at the registered  $k = 0$  and in the three registered sensitivities, and the discordant policy pairs. “Within noise”: the live difference is at most  $2\sqrt{2}\sigma_{\text{live}}$ .

|                                       | Mooncake open L2                 | Mooncake open L3 | Mooncake closed L3               | FAST25 conv. open L2 | FAST25 synth. open L2 | Sessions open L2                    |
|---------------------------------------|----------------------------------|------------------|----------------------------------|----------------------|-----------------------|-------------------------------------|
| $\tau_K$ -b, sim $k = 0$ (registered) | 0.87                             | 0.87             | 0.87                             | 0.87                 | 1.00                  | 0.87                                |
| $\tau_K$ -b, sim mean of $k = 0..2$   | 0.87                             | 1.00             | 0.87                             | 0.87                 | 1.00                  | 0.73                                |
| $\tau_K$ -b, sim lag 10 ms            | 0.87                             | 1.00             | 1.00                             | 1.00                 | 0.87                  | 0.97                                |
| $\tau_K$ -b, sim lag 50 ms            | 0.87                             | 1.00             | 1.00                             | 0.87                 | 0.87                  | 0.87                                |
| Discordant pairs ( $k = 0$ )          | ramjet (tuned)/M0 (within noise) | M1-v2/M1         | ramjet (tuned)/M0 (within noise) | M1-v2/M1             | none                  | ramjet (tuned)/M1-v2 (within noise) |

and its null of a random order of all six policies is one that round-robin and  $\pi_0$  nearly reject by themselves. Over all 30 (cell, policy) points  $\tau_K$ -b is 0.74, and per cell it is 0.87 to 1.00 (mean 0.89). The registered sensitivities, with the simulated side taken as the mean over replicates  $k = 0..2$  or from replay with router-state lag of 10 or 50 ms, all give a pooled  $\tau_K$ -b of 1.00 (round-robin keeps its  $k = 0$  value under lag). Two qualifications apply. The pooled 1.00 at  $k = 0$  rests on a simulated gap of 0.0009 between the mean gains of M1-v2 and M1; had the two swapped, it would read 0.87. Over replicates  $k = 0..2$  the gap is 0.0074 and the pooled value is still 1.00. And all five per-cell discordances are pairs that simulation separates by at most 0.011. Three of them are within live noise ( $2\sqrt{2}\sigma_{\text{live}}$ ); the other two put M1-v2 ahead of M1 live, on Mooncake open L3 (+0.032 live against -0.001 simulated) and on FAST25 conversation (+0.062 against -0.011).

**M1-v2 against ramjet.** On each of the five informative cells the live delta is positive, has the sign simulation predicts, and exceeds the registered noise bar  $2\sqrt{2}\sigma_{\text{live}} = 0.0256$  (Table 35). It averages +0.219 live against +0.162 in simulation. By the registered rule (positive on at least two thirds of the informative cells, and positive on average), M1-v2 > **ramjet** holds live. On the sessions cell, where the four tuned policies tie at ceiling, both values are ties (-0.0004 simulated, +0.0002 live). The bar inherits the uncertainty in  $\sigma_{\text{live}}$ : at its one-sided 95% upper bound (0.017) the bar would be 0.049, which FAST25 synthetic’s +0.035 does not clear; the other four cells clear it.

**Table 35:** M1-v2 minus tuned `ramjet` per live cell: simulated at  $k = 0$  and live. The registered sign test counts the five informative cells.

| Cell                                                          | Sim ( $k = 0$ ) | Live    | Same sign | $ \text{live}  > 2\sqrt{2}\sigma_{\text{live}}$ |
|---------------------------------------------------------------|-----------------|---------|-----------|-------------------------------------------------|
| Mooncake open L2                                              | +0.069          | +0.077  | yes       | yes                                             |
| Mooncake open L3                                              | +0.122          | +0.190  | yes       | yes                                             |
| Mooncake closed L3                                            | +0.141          | +0.133  | yes       | yes                                             |
| FAST25 conv. open L2                                          | +0.470          | +0.663  | yes       | yes                                             |
| FAST25 synth. open L2                                         | +0.011          | +0.035  | yes       | yes                                             |
| Sessions open L2 (not informative: $ \text{sim}  \leq 0.01$ ) | -0.0004         | +0.0002 | no        | no                                              |

**Where live and simulation differ.** Absolute goodput agrees closely on some cells and not on others (Table 36). Live goodput is 0.72 to 2.18 times the simulated value; on the three Mooncake cells every policy but round-robin is within 0.94 to 1.01 of it. The ratio differs between policies in the same cell, so the departures also move the gaps between policies, in both directions. We flag a live delta as departing from simulation when the two differ by more than twice the standard deviation of the live delta; the simulated counterpart replays the identical requests, so live noise is the only sampling term. The rule is descriptive and was added after the registered analysis, at the request of the campaign’s audit of the live results. Under it, 18 of the 30 deltas against  $\pi_0$  depart (9 higher live, 9 lower), and 13 of the 30 against `ramjet` (Table 37).

- **FAST25 conversation.**  $\pi_0$ , `ramjet`, M0 and round-robin reach 0.72 to 0.81 of their simulated goodput, M1 and M1-v2 0.91 to 0.98. Every gain over  $\pi_0$  is larger live (M1-v2 +0.833 against +0.526; `ramjet` +0.171 against +0.056), which widens M1-v2’s lead over `ramjet` (+0.663 against +0.470). It also nearly erases round-robin’s deficit against  $\pi_0$  (-0.010 live against -0.132, within live noise), because  $\pi_0$  loses the most live. vLLM preempted 18 to 24 requests per run under  $\pi_0$ , `ramjet` and M0 and 12 under round-robin, against 1 to 6 under M1 and M1-v2, and the non-learned policies’ median TFFT rose 1.43 to 1.76 times over simulation, the learned ones’ 1.08 to 1.13 times.
- **Sessions.**  $\pi_0$  runs at 1.07 times its simulated goodput, the four tuned policies at 0.997 to 1.000. Live, 97.8 to 98.4% of  $\pi_0$ ’s in-window requests are good, against 91.6% in simulation; the tuned policies serve 99.9 to 100% in both, at ceiling. So every tuned policy’s simulated gain over  $\pi_0$ , +0.084 to +0.086, is only +0.011 to +0.016 live, inside twice the standard deviation of a live delta (0.026): on this cell the simulated gain over  $\pi_0$  is not reproduced beyond live noise. The plan included this cell to check the default and round-robin gaps live; the default half of that check fails. Round-robin reaches 2.18 times its simulated goodput; it is still last by far, but its deficit against  $\pi_0$  shrinks from -1.099 (the clip at  $\ln 3$ ) to -0.770.
- **Mooncake and FAST25 synthetic.** Round-robin’s deficit against  $\pi_0$  is larger live on all four cells, by 0.033 to 0.086. The eleven other departures on these cells are 0.023 to 0.082 in size (Table 37).

A plausible reading, which these runs do not test, is engine timing. Live time to first token is longer than AIS’s and decoding is faster (Table 38), so the same cell runs at a different effective load or SLO point on hardware. FAST25 conversation would then run closer to prefill saturation live, where the policies that leave more prefill work queued lose more goodput. On sessions, live mean-ITL p50 is 0.84 to 0.85 of simulation for every policy, which loosens the SLO in effect for the policies not already at ceiling ( $\pi_0$  and round-robin). Cache reuse is not the cause: the live prefix-cache hit rate matches replay’s (below).

**Table 36:** Windowed goodput (good requests per second), live / simulated at  $k = 0$ . Live  $\pi_0$  is the mean of its two runs.

| Policy              | Mooncake<br>open L2 | Mooncake<br>open L3 | Mooncake<br>closed L3 | FAST25 conv.<br>open L2 | FAST25 synth.<br>open L2 | Sessions<br>open L2 |
|---------------------|---------------------|---------------------|-----------------------|-------------------------|--------------------------|---------------------|
| $\pi_0$             | 3.547 / 3.544       | 3.582 / 3.733       | 3.948 / 4.034         | 0.896 / 1.249           | 1.894 / 1.962            | 8.249 / 7.691       |
| tuned <b>ramjet</b> | 3.631 / 3.678       | 3.902 / 4.135       | 4.575 / 4.617         | 1.068 / 1.321           | 2.086 / 2.134            | 8.357 / 8.385       |
| M1-v2               | 3.921 / 3.939       | 4.720 / 4.671       | 5.225 / 5.318         | 2.072 / 2.115           | 2.159 / 2.157            | 8.359 / 8.381       |
| M1                  | 3.902 / 3.860       | 4.573 / 4.677       | 5.155 / 5.148         | 1.947 / 2.137           | 2.178 / 2.235            | 8.356 / 8.380       |
| M0                  | 3.636 / 3.649       | 3.859 / 3.961       | 4.497 / 4.641         | 0.995 / 1.303           | 1.917 / 2.037            | 8.365 / 8.367       |
| round-robin         | 3.087 / 3.187       | 2.442 / 2.774       | 3.037 / 3.326         | 0.891 / 1.095           | 1.541 / 1.663            | 3.831 / 1.757       |

**Table 37:** Live minus simulated paired delta, for every delta of Table 33. \*: a difference larger than twice the standard deviation of the live delta (a descriptive rule, not registered).

| Comparison                     | Mooncake<br>open L2 | Mooncake<br>open L3 | Mooncake<br>closed L3 | FAST25 conv.<br>open L2 | FAST25 synth.<br>open L2 | Sessions<br>open L2 |
|--------------------------------|---------------------|---------------------|-----------------------|-------------------------|--------------------------|---------------------|
| tuned <b>ramjet</b> vs $\pi_0$ | -0.014              | -0.017              | +0.012                | +0.114*                 | +0.012                   | -0.070*             |
| M1-v2 vs $\pi_0$               | -0.005              | +0.052*             | +0.004                | +0.307*                 | +0.036*                  | -0.070*             |
| M1 vs $\pi_0$                  | +0.010              | +0.019              | +0.023*               | +0.234*                 | +0.009                   | -0.070*             |
| M0 vs $\pi_0$                  | -0.004              | +0.015              | -0.010                | +0.058*                 | -0.026*                  | -0.073*             |
| round-robin vs $\pi_0$         | -0.033*             | -0.086*             | -0.069*               | +0.121*                 | -0.041*                  | +0.329*             |
| M1-v2 vs <b>ramjet</b>         | +0.008              | +0.068*             | -0.008                | +0.193*                 | +0.024                   | +0.001              |
| M1 vs <b>ramjet</b>            | +0.024              | +0.035*             | +0.011                | +0.120*                 | -0.003                   | +0.0004             |
| M0 vs <b>ramjet</b>            | +0.009              | +0.032*             | -0.022                | -0.056*                 | -0.038*                  | +0.003              |
| $\pi_0$ vs <b>ramjet</b>       | +0.014              | +0.017              | -0.012                | -0.114*                 | -0.012                   | +0.070*             |
| round-robin vs <b>ramjet</b>   | -0.019              | -0.070*             | -0.082*               | +0.007                  | -0.053*                  | +0.319*             |

### Registered secondaries.

- **Token-weighted goodput (A19).** The pooled  $\tau_{K-b}$  is 0.60, exactly at the registered threshold (mean per cell 0.58; permutation  $p$  0.068). M1-v2 minus **ramjet** is positive live on three of the five informative cells, with mean  $-0.035$ , so M1-v2  $>$  **ramjet** does not hold live on this metric. By the same rule it does not hold in simulation on these cells either (two of five positive, mean  $-0.044$ ; the signs agree on four of five), in line with the request-count limit of Section 10.5.
- **Live reference latency.** Re-scoring every run against an  $E_0$  fitted on the live idle runs (72 requests from four jobs) instead of AIS’s gives a pooled  $\tau_{K-b}$  of 1.00, and M1-v2 minus **ramjet** is positive on all five informative cells, mean  $+0.241$ . So the ranking and the sign do not hinge on the simulator’s  $E_0$ . The per-node fit has only two complete cells, since only some jobs ran an idle calibration; there  $\tau_{K-b}$  is 0.87 and the sign holds on both.
- **Engine timing** (Table 38). Live TTFT is longer than AIS’s (median ratios over the cells 1.14 to 1.39 at p50, 1.09 to 1.22 at p90), and the median request decodes faster (0.90 to 0.99; at p90 0.99 to 1.10), the offsetting pattern of the smoke run.
- **Prefix-cache reuse.** Each successful request’s usage record carries the server’s count of cached prompt tokens. The live hit rate computed from it (cached over prompt tokens) differs from replay’s reuse fraction by at most 0.015 in each of the 42 runs, and by at most 0.010 in 40, so routing reproduced replay’s cache reuse on hardware for every policy and cell. The registered descriptive used vLLM’s own hit counter instead, which is comparable in only 14 runs, all with little queueing; there it is within 0.009 of replay. By our reading of the vLLM 0.24.0 scheduler, the counter records a query each time a waiting request is retried, and preemption re-queries too, so queueing inflates it.

**Table 38:** Engine-level descriptives per policy: median over the six cells of the live / simulated ratio of TTFT and of per-request mean ITL, and the largest per-worker request share over the cells, live / simulated.

| Policy              | TTFT p50 | TTFT p90 | Mean-ITL p50 | Mean-ITL p90 | Max worker share, live / sim |
|---------------------|----------|----------|--------------|--------------|------------------------------|
| $\pi_0$             | 1.39     | 1.12     | 0.99         | 1.02         | 0.28 / 0.29                  |
| tuned <b>ramjet</b> | 1.37     | 1.09     | 0.97         | 1.02         | 0.33 / 0.34                  |
| M1-v2               | 1.14     | 1.09     | 0.90         | 0.99         | 0.82 / 0.81                  |
| M1                  | 1.22     | 1.10     | 0.93         | 1.01         | 0.81 / 0.78                  |
| M0                  | 1.35     | 1.09     | 0.99         | 1.05         | 0.27 / 0.28                  |
| round-robin         | 1.32     | 1.22     | 0.97         | 1.10         | 0.25 / 0.25                  |

- **Concentration.** The LR-13 worker-share cap (0.5 at  $N = 4$ ) is exceeded live by M1-v2 on Mooncake open L3 and closed L3 and by M1 on Mooncake closed L3 and FAST25 conversation. In simulation it is exceeded by M1-v2 on Mooncake open L3, Mooncake closed L3 and FAST25 conversation, and by M1 on Mooncake closed L3 and FAST25 conversation. The concentration of Section 10 also happens on hardware. The long-prompt guards of Section 10 were not part of the registered live analysis; the token-weighted goodput above is its size-aware check.
- **Load generation.** On the 35 open-loop runs, requests left at most 2.6 ms late at the 99th percentile (15.2 ms at most). In 16 runs, 18 requests failed in AIPerf because an output of 1 to 3 tokens carried no text; the scorer counts them not good, as registered.

#### 11.4 What the live runs can and cannot show

They show that on six  $N = 4$  test cells, on real H100 GPUs serving the same requests, the live ranking of these six policies agrees with the simulator’s, and that M1-v2 leads **ramjet** live in every cell where simulation predicts a lead of more than 0.01, each time by more than the registered noise bar. They cannot show the following.

- **The headline’s magnitude or significance.** There is one run per cell and policy, at one replicate, on six cells, and no test was registered.
- **A representative effect.** The cells are not a random sample. They were chosen after the frozen test pass, with its per-cell results in the registration, and they include the three largest of the 13 simulated  $N = 4$  gaps between M1-v2 and **ramjet** (FAST25 conversation +0.470, Mooncake closed L3 +0.141, Mooncake open L3 +0.122). Their simulated gap averages +0.135 at  $k = 0$ , against +0.071 over all 13  $N = 4$  test cells at  $k = 0$  and the  $N = 4$  stratum’s segment mean of +0.055 over  $k = 0..2$  (Table 21). FAST25 conversation alone contributes 58% of the six-cell sum; the other five average +0.068. Live deltas exceed the simulated ones on four of the five informative cells, but with single runs that is not evidence that the effect is larger on hardware.
- **The size of the gaps between policies.** 18 of the 30 live deltas against  $\pi_0$  and 13 of the 30 against **ramjet** depart from simulation, in both directions. On sessions, the simulated gain over  $\pi_0$  is not reproduced beyond live noise, so the live runs do not support the size of the simulated sessions-family gains over  $\pi_0$  (Table 42); on FAST25 conversation every gain over  $\pi_0$  is larger live.
- **Close pairs.** M1-v2 minus M1 is within live noise on four of the six cells, and so is M0 minus **ramjet**; the live order inside those pairs is mostly unresolved.
- **Absolute goodput.** Live over simulated goodput ranges from 0.72 to 2.18 and differs between policies in a cell; AIS’s absolute accuracy is not validated here.

- **Session context.** The session header reaches the frontend exactly when replay sends a session context (no mismatch in 42 runs), and the code path to the policy exists, but no run shows `learned-choice`’s session feature receiving a session live. Only the sessions cell could exercise it, and that cell cannot inform M1-v2 minus `ramjet`.
- **Scope.** One model, one engine version, one GPU type and  $N = 4$  only; no AgentX; one closed-loop cell; FAST25 and sessions in open loop only. Nothing here tests worker-count extrapolation, the AIS league, controlled router-state lag on hardware, or faithful LMetric, which was not in the live matrix.
- **A token-weighted gain.** M1-v2  $>$  `ramjet` does not hold live on the token-weighted goodput, as in simulation.

## 12 Discussion, limitations and threats to validity

Every claim this study makes is a claim about AIS-timed offline replay of one deployment. This section lists what could make the headline wrong inside that setting, or keep it from transferring to a live router, with what the campaign did about each and what stays open. Table 39 summarizes; the subsections give the evidence.

**What the results say.** Within replay, the picture is consistent. Cache-aware routing matters a great deal on this deployment (round-robin trails the default router by a segment-mean clipped log-ratio of  $-0.56$ ), and the strong cache-aware heuristics, once tuned, are close to each other. A conditional-logit policy over the router’s own signals, started from the default router, beats the branch’s ported heuristics, each tuned with an equal budget; the step that passes is the extended feature set, whose largest learned term makes uncached prefill more expensive the busier a worker is. The margin is small, below the pre-registered minimum detectable effect, and stable in sign under lag, timing, SLO-definition and window perturbations. Three findings bound it. Most of the validation margin is reproduced by the default cost plus a faithful LMetric term, which the baseline pool did not contain. The gain is concentrated on short and mid prompts, paid for by long prompts. And an unconstrained search found a degenerate policy that only a sign constraint removed. On GPUs, six test cells served live agree with the simulated ranking and with the sign of M1-v2 minus `ramjet`, but not with absolute goodput or with the size of the gaps between policies (Section 11).

### 12.1 Simulator validity

**Simulated timing only.** Every latency in this paper comes from AIS’s forward-pass model; none was measured on GPUs, and the study does not validate AIS’s absolute accuracy. What matters here is narrower: whether replay ranks policies as GPUs would. That is not implied by aggregate accuracy. In a study of learned routing for disaggregated serving, a simulator re-fitted to match measured goodput to a mean absolute residual of 0.068 still did not reproduce the measured winner on any workload [52]; that simulator was a hand-built discrete-event model, so how far the lesson carries to AIS-timed engine replay is an open question. Controllers trained in emulation have also lost to simple ones in the real world [59]. Two features of this campaign raise the stakes. The calibrated loads sit in the default router’s transition region, where performance-model errors grow fastest [3]. And AgentX requests reach 128K tokens, where AIS’s accuracy has not been checked against hardware; the setup only showed that replay’s 120K-token prefill time equals an independent sum of AIS’s chunk estimates (17,815.54 ms, about 64% model FLOPs utilization), an internal-consistency check. No AgentX play fits a 32K context (caveat 10 of Table 2), so this

**Table 39:** Threats to validity. “Direction” is the likely effect on the learned policy’s measured advantage, where one can be argued.

| Threat                                               | Likely direction                         | Mitigation                                             | Status                                                                                  |
|------------------------------------------------------|------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------|
| AIS timing differs from GPUs                         | unknown                                  | live ranking check; timing $\times 0.8$ , $\times 1.2$ | timing: sign holds; live: ranking and sign agree, absolute goodput and gap sizes do not |
| Live check on few, chosen cells                      | inflates live gaps                       | registered analysis; validation only                   | six cells, chosen after the test pass; one run each                                     |
| Router state perfectly fresh in replay               | favors policies that lean on exact state | lag 10–200 ms on test                                  | ranking holds                                                                           |
| No host, disk or global KV tier                      | excludes tier-dependent wins             | scope statement                                        | inherent                                                                                |
| AgentX recorded with another model, closed loop only | unknown                                  | family reported separately                             | inherent                                                                                |
| SLOs and loads set with $\pi_0$                      | unknown                                  | SLO scale and definition re-scoring                    | sign holds; significance only at scales 1–3                                             |
| Objective gaming by concentration                    | inflates                                 | guards; audit; sign constraint                         | lost-requests worker removed; segregation reported                                      |
| 12 independent test segments                         | low power at small gaps                  | pre-registered test; MDE                               | effect below MDE: sign only                                                             |
| Selection and redesign on validation                 | inflates validation numbers              | single frozen test pass; symmetric selection           | disclosed                                                                               |
| Ports differ from papers                             | inflates                                 | stated per port; in-class diagnostic                   | headline scoped to the ports                                                            |
| Equal budget, unequal search                         | unknown                                  | equal $B$ and restarts; disclosed starts               | no baseline under-tuned                                                                 |

family cannot be moved to a better-validated range. What exists: engine speed scaled by 0.8 and 1.2 on test, four simulator-model counterfactuals that act on M1-v2’s concentration mechanism (Section 9.9), and one live smoke run of the default router on one cell, which matched replay’s windowed goodput within 0.25% while live prefill was slower and live decode faster than AIS’s, two errors that offset there (Section 11.1). The finalist comparison on GPUs compares rankings rather than absolute goodput (Section 11). On six  $N = 4$  test cells the live ranking agrees with simulation and M1-v2 leads **ramjet** wherever simulation predicts it, but the two errors no longer offset everywhere: live goodput ranges from 0.72 to 2.18 times the simulated value, differently for different policies in a cell, and 18 of the 30 live deltas against  $\pi_0$  depart from simulation beyond live noise, in both directions. So replay’s ranking held on hardware where it was checked, and its gap sizes did not.

**Perfectly fresh router state.** Replay applies KV events to the router’s index synchronously, so a policy always sees exactly which blocks each worker holds; a live router sees those events late. Load counters (active prefill tokens, decode blocks, active requests) come from the router’s own request tracking in both settings, but that their update timing matches is an assumption. Stale state matters most for a deterministic argmax: routing to the least-loaded server on old information can perform much worse than a randomized choice [33], and greedy dispatch on delayed state herds [49]. The expected direction was that a policy leaning harder on exact overlap and on small load differences, as the learned arms do, would lose more of its replay advantage under lag than the baselines. Training never sees lag, by operator decision; the test stage re-ran 17 finalists at

$\delta \in \{10, 50, 200\}$  ms on a separate build that delays KV events, prefill completions and request completions by  $\delta$  while keeping the router’s own dispatch bookkeeping immediate. The expectation was not borne out: M1-v2 kept first place at every lag with its lead inside the nominal interval, and on validation at 1,000 ms the baselines lost more to staleness than M1-v2 did (Section 9.9). Two limits remain. The lag model is uniform and deterministic, while live staleness varies with load and with router replicas; and a concentrated policy is the kind most exposed to herding, which only the live runs can test.

**One cache tier, and no global KV store.** Replay models only the GPU tier. A live Dynamo router’s accounting estimate blends host and disk tiers into the effective overlap, so feature 1 (`overlap_frac`) is the device overlap in replay but a weighted blend live, a distribution the policy never saw. Replay also has no global KV store, which changes which published routers can win. SMetric’s advantage over the best baseline holds “except for setups without any global KV\$ store”, and in SMetric’s own measurements on an agentic workload without a global tier, LMetric’s product score served 36% less traffic within SLO than a tuned linear cache-plus-load score [54, 63]. Session migration is cheap only when a global tier holds the context, so gains of that kind cannot appear here, and SMetric was not ported. The results speak to deployments whose KV cache lives on the serving GPUs.

**Replay-specific inputs.** Two replay artifacts could leak into a policy and were checked. Replay fills a request’s expected output length with the true one; `learned-choice` never reads that field, and a leakage audit confirmed it (Section 5.1). Replay also gives every request of a closed-loop flat trace a synthetic single-use session identifier that a live client would not send. For feature 6 this is harmless, since a single-use session never has a previous worker, but it made a session-keyed home feature of the optional set `v2` behave differently in open and closed loop until that feature was re-keyed on the prompt prefix. The upstream fix is not in the simulator version the campaign pins.

## 12.2 Workload and objective validity

**AgentX is another model’s traffic.** The 82 AgentX plays were recorded with Opus, not Qwen3-32B, and are replayed as Qwen3-32B traffic at up to 128K tokens. Request lengths, tool-call gaps and the dependency structure between calls are the recorded ones, so the family has realistic agent structure but not Qwen3-32B’s own output lengths or stopping behavior. Idle gaps longer than 300 s are capped: 140 of the 2,666 gaps, in 56 of the 82 plays, exceed it, and uncapped the plays would span 399,658 s of simulated time instead of 11,942 s. Plays are recycled with disjoint hash ranges so that worker counts up to 32 stay loaded, which by construction rules out KV reuse across copies. AgentX runs only in closed-loop lanes (caveat 7 of Table 2) and has no live load generator, so it is the one family the live check cannot cover. It is also the family on which the pilot M1 trailed and on which M1-v2 now leads on all four segments, so these choices bear directly on the headline.

**Generated and closely related traces.** Session traffic is generated: seeded multi-turn conversations whose turns extend the whole context, with lognormal lengths and exponential think times measured from the previous turn’s completion. Replay’s native session source could not express a growing conversation. The flat traces, Mooncake and the two FAST25 traces, come from one provider, and a third public trace was excluded as a relabeled copy of Mooncake. The FAST25 conversation trace shares the arrival and length skeleton of two Mooncake windows, so it adds cells, not independent workloads.

**Arrival burstiness is a modeling choice.** Mooncake and FAST25 conversation timestamps are logged on a grid of about 3s (the generated sessions on 1s), so a replay that keeps them puts whole bursts on one instant at any speed-up. Each replicate instead re-draws every session’s first arrival uniformly inside its logging slot. The choice is large: at  $N = 2$  and light load,  $\pi_0$ ’s 85th-percentile slowdown was 8.32 with the logged bursts and 1.46 with spreading. Absolute goodput levels therefore depend on this decision; paired comparisons on the same replicate are less exposed.

**Open-loop arrivals do not react.** In open-loop cells requests arrive on the trace’s schedule whatever the latency, while a real client slows down when served slowly. Trace-driven simulation is biased when the policy changes the traffic it replays [4]. Closed-loop and agentic-lane cells release work causally and are reported separately.

**SLOs and loads are defined by the default router.** Each family’s  $(I_{\max}, s_{\max})$  was fixed from  $\pi_0$ ’s runs at  $N = 8$ , and each cell’s load is where  $\pi_0$ ’s good fraction falls to 0.95, 0.85 or 0.65. That puts every cell in the default router’s transition region, which is where routing matters, but the region is defined relative to one baseline, and every ratio is normalized by the same baseline. The SLOs also differ widely across families ( $s_{\max}$  from 1.16 for AgentX to 3.47 for Mooncake), and the end-to-end slowdown clause does most of the work: the ITL clause alone fails 0.6–4.5% of  $\pi_0$ ’s requests at L2. The objective is therefore close to “finish near the uncontended latency”, and a different SLO could rank policies differently. The re-scoring of the test records under four other definitions of a good request kept M1-v2 first under each of them; under scaled SLOs it stayed first among the router-observable policies at every scale, though at scales 2 and 3 the AIS-league M2-ais edged it overall. Significance holds only at the calibrated scale and looser, and at a tighter SLO the session segments reverse (Section 9.9).

**Worker counts are compared at matched stress.** Loads were matched to  $\pi_0$ ’s knee at each  $N$ , not to a fixed per-worker rate; across worker counts the per-worker load differs by  $1.2\text{--}1.9\times$  ( $2.4\times$  in AgentX prefill). The held-out- $N$  result therefore compares policies at equally stressed operating points of the default, which is the intended question, but it does not show behavior at a matched per-worker rate. At large  $N$  capacity pools and simple policies catch up, and at small  $N$  scarcity can favor randomization, so per- $N$  differences need that context [52, 53].

### 12.3 Objective gaming and concentration

Black-box search exploits whatever its objective leaves out [22]. Three failure modes are documented for policies like this one. A binary per-request goodput rewards giving up on requests that will miss their SLO anyway [55]. An evolution strategy that tuned a routing cost for p95 TTFT alone set the queue weight to zero and sent all traffic to one replica [42]. And learned load balancers can drift into reserving or segregating servers in ways that break under workload shift [29]. Here, the ITL clause bounds the second failure, but only weakly, because a mean ITL averages out a long stall [55].

The audit found the second reading, sacrifice, for the first tuned M1, and the sign constraint removed its mechanism (Section 10). What remains is segregation by request size, which the request-count objective rewards: M1-v2 packs short cached requests onto a hot worker and gives up some long-prompt goodput. Whether that is acceptable depends on what the deployment values. Under the operator’s objective it is a legitimate win; under a size-aware objective it is partly a transfer, as the mixed token-weighted goodput shows (Section 10.5). Concentration also interacts with stale state: a policy that funnels traffic to a hot worker is the kind most exposed to herding, and only uniform, deterministic lag was tested (Section 12.1).

## 12.4 Statistical validity

**Few independent units.** The headline rests on 12 test segments, the unit of independence, and per-family claims rest on 2–4 and are descriptive only [10]. The segment structure is uneven: Mooncake and FAST25 conversation contribute 28 of the 60 test cells but 2 of the 12 segments, AgentX 14 cells and 4 segments (Table 19). Workload replicates and duplicated trace copies lengthen and re-draw a segment; they do not add independent segments.

**Effects near the detection limit.** The headline effect, +0.0316 per segment, is below the minimum detectable effect of 0.038 that the pilot fixed, and below the 0.060 spread across timing perturbations. The test still rejects, because 10 of 12 segments favor M1-v2 and the two that do not are ties at ceiling; but a significant small effect is not a measured size, so only the sign is claimed. The MDE is also measured in units of (ratio – 1), while the headline statistic is a clipped log-ratio; at these sizes the two agree to within a few percent of the effect ( $\log 1.038 = 0.0373$ ).

**Selection on validation.** Many configurations are compared on validation, and selection overfits what it selects on [7]. Disjoint uses of data contain this (Section 8.3): each policy’s configuration is chosen on validation replicates  $k = 0, 1, 2$ ; the best baseline, and by the same rule the headline learned arm, are chosen on fresh validation replicates  $k = 3, \dots, 10$ ; and the test set is evaluated once. The cross-policy step reuses the same fresh replicates for both choices, which keeps the comparison symmetric but makes both selected validation scores optimistic; only the test numbers are clean. Every learned variant scored on validation is counted and reported [11], and the 7 test cells at  $N = 6$ , a worker count validation also used, are labeled selection-exposed. With 6 validation segments, selection itself is noisy: the choice of `ramjet` over `llm-d-precise-prefix` flipped between the cell and the segment mean, and on test `llm-d-precise-prefix` was the stronger baseline. The learned side was also redesigned on validation evidence while the baseline side was not (Section 9.11); the single test pass keeps the headline’s  $p$ -value valid for the selected arm, but the originally planned M1 and M2 would not have passed.

## 12.5 Comparison validity and fairness

**Ports, not papers.** Baselines run as implemented on the campaign branch; by operator decision there are no paper-faithful variants, although different implementations of one method can perform materially differently [15]. The known gaps and whether tuning can close them:

- `lmetric` scores uncached prompt tokens times active requests plus one, without the queued-prefill term the paper credits for its advantage [63]. Tuning cannot add it: the port’s tunable parameters are its hot-spot detector’s, and at the largest allowed step they change 0.06% of decisions. The queued-prefill product is representable in the learned model’s feature set `v2`, which the headline arm uses; this asymmetry is why the headline is scoped to the branch’s ports: the default cost plus faithful LMetric carries most of the validation margin, and faithful LMetric alone, untuned, beats every tuned baseline on validation (Section 9.11).
- `dualmap` ships a pending-prefill budget of 65,536 tokens and `llm-d-optimized-baseline` a peak prefill rate of 15,928 tokens/s, both from other deployments [61]. Tuning reaches both: the budget is a free parameter, and the rate is pinned only because its product with the free TTFT-penalty parameter is what determines decisions.

A brief sanity check found that every port’s decisions respond to its inputs in the documented direction; two of them (`dualmap`, `chwb1`) lost to the default router at shipped defaults on the pilot’s

train subset, plausibly because hash-pinned policies suffer when a few hot prefixes carry most of the traffic.

**Equal budget, unequal search.** Every tuned policy gets  $B = 400$  evaluations per restart and  $R = 3$  restarts, but the policies differ in what that budget can buy. The search spaces range from 2 free parameters (`lmetric`, `chwb1`) to 7 (M1) and 22 (M1-v2). Each space’s initial step size was set so that one step changes about 15–20% of routing decisions; five baselines cannot reach that even at the largest allowed step, where they change 0.06% (`lmetric`), 1.8% (`dualmap`), 3.4% (`llm-d-optimized-baseline`), 8.3% (`two-tier`) and 13.3% (`ramjet`) of decisions. For those policies tuning cannot move far from the shipped behavior; that is a property of their parameterization, and a learned model with a richer one is expected to gain more from the same budget. The equal-budget rule fixes what each policy may spend, not how far each can get. The fairness audit’s validation-oracle probes found no baseline that more search would have improved (Section 6.7).

**Initialization.** Baselines restart from their shipped defaults. The learned arms’ second and third restarts start from a behavior clone of `llm-d-precise-prefix` and from a third informed point, for M1-v2 the faithful LMetric point. These starts use train-split decisions, not extra objective evaluations, so the budget is unchanged, but they hand the learned arms information about strong heuristics. Both selected configurations, M1’s and M1-v2’s, came from the default-router start `s1`, and M1 restricted to its default start is the same policy as M1 (Section 9.7). The faithful LMetric start is nevertheless strong on its own, which is the scope issue of Section 9.11.

**Excluded and separated baselines.** ThunderAgent’s routing half needs a request classifier that replay cannot run and is not evaluated [19]. Policies that read AIS estimates at decision time, including `llm-d`’s optimized baseline with modeled TTFT, compete in a separate, secondary league against the same normalizer (Section 9.10). SMetric and faithful LMetric were not tuned as baselines; an equal-budget comparison with them on fresh data is the open follow-up of Section 9.11.

## 12.6 Deployment scope

The study covers one model (Qwen3-32B), one engine (vLLM 0.24.0), one GPU type (H100 SXM), tensor parallelism 2, aggregated serving, identical workers, and 2 to 32 workers. Disaggregated prefill and decode, other models and hardware, and heterogeneous workers are out of scope. The coefficients are specific to this deployment; moving to another means re-tuning, and the cost of doing so (Table 29) is part of what a team would weigh.

The headline policy reads only inputs the live router already computes, so its YAML runs in the live router unchanged; whether those inputs carry the same values and timing live is what the lag check and the live run test. It decides by argmax ( $\tau = 0$ ) and leaves the router’s queue threshold unset, so it never holds a request back; a randomized variant and a jointly tuned queue threshold were tested and added nothing. Shipping it needs four settings to match replay: the router’s structural knobs (`router_track_prefill_tokens`, `router_assume_kv_reuse`, `router_track_active_blocks`) at their defaults, since they change what the features mean (caveat 3 of Table 2); host session affinity off, or accepted as a replacement for  $\theta_6$ , since the policy asks the host to treat an affinity target as exclusive; no `seed`, so that router replicas do not break ties in lockstep; and monitoring of the concentration guards of Section 10, since the policy’s expected behavior includes a hot worker at about twice the mean in-flight load. A live router’s overlap estimate also blends host and disk tiers, which replay does not have. The simulator-informed league would need AIS estimates, or a closed form calibrated offline, inside the live router.

## 13 Related work

Five lines of work meet in this study. For each we say what we take from it and where we differ.

**KV-cache-aware routing.** Most of our baselines come from this line (Section 7), and the published rules weigh prefix reuse against load in different ways. Preble sends a request to the GPU holding its prefix when the shared part outweighs the rest of the prompt, and otherwise to the GPU with the lowest cost, which counts recent load, the eviction the request would cause and the prefill of its missed tokens [48]. LMetric replaces weighted sums with the product of a worker’s new prefill tokens and its batch size, arguing that the best weight of a linear score changes with the workload: 0.7 on one of its traces and 0.55 on another [63]. DualMap restricts each prefix to two hashed candidates; it argues that picking the best of all workers amounts to a power-of- $N$ -choices rule that scatters shared prefixes, and that a per-request minimum-TTFT rule oscillates between cache-aware and load-aware decisions [61]. CacheRoute periodically plans which replicas serve the most popular prefixes, and gates each plan on a shadow replay because a higher cache-hit rate alone did not guarantee lower tail latency or higher capacity [8]. llm-d scores workers with a prefix scorer fed by an index of the engines’ KV events [25]. For agent traffic, SMetric sends a session’s follow-up turns to the worker with the most cached context unless that worker’s estimated load would break a slack multiple of the TTFT SLO [54]; Autellix keeps an agent program’s long calls on the program’s engine and sends short calls to the least-used engine [26]; and ThunderAgent schedules whole programs, observing that a router-side prefix tree that approximates worker caches goes stale under eviction thrashing [19]. A workload study simulated cache-first, load-first and sticky routing at several instance counts and tracked hit ratio, per-session replication and load imbalance [35].

These rules disagree about which signal should dominate. On an agentic workload without a global KV tier, which is the setting of our replay, SMetric measured LMetric serving 36% less SLO-meeting throughput than a tuned linear cache-plus-load score, with prefix reuse of 45% against 74% [54]. We do not propose another heuristic. We tune these rules on equal terms, and the learned policy’s features draw on their signals: uncached prefill, active prefill, batch size, KV utilization and session affinity, and, in feature set v2, LMetric’s log-product and a hashed home-worker indicator in the spirit of DualMap (Tables 8 and 9).

**Learned routers for LLM serving.** Learned routers differ in what they learn and in where they learn it. Jain et al. train a reinforcement-learning router with a heuristic-guided reward; its action picks one of  $m$  instances or none, so the network’s shape is tied to the instance count [17]. Lodestar learns online an MLP that predicts each instance’s reward from per-instance features, with parameters shared across instances and no instance identity, so it does not depend on the instance count. It also shows that a latency predictor trained offline on logs of other routers loses its ranking signal once it routes, because the routing policy changes the states the predictor is evaluated on [23]. GORGO tunes the weights of an additive routing cost with an evolution strategy; optimizing p95 TTFT alone drove the queue weight to zero and sent every request to the closest replica [42]. Wu et al. analyze KV eviction and learning-based load balancing together [57]. Routers that choose among models or service-quality levels, rather than among replicas of one model, address a different decision [9, 18]. The closest study in method is Calibrate-then-Route [52], a router with hand-set weights and fitted constants that was developed in a simulator and then measured on GPUs in disaggregated serving. With simulator-derived constants it lost 4.5 goodput points to the same router with measured constants (0.819 against 0.864). After re-fitting, the simulator matched measured goodput to a mean absolute residual of 0.068 but did not reproduce the measured winner on any workload.

In a starved pool, greedy cost minimization collapsed to 0.292 goodput while round-robin held 0.680. AgentServeSim releases agent turns causally in simulation and searches policies with an LLM-driven evolutionary loop; its best evolved policies improved mean job completion time by 0.5% for KV retention and 2.8% for scheduling over hand-written starting policies [41]. Our policy differs from these in three ways. It is trained on-policy, since every candidate is scored by replaying it, which avoids the logged-policy mismatch Lodestar describes, and it is frozen afterwards. Its objective counts a request as good only if both its inter-token latency and its end-to-end slowdown are within bounds, where GORGO’s objective was p95 TTFT alone. And Calibrate-then-Route’s findings are why ranking stability under router-state lag and on GPUs is part of the protocol (Sections 8.7 and 11). Among the learned routers we read, none trains at some worker counts and tests at unseen ones; Lodestar’s shared predictor is independent of the instance count by construction.

**Learned systems policies and black-box search.** Decima learns cluster scheduling with reinforcement learning in a simulator, and it fixes the job-arrival sequence across episodes to cut the variance that stochastic arrivals add [30, 31]. It shows both sides of transfer: an agent trained on a cluster ten times smaller was 3% worse in average job completion time (630 against 610 s), while one trained on a mismatched load did worse than the tuned heuristic (104.8 against 91.2 s) [30]. Pensieve trains a bitrate controller in a chunk-level simulator and disabled TCP slow-start restart on its video server so that deployment matched a simulator assumption [28]. Park catalogs the difficulties of learning systems policies; one of its load balancers learned to keep a server free for small jobs and was not robust when the job mix changed, and it suggests bootstrapping search from existing policies [29], as M1’s heuristic-derived starting points do (Section 6.4). The cautions are empirical. Puffer’s randomized trial found it difficult for sophisticated or learned bitrate controllers to outperform a simple buffer-based scheme despite good results in emulation, and a controller whose predictor was trained in emulation performed poorly in deployment [59]. Trace-driven simulation is biased when the policy changes the traces it replays [4]. Learned policies that win on average can still lose to the baselines in a substantial fraction of environments [58]. Randomizing latency and observation noise was necessary for simulation-trained robot control to transfer [39]. And search exploits simulator bugs [22]. These findings motivate the closed-loop and agent-lane cells, the per-cell winner map, the timing and lag perturbations, the concentration audit and the live check (Section 8).

On method, static linear policies trained by random search matched reinforcement learning’s sample efficiency on the MuJoCo benchmarks, given normalized inputs [27]; evolution strategies’ gradient estimates do not depend on episode length, which suits long horizons with delayed reward [44]. We use CMA-ES [12], score every candidate of a generation on the same random scenarios, as PEGASUS fixes scenarios to make policy evaluation deterministic [34], and borrow UH-CMA-ES’s rank-change measurement as a diagnostic, without its treatment of the measured uncertainty [13]. Load-balancing theory frames two risks of our setting: sending work to the least-loaded server on old load information can significantly hurt performance [33], and delayed information makes join-the-shortest-queue dispatch herd [49]; how balancing rules compare also changes with the number of servers and the load regime [53].

**Discrete choice with context effects.** With rank 0, **learned-choice** is a conditional logit, the base case of the generalized extreme-value family [51]: one coefficient vector scores every candidate, so the same parameters serve any worker count, and at  $\tau = 0$  the policy takes the logit’s most likely choice. Halo models add pairwise terms between specific items [60]; **learned-choice** has no worker-identity terms, so its parameters apply unchanged as workers come and go. A plain logit

satisfies the independence of irrelevant alternatives: the preference between two candidates never depends on which others are present, however rich the item score [43]. M2’s context terms and v2’s set-relative features add that dependence. In the linear context logit (LCL) of Tomlinson and Benson [50] the coefficient vector shifts linearly with the mean features of the choice set,  $\theta + A\bar{x}_S$ . M2’s pooled form is an LCL with a low-rank  $A$ ; its named-source form is an LCL over a vector of named set statistics, with one free column of  $A$  per statistic. Neither is their DLCL, a mixture of logits with separate intercepts. Three of their results shaped our design. Choice depends only on deviations from the set mean, so terms that are constant across the set cancel.  $A$  is identifiable only from at least  $d + 1$  choice sets with affinely independent mean features. And LCL did not beat the plain logit within a fixed training budget although it nests it, one reason the ladder adds context in a small step that starts from the selected M1 and is compared with M1 trained for the same added evaluations (Section 5.9).

Other set-dependent models use pairwise context vectors summed over the set [47], set-dependent aggregation [43], pooled set embeddings [40], mixtures of random-utility networks [5], self-attention [20] and stacked interaction layers [64]; divisive normalization fits human choices across set sizes from 2 to 12 [56] and inspired v2’s ratio features. Low-rank choice parameters are unique only up to a multiplicative factor unless a penalty fixes the scale [20]. M2’s pooled form has the same freedom, since scaling  $p_\ell$  up and  $q_\ell$  down by one factor changes nothing, while the named-source form, which is linear in  $p_\ell$ , has none. The nested logit groups alternatives whose unobserved utilities are correlated [51]; under a deterministic argmax its nest parameters have no effect (our inference), so node structure would enter as nest-level features rather than as a GEV model (Section 5.7). In reinforcement learning, Jain et al. [16] use a shared utility followed by a softmax over the available actions, a conditional logit over a variable action set; generalizing it to unseen action sets took subsampled action sets, entropy regularization and model selection on held-out action sets. Our analogue of the last is validation at  $N = 6$ , a worker count training never sees. The named context sources are request-level quantities or means and maxima over candidates, a permutation-invariant pooling [62], chosen so that duplicating every worker leaves them unchanged.

**Goodput, SLOs and evaluation methodology.** DistServe defines goodput as the highest request rate that meets an SLO-attainment target, and sweeps a single scale factor over its SLOs [65]. We measure good requests per second at a fixed load inside a window, and judge a request by its mean inter-token latency and its end-to-end slowdown against an uncontended latency (Section 3.3); as our analogue of DistServe’s sweep, the finalists’ test runs are re-scored at several SLO scales. Binary SLO goodput rewards giving up on requests that will miss anyway, and a TPOT-style bound averages out stalls in the middle of a request [55]; both are why the audits look for sacrificed requests and concentrated load (Section 8.8). Open and closed load generators behave differently: scheduling helps a closed system only at moderate load, and, as a rule of thumb, sessions of more than ten requests make a partly open system behave as closed [45]; we therefore stratify by load mode. Simulator error grows near capacity: in Vidur, small prediction errors near the capacity point blow up into large latency errors [3]. A load-testing study reports that handling warm-up properly improved the accuracy of capacity estimates by 22.2% across 14 industrial cases [1]. Our statistics follow the guidance for comparisons over few independent tasks: a Wilcoxon signed-rank test over independent units [10], bootstrap intervals and interquartile means over tasks [2], a probability-of-improvement effect size with thresholds for significant and meaningful [6], several seeds and restarts [15], budget-aware reporting [11], and selection on data the final test never sees [7].

**Position of this work.** The study combines four choices that we did not find together in the work above. The policy is a deployable choice rule, linear in inputs the live router already has, rather than a latency predictor. Every baseline is tuned with the same budget and selection rule as the learned model, and the headline comparison is against the validation-selected best of them. Worker counts are held out: training uses  $N \in \{4, 8\}$ , and the test adds 2, 6, 16 and 32. And the headline test is pre-registered over independent trace segments. Two of our findings echo this literature. The learned policy’s advantage over the ported heuristics is largely an LMetric-style interaction between uncached prefill and load (Sections 9.8 and 9.11), which supports LMetric’s point that the two should interact, though here as a correction added to the default’s additive cost rather than in place of it; and the first unconstrained search found the kind of objective gaming GORGO and Lehman et al. document, which a sign constraint on load coefficients removed (Section 10). The price is that every result in this paper is simulated; as Calibrate-then-Route and Puffer show, a simulated ranking need not survive deployment. Our live check on six test cells found the simulated ranking intact on GPUs, but not the simulated gap sizes (Section 11).

## 14 Conclusion

Dynamo’s worker-selection API makes every routing rule, from the default cost function to the ported heuristics, a small composition of scorers and a picker over a few declared inputs. That uniformity made it possible to tune every rule with the same budget in AIS-timed replay, and to learn a generalization of them, a conditional-logit policy over the router’s own signals that starts from the default router. In simulation, the learned policy M1-v2 beats the branch’s ported heuristics, each tuned with an equal budget, on a test set that holds out time windows, agent plays, session seeds, a trace family and worker counts, and it stays first, ahead of Ramjet, under router-state lag, timing perturbations and alternative definitions of a good request. The effect is small, and only its sign is claimed; the per-axis results, such as those at unseen worker counts, are descriptive. A pre-registered check on H100 GPUs, on six of the test cells, found the same policy ranking and the same sign of M1-v2 against Ramjet as replay, while absolute goodput and the size of the gaps between policies departed from replay.

For a deployment, the evidence supports three practical statements and one caution.

- **Prefer a cache-aware rule over the shipped default, and tune it.** Every strong heuristic beats the default router by a wide margin here, and tuning the default’s own knobs closes only part of the gap.
- **Make uncached prefill more expensive on busy workers.** The learned policy’s largest term is an interaction between uncached prefill and a worker’s active requests; zeroing it costs about half of M1-v2’s validation margin over the best tuned baseline. Faithful LMetric multiplies the same two quantities, and the default cost plus faithful LMetric reproduces 53 to 66% of that margin on validation, so it is the simpler candidate to validate next to M1-v2.
- **Constrain what search may learn.** Tuning a request-count objective without constraints produced a policy that parked long requests on a busy worker. A sign constraint on load coefficients removed it with no loss (the constrained M1 is +0.004 per test segment against the unconstrained one), and should be the default for any learned router of this kind; the remaining size segregation should be monitored with the window-basis guards.
- **Caution.** Long prompts wait longer under M1-v2 than under `ramjet`; a deployment that values served tokens, or long-context latency, may prefer the heuristic.

### Next steps.

- A larger live check: repeated runs per policy, randomly drawn cells, other worker counts and an AgentX load generator, which could test the size of the effect on hardware and not only its sign (Section 11.4).
- An equal-budget comparison with faithful LMetric and an SMetric-style baseline on fresh segments or on live runs over non-test cells, which would measure the learned increment the test set could not.
- A size-aware objective or constraint, if a deployment values tokens rather than requests.
- Disaggregated prefill and decode, other models, heterogeneous workers, and a live load generator for agentic traffic.

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## A Full result tables

The tables in this appendix are generated from the final report’s table fragments, like those in Section 9. Intervals are 95% segment-bootstrap intervals ( $\dagger$ : within-segment cell bootstrap for strata with fewer than three segments); only the headline row of Table 40 is a test.

**Table 40:** Every test policy against  $\pi_0$  and against the validation-selected best baseline (tuned ramjet), 60 cells in 12 segments, segment means with 95% segment-bootstrap intervals. Selected config: restart, generation and candidate (best-so-far sample or CMA-ES mean) chosen on validation replicates  $k_{0..2}$ . Val  $k_{3..10}$ : mean clipped log-ratio against  $\pi_0$  on fresh validation replicates, the cross-policy selection score; policies at shipped defaults have none. The one-sided  $p$  is a test only in the headline row; elsewhere it is descriptive.

| Policy                                | Selected config                       | Val $k_{3..10}$ | Test vs $\pi_0$ [95% CI]   | Test vs ramjet [95% CI]    | Ahead | $p$    |
|---------------------------------------|---------------------------------------|-----------------|----------------------------|----------------------------|-------|--------|
| M1-v2 (headline learned arm)          | m1v2-s1 gen 20                        | 0.1783          | +0.1758 [+0.1270, +0.2218] | +0.0316 [+0.0150, +0.0501] | 10/12 | 0.0017 |
| M2-ais                                | best_so_far<br>m2ais-s3 gen 10        | 0.1735          | +0.1729 [+0.1239, +0.2191] | +0.0286 [+0.0138, +0.0458] | 11/12 | 0.0005 |
| M1-noaff                              | mean<br>m1noaff-s1 gen 25             | 0.1692          | +0.1712 [+0.1218, +0.2178] | +0.0270 [+0.0109, +0.0443] | 8/12  | 0.0134 |
| M1 + queue threshold (abl. a)         | best_so_far<br>ablaml-s1 gen 25       | 0.1684          | +0.1714 [+0.1222, +0.2179] | +0.0272 [+0.0115, +0.0444] | 8/12  | 0.0171 |
| M1 continued (secondary)              | mean<br>m1cont-s3 gen 25              | 0.1672          | +0.1699 [+0.1211, +0.2158] | +0.0256 [+0.0090, +0.0437] | 8/12  | 0.0212 |
| M1-ais                                | best_so_far<br>m1ais-s1 gen 25        | 0.1718          | +0.1699 [+0.1208, +0.2163] | +0.0256 [+0.0113, +0.0423] | 11/12 | 0.0012 |
| M2 rank 2                             | best_so_far<br>m2r2-s3 gen 25         | 0.1677          | +0.1653 [+0.1176, +0.2107] | +0.0211 [+0.0043, +0.0406] | 7/12  | 0.0549 |
| M1 (= A12 M1-default-init)            | mean<br>m1c-s1 gen 20                 | 0.1656          | +0.1628 [+0.1156, +0.2072] | +0.0186 [+0.0013, +0.0378] | 7/12  | 0.0549 |
| M1 unconstrained (gaming-flagged)     | mean<br>m1-s2 gen 15                  | n/a             | +0.1590 [+0.1114, +0.2047] | +0.0147 [+0.0032, +0.0284] | 7/12  | 0.0386 |
| llm-d-precise-prefix                  | mean<br>llmdpp-s2 gen 25              | 0.1393          | +0.1464 [+0.0986, +0.1939] | +0.0022 [−0.0089, +0.0139] | 4/12  | 0.4548 |
| ramjet (val-best baseline)            | ramjet-s3 gen 25                      | 0.1411          | +0.1442 [+0.0983, +0.1894] | (reference)                |       |        |
| ramjet + queue threshold (abl. a)     | best_so_far<br>ablabase-s3 gen 5      | 0.1405          | +0.1426 [+0.0970, +0.1875] | −0.0016 [−0.0050, +0.0010] | 4/12  | 0.8833 |
| M0-ais                                | best_so_far<br>m0ais-s2 gen 20        | 0.1397          | +0.1423 [+0.0967, +0.1874] | −0.0019 [−0.0086, +0.0048] | 6/12  | 0.6890 |
| llm-d-precise-prefix@defaults         | best_so_far<br>shipped                | n/a             | +0.1442 [+0.0952, +0.1932] | −0.0000 [−0.0137, +0.0139] | 4/12  | 0.5452 |
| sticky-session bounded                | parameters<br>stickybounded-s2 gen 25 | 0.1338          | +0.1357 [+0.0893, +0.1818] | −0.0086 [−0.0157, −0.0024] | 3/12  | 0.9866 |
| lmetric@defaults                      | mean<br>shipped                       | n/a             | +0.1392 [+0.0889, +0.1901] | −0.0050 [−0.0174, +0.0068] | 8/12  | 0.5750 |
| two-tier                              | parameters<br>twotier-s1 gen 15       | 0.1345          | +0.1374 [+0.0879, +0.1866] | −0.0068 [−0.0174, +0.0029] | 6/12  | 0.7407 |
| ramjet@defaults                       | mean<br>shipped                       | n/a             | +0.1312 [+0.0926, +0.1706] | −0.0130 [−0.0227, −0.0036] | 3/12  | 0.9788 |
| two-tier@defaults                     | parameters<br>shipped                 | n/a             | +0.1371 [+0.0867, +0.1875] | −0.0071 [−0.0184, +0.0036] | 6/12  | 0.7881 |
| lmetric (port)                        | parameters<br>lmetric-s2 gen 10       | 0.1321          | +0.1356 [+0.0860, +0.1857] | −0.0086 [−0.0192, +0.0011] | 6/12  | 0.8303 |
| M0 (default cost fn, tuned)           | mean<br>m0-s3 gen 20                  | 0.1326          | +0.1162 [+0.0777, +0.1562] | −0.0281 [−0.0439, −0.0137] | 1/12  | 0.9993 |
| sticky-session hard                   | mean<br>stickyhard-s1 gen 25          | 0.1125          | +0.1127 [+0.0682, +0.1609] | −0.0316 [−0.0603, −0.0073] | 4/12  | 0.9539 |
| llm-d-optimized-baseline (throughput) | mean<br>llmdob-s1 gen 5               | 0.1035          | +0.0844 [+0.0377, +0.1341] | −0.0598 [−0.0946, −0.0277] | 0/12  | 1.0000 |
| llm-d-optimized-baseline@throughput   | mean<br>shipped                       | n/a             | +0.0809 [+0.0325, +0.1314] | −0.0633 [−0.0979, −0.0321] | 0/12  | 1.0000 |
| llm-d-optimized-baseline (modeled)    | parameters<br>llmdobm-s2 gen 15       | 0.0973          | +0.0724 [+0.0241, +0.1232] | −0.0718 [−0.1057, −0.0392] | 0/12  | 1.0000 |

| Policy                  | Selected config        | Val $k_{3..10}$ | Test vs $\pi_0$ [95% CI]   | Test vs ramjet [95% CI]    | Ahead | $p$    |
|-------------------------|------------------------|-----------------|----------------------------|----------------------------|-------|--------|
| sticky-bounded@defaults | shipped parameters     | n/a             | +0.0982 [+0.0475, +0.1514] | -0.0460 [-0.0760, -0.0201] | 1/12  | 0.9983 |
| sticky-hard@defaults    | shipped parameters     | n/a             | +0.0806 [+0.0305, +0.1368] | -0.0637 [-0.0970, -0.0311] | 2/12  | 0.9976 |
| chwbl                   | chwbl-s1 gen 5 mean    | 0.0218          | -0.0299 [-0.0711, +0.0132] | -0.1741 [-0.2378, -0.1115] | 0/12  | 1.0000 |
| dualmap                 | dualmap-s3 gen 10 mean | 0.0263          | -0.0044 [-0.0366, +0.0297] | -0.1486 [-0.1794, -0.1148] | 0/12  | 1.0000 |
| dualmap@defaults        | shipped parameters     | n/a             | -0.0817 [-0.1200, -0.0428] | -0.2257 [-0.2779, -0.1769] | 0/12  | 1.0000 |
| chwbl@defaults          | shipped parameters     | n/a             | -0.1866 [-0.2557, -0.1258] | -0.3254 [-0.4185, -0.2435] | 0/12  | 1.0000 |
| round_robin             | shipped parameters     | n/a             | -0.5631 [-0.7147, -0.4031] | -0.6657 [-0.8234, -0.5019] | 0/12  | 1.0000 |

Table 41: By worker count, against  $\pi_0$  (generated; selected rows).

| Policy                            | $N = 2$                 | 4                       | 6                       | 8                       | 16                      | 32                      |
|-----------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.105 [+0.060, +0.166] | +0.152 [+0.079, +0.233] | +0.167 [+0.104, +0.222] | +0.173 [+0.106, +0.241] | +0.267 [+0.140, +0.399] | +0.265 [+0.146, +0.379] |
| M2-ais                            | +0.091 [+0.041, +0.158] | +0.151 [+0.076, +0.237] | +0.166 [+0.101, +0.223] | +0.169 [+0.102, +0.237] | +0.266 [+0.141, +0.397] | +0.267 [+0.147, +0.381] |
| M1 (= A12 M1-default-init)        | +0.099 [+0.056, +0.161] | +0.139 [+0.065, +0.221] | +0.149 [+0.093, +0.202] | +0.163 [+0.094, +0.232] | +0.245 [+0.133, +0.354] | +0.252 [+0.143, +0.356] |
| M1 unconstrained (gaming-flagged) | +0.083 [+0.038, +0.151] | +0.139 [+0.066, +0.229] | +0.151 [+0.091, +0.207] | +0.154 [+0.090, +0.219] | +0.245 [+0.130, +0.362] | +0.254 [+0.142, +0.362] |
| llm-d-precise-prefix              | +0.072 [+0.031, +0.128] | +0.102 [+0.041, +0.198] | +0.136 [+0.075, +0.198] | +0.131 [+0.079, +0.188] | +0.242 [+0.121, +0.379] | +0.244 [+0.129, +0.356] |
| ramjet (val-best baseline)        | +0.067 [+0.021, +0.141] | +0.098 [+0.044, +0.182] | +0.137 [+0.077, +0.198] | +0.130 [+0.075, +0.191] | +0.238 [+0.124, +0.357] | +0.241 [+0.137, +0.338] |
| lmetric (port)                    | +0.070 [+0.018, +0.148] | +0.089 [+0.031, +0.184] | +0.124 [+0.067, +0.181] | +0.114 [+0.057, +0.180] | +0.228 [+0.116, +0.344] | +0.229 [+0.129, +0.327] |
| M0 (default cost fn, tuned)       | +0.055 [+0.014, +0.118] | +0.047 [+0.031, +0.063] | +0.107 [+0.068, +0.150] | +0.108 [+0.058, +0.167] | +0.195 [+0.113, +0.266] | +0.212 [+0.129, +0.292] |
| round_robin                       | -0.325 [-0.455, -0.185] | -0.388 [-0.534, -0.263] | -0.684 [-0.964, -0.397] | -0.477 [-0.690, -0.277] | -0.675 [-0.950, -0.376] | -0.640 [-0.940, -0.322] |

Table 42: By family, against  $\pi_0$  (generated; selected rows).

| Policy                            | Mooncake                             | FAST25 conv.                         | FAST25 synth.                        | Sessions                | AgentX                  |
|-----------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.154 [+0.126, +0.181] <sup>†</sup> | +0.587 [+0.486, +0.688] <sup>†</sup> | +0.120 [+0.080, +0.159] <sup>†</sup> | +0.178 [+0.089, +0.264] | +0.192 [+0.096, +0.257] |
| M2-ais                            | +0.150 [+0.122, +0.178] <sup>†</sup> | +0.562 [+0.471, +0.653] <sup>†</sup> | +0.118 [+0.079, +0.157] <sup>†</sup> | +0.179 [+0.089, +0.265] | +0.186 [+0.089, +0.252] |
| M1 (= A12 M1-default-init)        | +0.144 [+0.116, +0.172] <sup>†</sup> | +0.565 [+0.463, +0.667] <sup>†</sup> | +0.122 [+0.086, +0.158] <sup>†</sup> | +0.176 [+0.088, +0.261] | +0.159 [+0.065, +0.221] |
| M1 unconstrained (gaming-flagged) | +0.132 [+0.104, +0.159] <sup>†</sup> | +0.543 [+0.445, +0.641] <sup>†</sup> | +0.092 [+0.057, +0.127] <sup>†</sup> | +0.176 [+0.088, +0.260] | +0.170 [+0.074, +0.233] |
| llm-d-precise-prefix              | +0.090 [+0.065, +0.116] <sup>†</sup> | +0.296 [+0.289, +0.302] <sup>†</sup> | +0.069 [+0.034, +0.103] <sup>†</sup> | +0.168 [+0.084, +0.251] | +0.184 [+0.089, +0.249] |
| ramjet (val-best baseline)        | +0.104 [+0.077, +0.132] <sup>†</sup> | +0.200 [+0.200, +0.200] <sup>†</sup> | +0.089 [+0.064, +0.115] <sup>†</sup> | +0.178 [+0.088, +0.264] | +0.155 [+0.063, +0.219] |
| lmetric (port)                    | +0.077 [+0.051, +0.104] <sup>†</sup> | +0.103 [+0.102, +0.105] <sup>†</sup> | +0.055 [+0.017, +0.093] <sup>†</sup> | +0.180 [+0.090, +0.266] | +0.159 [+0.067, +0.224] |
| M0 (default cost fn, tuned)       | +0.094 [+0.070, +0.120] <sup>†</sup> | +0.049 [+0.045, +0.053] <sup>†</sup> | +0.053 [+0.026, +0.081] <sup>†</sup> | +0.165 [+0.082, +0.248] | +0.111 [+0.051, +0.144] |
| round_robin                       | -0.168 [-0.208, -0.133] <sup>†</sup> | -0.229 [-0.232, -0.227] <sup>†</sup> | -0.270 [-0.283, -0.258] <sup>†</sup> | -0.811 [-0.904, -0.687] | -0.656 [-0.742, -0.528] |

Table 43: By load mode, against tuned ramjet (top) and against  $\pi_0$  (bottom) (generated; selected rows).

| Policy                            | Open loop (31 cells, 8 seg.) | Closed loop (15, 6)     | Agentic lanes (14, 4)   |
|-----------------------------------|------------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.028 [+0.003, +0.061]      | +0.035 [+0.011, +0.063] | +0.037 [+0.033, +0.041] |
| M2-ais                            | +0.026 [+0.003, +0.055]      | +0.037 [+0.013, +0.064] | +0.031 [+0.025, +0.034] |
| M1 (= A12 M1-default-init)        | +0.027 [+0.005, +0.056]      | +0.030 [+0.006, +0.059] | +0.004 [-0.019, +0.025] |
| M1 unconstrained (gaming-flagged) | +0.016 [-0.003, +0.042]      | +0.017 [+0.001, +0.033] | +0.015 [+0.010, +0.020] |
| llm-d-precise-prefix              | -0.008 [-0.019, -0.000]      | -0.014 [-0.021, -0.008] | +0.029 [+0.025, +0.034] |
| lmetric (port)                    | -0.021 [-0.037, -0.006]      | -0.006 [-0.016, +0.004] | +0.004 [-0.004, +0.012] |
| M0 (default cost fn, tuned)       | -0.016 [-0.032, -0.002]      | -0.024 [-0.034, -0.014] | -0.044 [-0.079, -0.010] |
| round_robin                       | -0.597 [-0.844, -0.372]      | -0.631 [-0.867, -0.400] | -0.740 [-0.865, -0.562] |
| default@defaults (reference)      | -0.098 [-0.162, -0.051]      | -0.205 [-0.258, -0.154] | -0.155 [-0.219, -0.063] |

| Policy                            | Open loop (31 cells, 8 seg.) | Closed loop (15, 6)     | Agentic lanes (14, 4)   |
|-----------------------------------|------------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.126 [+0.065, +0.199]      | +0.241 [+0.208, +0.273] | +0.192 [+0.096, +0.257] |
| M2-ais                            | +0.124 [+0.064, +0.196]      | +0.243 [+0.211, +0.274] | +0.186 [+0.089, +0.252] |
| M1 (= A12 M1-default-init)        | +0.125 [+0.065, +0.195]      | +0.236 [+0.203, +0.269] | +0.159 [+0.065, +0.221] |
| M1 unconstrained (gaming-flagged) | +0.114 [+0.056, +0.186]      | +0.222 [+0.187, +0.259] | +0.170 [+0.074, +0.233] |
| llm-d-precise-prefix              | +0.090 [+0.040, +0.157]      | +0.191 [+0.145, +0.236] | +0.184 [+0.089, +0.249] |
| ramjet (val-best baseline)        | +0.098 [+0.051, +0.162]      | +0.205 [+0.154, +0.258] | +0.155 [+0.063, +0.219] |
| lmetric (port)                    | +0.078 [+0.030, +0.146]      | +0.200 [+0.140, +0.259] | +0.159 [+0.067, +0.224] |
| M0 (default cost fn, tuned)       | +0.082 [+0.037, +0.148]      | +0.182 [+0.135, +0.230] | +0.111 [+0.051, +0.144] |
| round_robin                       | -0.550 [-0.825, -0.302]      | -0.426 [-0.623, -0.230] | -0.656 [-0.742, -0.528] |

**Table 44:** Hold-out strata, against tuned **ramjet** (top) and against  $\pi_0$  (bottom) (generated; selected rows).

| Policy                            | Unseen $N$ 2, 16, 32 (23 cells) | Seen $N$ 4, 8 (30)      | Transforms (10)         |
|-----------------------------------|---------------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.034 [+0.016, +0.052]         | +0.035 [+0.012, +0.061] | +0.025 [+0.007, +0.044] |
| M2-ais                            | +0.030 [+0.014, +0.046]         | +0.033 [+0.013, +0.057] | +0.022 [+0.006, +0.039] |
| M1 (= A12 M1-default-init)        | +0.015 [-0.001, +0.032]         | +0.022 [+0.000, +0.048] | +0.010 [+0.001, +0.018] |
| M1 unconstrained (gaming-flagged) | +0.012 [+0.003, +0.021]         | +0.023 [+0.006, +0.043] | +0.011 [-0.000, +0.023] |
| llm-d-precise-prefix              | +0.009 [-0.009, +0.028]         | +0.004 [-0.006, +0.016] | -0.001 [-0.019, +0.018] |
| lmetric (port)                    | -0.007 [-0.016, +0.001]         | -0.007 [-0.021, +0.005] | -0.003 [-0.021, +0.015] |
| M0 (default cost fn, tuned)       | -0.042 [-0.086, -0.012]         | -0.037 [-0.085, -0.007] | -0.007 [-0.017, -0.001] |
| round_robin                       | -0.718 [-0.928, -0.470]         | -0.597 [-0.753, -0.453] | -0.576 [-0.815, -0.367] |
| default@defaults (reference)      | -0.212 [-0.317, -0.120]         | -0.118 [-0.177, -0.066] | -0.091 [-0.166, -0.043] |

| Policy                            | Unseen $N$ 2, 16, 32 (23 cells) | Seen $N$ 4, 8 (30)      | Transforms (10)         |
|-----------------------------------|---------------------------------|-------------------------|-------------------------|
| M1-v2 (headline learned arm)      | +0.246 [+0.145, +0.362]         | +0.152 [+0.097, +0.212] | +0.116 [+0.070, +0.183] |
| M2-ais                            | +0.241 [+0.141, +0.356]         | +0.151 [+0.094, +0.213] | +0.113 [+0.065, +0.182] |
| M1 (= A12 M1-default-init)        | +0.226 [+0.137, +0.322]         | +0.140 [+0.083, +0.200] | +0.101 [+0.053, +0.171] |
| M1 unconstrained (gaming-flagged) | +0.223 [+0.132, +0.326]         | +0.140 [+0.085, +0.203] | +0.102 [+0.057, +0.171] |
| llm-d-precise-prefix              | +0.221 [+0.119, +0.341]         | +0.122 [+0.069, +0.185] | +0.090 [+0.049, +0.155] |
| ramjet (val-best baseline)        | +0.212 [+0.120, +0.317]         | +0.118 [+0.066, +0.177] | +0.091 [+0.043, +0.166] |
| lmetric (port)                    | +0.205 [+0.111, +0.310]         | +0.111 [+0.057, +0.175] | +0.088 [+0.038, +0.166] |
| M0 (default cost fn, tuned)       | +0.170 [+0.107, +0.234]         | +0.081 [+0.047, +0.126] | +0.084 [+0.041, +0.149] |
| round_robin                       | -0.605 [-0.840, -0.356]         | -0.498 [-0.640, -0.364] | -0.498 [-0.732, -0.312] |

**Table 45:** Coefficients of the **v1** learned arms,  $\theta_f$  (with  $\theta_f \text{sd}_f$  in parentheses);  $\theta_0 = -1$  throughout (generated). M2 rank 2 also has a context term, whose directions are in the finalists file.

| Arm                               | $\theta_1$     | $\theta_2$     | $\theta_3$     | $\theta_4$     | $\theta_5$    | $\theta_6$       | $\theta_7$      |
|-----------------------------------|----------------|----------------|----------------|----------------|---------------|------------------|-----------------|
| M1 (= A12 M1-default-init)        | +16.5 (+4.24)  | -2.27 (-0.80)  | +0 (+0.00)     | +0 (+0.00)     | -39.2 (-2.35) | +0.206 (+0.07)   | +0 (+0.00)      |
| M1-noaff                          | +18.9 (+4.84)  | +0.996 (+0.35) | +0 (+0.00)     | +0 (+0.00)     | -44.2 (-2.65) | +0 (+0.00)       | +0 (+0.00)      |
| M1 + queue threshold (abl. a)     | +19.1 (+4.88)  | +1.04 (+0.36)  | +0 (+0.00)     | +0 (+0.00)     | -38.3 (-2.30) | +0.00905 (+0.00) | +0 (+0.00)      |
| M2 rank 2                         | +16.2 (+4.16)  | -0.954 (-0.34) | +0 (+0.00)     | +0 (+0.00)     | -34.6 (-2.07) | -0.493 (-0.16)   | +0 (+0.00)      |
| M1 continued (secondary)          | +17.9 (+4.58)  | +0.748 (+0.26) | +0 (+0.00)     | +0 (+0.00)     | -47.5 (-2.85) | -0.794 (-0.26)   | +0 (+0.00)      |
| M1 unconstrained (gaming-flagged) | +61.3 (+15.71) | +1.98 (+0.70)  | -0.724 (-0.93) | -0.222 (-0.02) | -146 (-8.74)  | -2.85 (-0.92)    | +0.0878 (+0.19) |

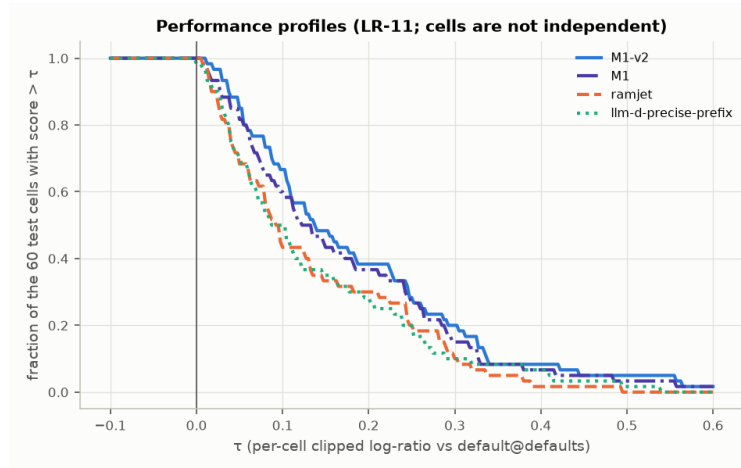
**Table 46:** Secondary metrics against ramjet on test, part 1 (latency) (descriptive, computed after the test pass from its records, no replay). Each entry is the segment-mean log-ratio, with the number of segments on which the policy's value is higher. Lower is better for latencies, higher for throughput and prefix reuse.

| Policy                                | TTFT p50       | TTFT p90       | TTFT p99       | TTFT p90, 32-64K | E2E p90 (in window) |
|---------------------------------------|----------------|----------------|----------------|------------------|---------------------|
| M1-v2 (headline learned arm)          | +0.006 (9/12)  | -0.002 (6/12)  | -0.082 (4/12)  | +0.169 (8/8)     | -0.048 (3/12)       |
| M2-ais                                | +0.018 (7/12)  | -0.018 (3/12)  | -0.105 (4/12)  | +0.111 (6/8)     | -0.054 (0/12)       |
| M1-noaff                              | +0.023 (9/12)  | +0.036 (5/12)  | -0.057 (5/12)  | +0.123 (5/8)     | -0.043 (4/12)       |
| M1 + queue threshold (abl. a)         | +0.025 (9/12)  | +0.028 (6/12)  | -0.069 (4/12)  | +0.104 (6/8)     | -0.042 (4/12)       |
| M1 continued (secondary)              | +0.025 (7/12)  | +0.060 (5/12)  | -0.047 (4/12)  | +0.147 (7/8)     | -0.037 (4/12)       |
| M1-ais                                | +0.012 (7/12)  | -0.019 (2/12)  | -0.108 (4/12)  | +0.103 (5/8)     | -0.047 (1/12)       |
| M2 rank 2                             | +0.028 (9/12)  | +0.054 (7/12)  | +0.013 (5/12)  | +0.139 (5/8)     | -0.034 (5/12)       |
| M1 (= A12 M1-default-init)            | +0.020 (9/12)  | +0.046 (7/12)  | +0.032 (5/12)  | +0.105 (5/8)     | -0.032 (5/12)       |
| M1 unconstrained (gaming-flagged)     | +0.015 (9/12)  | +0.053 (8/12)  | +0.065 (7/12)  | +0.207 (5/8)     | -0.031 (4/12)       |
| llm-d-precise-prefix                  | +0.125 (12/12) | +0.144 (9/12)  | +0.002 (7/12)  | +0.075 (7/8)     | -0.005 (6/12)       |
| ramjet + queue threshold (abl. a)     | +0.006 (9/12)  | +0.015 (8/12)  | +0.010 (9/12)  | +0.004 (5/8)     | +0.001 (8/12)       |
| M0-ais                                | +0.019 (9/12)  | -0.006 (5/12)  | -0.091 (1/12)  | -0.010 (2/8)     | +0.007 (10/12)      |
| llm-d-precise-prefix@defaults         | +0.106 (11/12) | +0.119 (8/12)  | -0.008 (7/12)  | +0.039 (7/8)     | -0.006 (6/12)       |
| sticky-session bounded                | -0.007 (4/12)  | -0.031 (2/12)  | -0.062 (2/12)  | +0.024 (5/8)     | +0.010 (7/12)       |
| lmetric@defaults                      | +0.115 (8/12)  | +0.101 (6/12)  | +0.046 (7/12)  | +0.079 (8/8)     | -0.005 (4/12)       |
| two-tier                              | +0.103 (8/12)  | +0.139 (12/12) | +0.119 (9/12)  | +0.108 (8/8)     | -0.000 (5/12)       |
| ramjet@defaults                       | +0.070 (8/12)  | +0.114 (10/12) | -0.041 (6/12)  | +0.011 (6/8)     | +0.021 (9/12)       |
| two-tier@defaults                     | +0.125 (8/12)  | +0.118 (8/12)  | +0.103 (10/12) | +0.066 (8/8)     | -0.002 (5/12)       |
| lmetric (port)                        | +0.125 (8/12)  | +0.134 (7/12)  | +0.058 (8/12)  | +0.130 (8/8)     | -0.004 (4/12)       |
| M0 (default cost fn, tuned)           | +0.090 (10/12) | +0.047 (10/12) | -0.071 (4/12)  | +0.036 (7/8)     | +0.038 (12/12)      |
| sticky-session hard                   | +0.014 (7/12)  | +0.078 (6/12)  | +0.182 (7/12)  | +0.079 (6/8)     | +0.040 (7/12)       |
| llm-d-optimized-baseline (throughput) | +0.011 (5/12)  | +0.027 (5/12)  | +0.036 (5/12)  | +0.079 (6/8)     | +0.066 (12/12)      |
| llm-d-optimized-baseline@throughput   | +0.023 (6/12)  | +0.071 (7/12)  | +0.097 (8/12)  | +0.087 (7/8)     | +0.069 (12/12)      |
| llm-d-optimized-baseline (modeled)    | -0.015 (5/12)  | +0.075 (6/12)  | +0.193 (10/12) | +0.231 (8/8)     | +0.099 (12/12)      |
| sticky-bounded@defaults               | +0.198 (12/12) | +0.207 (12/12) | -0.015 (5/12)  | +0.147 (7/8)     | +0.032 (10/12)      |
| sticky-hard@defaults                  | +0.134 (8/12)  | +0.109 (12/12) | +0.138 (5/12)  | +0.096 (7/8)     | +0.064 (9/12)       |
| chwbl                                 | +0.618 (12/12) | +0.714 (12/12) | +0.386 (11/12) | +0.327 (8/8)     | +0.136 (12/12)      |
| dualmap                               | +0.143 (12/12) | +0.230 (12/12) | +0.163 (9/12)  | +0.205 (8/8)     | +0.158 (12/12)      |
| dualmap@defaults                      | +0.291 (12/12) | +0.483 (12/12) | +0.562 (12/12) | +0.356 (8/8)     | +0.241 (12/12)      |
| chwbl@defaults                        | +0.664 (12/12) | +0.858 (12/12) | +0.510 (11/12) | +0.430 (8/8)     | +0.243 (12/12)      |

| Policy                       | TTFT p50       | TTFT p90       | TTFT p99       | TTFT p90, 32–64K | E2E p90 (in window) |
|------------------------------|----------------|----------------|----------------|------------------|---------------------|
| round_robin                  | +1.705 (12/12) | +1.340 (12/12) | +1.037 (12/12) | +0.897 (8/8)     | +0.512 (12/12)      |
| default@defaults (reference) | +0.579 (12/12) | +0.451 (12/12) | +0.034 (5/12)  | +0.180 (6/8)     | +0.116 (12/12)      |

**Table 47:** Secondary metrics against ramjet on test, part 2 (decode, throughput, reuse) (descriptive, computed after the test pass from its records, no replay). Each entry is the segment-mean log-ratio, with the number of segments on which the policy’s value is higher. Lower is better for latencies, higher for throughput and prefix reuse.

| Policy                                | Mean-ITL p90   | Per-token ITL p99 | Output throughput | Prefix reuse  | AgentX trajectory |
|---------------------------------------|----------------|-------------------|-------------------|---------------|-------------------|
| M1-v2 (headline learned arm)          | +0.006 (3/12)  | −0.077 (4/12)     | +0.006 (7/12)     | −0.024 (1/12) | −0.012 (0/4)      |
| M2-ais                                | +0.020 (1/12)  | −0.107 (0/12)     | +0.007 (10/12)    | −0.013 (6/12) | −0.010 (0/4)      |
| M1-noaff                              | +0.017 (5/12)  | −0.090 (4/12)     | +0.005 (7/12)     | −0.015 (6/12) | −0.011 (0/4)      |
| M1 + queue threshold (abl. a)         | +0.012 (5/12)  | −0.084 (4/12)     | +0.006 (7/12)     | −0.014 (7/12) | −0.011 (0/4)      |
| M1 continued (secondary)              | +0.028 (5/12)  | −0.070 (4/12)     | +0.004 (7/12)     | −0.018 (7/12) | −0.010 (0/4)      |
| M1-ais                                | +0.029 (1/12)  | −0.127 (0/12)     | +0.007 (10/12)    | −0.013 (8/12) | −0.009 (0/4)      |
| M2 rank 2                             | +0.011 (5/12)  | −0.082 (5/12)     | +0.004 (5/12)     | −0.012 (5/12) | −0.004 (1/4)      |
| M1 (= A12 M1-default-init)            | +0.023 (6/12)  | −0.080 (5/12)     | +0.006 (7/12)     | −0.011 (5/12) | −0.002 (1/4)      |
| M1 unconstrained (gaming-flagged)     | +0.035 (5/12)  | −0.091 (5/12)     | −0.006 (4/12)     | −0.010 (7/12) | −0.006 (0/4)      |
| llm-d-precise-prefix                  | −0.010 (5/12)  | +0.027 (4/12)     | −0.000 (4/12)     | −0.008 (3/12) | −0.012 (0/4)      |
| ramjet + queue threshold (abl. a)     | −0.005 (5/12)  | +0.010 (7/12)     | −0.001 (4/12)     | +0.001 (4/12) | −0.000 (2/4)      |
| M0-ais                                | +0.020 (9/12)  | +0.034 (10/12)    | −0.001 (4/12)     | −0.031 (0/12) | −0.003 (0/4)      |
| llm-d-precise-prefix@defaults         | −0.010 (5/12)  | +0.007 (5/12)     | +0.001 (7/12)     | −0.005 (6/12) | −0.012 (0/4)      |
| sticky-session bounded                | +0.026 (7/12)  | +0.003 (4/12)     | −0.002 (5/12)     | −0.052 (4/12) | +0.000 (3/4)      |
| lmetric@defaults                      | −0.001 (2/12)  | −0.034 (0/12)     | −0.000 (5/12)     | +0.002 (5/12) | −0.006 (0/4)      |
| two-tier                              | +0.001 (4/12)  | −0.024 (0/12)     | −0.001 (4/12)     | −0.013 (2/12) | −0.004 (0/4)      |
| ramjet@defaults                       | +0.027 (9/12)  | +0.091 (8/12)     | −0.005 (2/12)     | −0.049 (0/12) | +0.004 (3/4)      |
| two-tier@defaults                     | −0.002 (3/12)  | −0.028 (1/12)     | −0.001 (5/12)     | −0.002 (6/12) | −0.003 (2/4)      |
| lmetric (port)                        | −0.003 (2/12)  | −0.037 (0/12)     | −0.001 (4/12)     | −0.004 (5/12) | −0.003 (1/4)      |
| M0 (default cost fn, tuned)           | +0.066 (11/12) | +0.103 (10/12)    | −0.006 (1/12)     | −0.133 (0/12) | +0.009 (3/4)      |
| sticky-session hard                   | +0.041 (7/12)  | −0.005 (5/12)     | −0.013 (3/12)     | −0.029 (4/12) | +0.036 (4/4)      |
| llm-d-optimized-baseline (throughput) | +0.115 (12/12) | +0.083 (11/12)    | −0.010 (1/12)     | −0.023 (4/12) | +0.026 (4/4)      |
| llm-d-optimized-baseline@throughput   | +0.123 (12/12) | +0.082 (10/12)    | −0.011 (0/12)     | −0.011 (4/12) | +0.029 (4/4)      |
| llm-d-optimized-baseline (modeled)    | +0.217 (12/12) | +0.078 (9/12)     | −0.015 (1/12)     | −0.055 (4/12) | +0.037 (4/4)      |
| sticky-bounded@defaults               | +0.151 (9/12)  | +0.043 (8/12)     | −0.006 (1/12)     | −0.139 (0/12) | +0.001 (2/4)      |
| sticky-hard@defaults                  | +0.185 (11/12) | +0.032 (5/12)     | −0.017 (2/12)     | −0.117 (4/12) | +0.039 (4/4)      |
| chwbl                                 | +0.330 (11/12) | +0.297 (10/12)    | −0.030 (0/12)     | −0.252 (0/12) | +0.052 (4/4)      |
| dualmap                               | +0.309 (12/12) | +0.135 (11/12)    | −0.028 (0/12)     | −0.050 (0/12) | +0.044 (4/4)      |
| dualmap@defaults                      | +0.405 (12/12) | +0.260 (12/12)    | −0.043 (0/12)     | −0.033 (0/12) | +0.049 (4/4)      |
| chwbl@defaults                        | +0.458 (11/12) | +0.473 (10/12)    | −0.046 (0/12)     | −0.255 (0/12) | +0.056 (4/4)      |
| round_robin                           | +0.759 (12/12) | +1.045 (12/12)    | −0.115 (0/12)     | −0.960 (0/12) | +0.248 (4/4)      |
| default@defaults (reference)          | +0.277 (11/12) | +0.387 (10/12)    | −0.025 (1/12)     | −0.262 (0/12) | +0.030 (3/4)      |



**Figure 14:** Performance profiles over the 60 test cells (copied from the final report): the fraction of cells on which a policy’s clipped log-ratio against  $\pi_0$  exceeds  $\tau$ . Cells are not independent, so the profiles are descriptive.

## B The pilot (preliminary)

The pilot asked whether tuning moves any policy away from the default router by more than replicate noise, and how large the learned model’s margin over the best heuristics might be, before the campaign committed to phase 2. Its numbers come from the validation split and from a smaller tuning setup than phase 2, so they are *preliminary (pilot, validation split)* and were superseded by the phase-2 results of Section 9. They are kept because they shaped phase 2: the budget, the step-size cap, the multi-start initialization, the extended feature set and the concentration audit all trace back to them.

The setup was smaller than phase 2’s in four ways.

- M0 (the default cost, tuned) and M1 (**learned-choice** on feature set **v1**) were tuned on 12 of the 34 train cells, chosen to cover every family, both train worker counts and every load mode, with 480 evaluations (30 generations of 16) per run.
- Each model was tuned twice with different CMA-ES seeds from the *same* start (M1 from  $\theta = -\mathbf{e}_0$ , M0 from the shipped defaults). Below they are called seed 1 and seed 2; they are not phase 2’s restarts s1–s3, which differ in their start.
- The configuration of each model was selected on validation replicates  $k = 0, 1, 2$  over both seeds and their checkpoints: seed 1 at generation 25 for M0, seed 2 at generation 30 for M1.
- The heuristics were not tuned. A screen of all nine catalog heuristics at shipped defaults on the 12 train cells ( $k = 0, 1$ ) ranked **llm-d-precise-prefix** (+0.169), **ramjet** (+0.166) and **lmetric** (+0.165) first by mean clipped log-ratio; the last two are tied, so all three went to validation. **dualmap** (−0.150) and **chwbl** (−0.214) lost to the default on that subset.

All policies were then scored on the 14 validation cells with 8 fresh replicates ( $k = 3, \dots, 10$ ) that no selection step had used, paired with  $\pi_0$  on the same replicate. The pilot ran 56,552 fresh replays with 0 errors, and an independent judge, working from the raw records without importing the harness, reproduced every comparison against  $\pi_0$  and against the heuristics in Tables 48 and 49 to four decimals.

“Beyond noise” below is the campaign’s two-clause rule (Section 8.2): on some cell the replicate-mean goodput ratio to the reference departs from 1 by more than three pooled replicate standard deviations, and over at least three independent segments the mean of the per-segment mean of (ratio − 1) exceeds twice its standard error in absolute value. The rule is two-sided, so a policy worse than the reference also passes it.

**Everything beat the default router; round-robin collapsed.** Every tuned and heuristic policy in Table 48 differed from  $\pi_0$  beyond noise, in the positive direction, overall and separately in each load mode; round-robin differed in the negative direction. The three cache-aware heuristics at shipped defaults already reached a mean per-cell goodput ratio over  $\pi_0$  of 1.150–1.153, and M1 reached 1.191. M0 was the weakest of the tuned and heuristic policies: tuning the default cost’s five knobs on 12 cells gained less than switching to an untuned cache-aware heuristic. The bar M1 had to clear in phase 2 was therefore set by those heuristics, after they got the same tuning budget, not by the default.

**M1 against the strongest heuristic: not beyond noise.** Table 49 pairs the selected M1 and M0 directly with each heuristic. Against **llm-d-precise-prefix**, the validation-best heuristic, M1’s mean clipped log-ratio was +0.030 (mean ratio 1.032, 10 of 14 cells better). The pre-registered minimum detectable effect (MDE) is  $\max(1\%, 2\text{ SE})$ , where SE is the standard error of the paired

**Table 48:** Pilot, validation split, every policy against  $\pi_0$  (*preliminary (pilot, validation split)*). Mean  $\Delta$ : mean over cells of the replicate-mean clipped log-ratio, overall and per family (Mooncake 5 cells, sessions 5, AgentX 4). Ratio: mean over cells of the replicate-mean goodput ratio  $G(\pi)/G(\pi_0)$ . Better: cells whose mean ratio exceeds 1. Seg. mean (SE): mean over the 6 validation segments of the per-segment mean of (ratio  $- 1$ ), with its standard error. Over cap: cells on which, in at least one replicate, one worker received more than  $\min(2/N, 1/N + 0.25)$  of the requests.

| Policy                | Mean $\Delta$ |          |          |        | Ratio | Better | Seg. mean (SE) | Over cap |
|-----------------------|---------------|----------|----------|--------|-------|--------|----------------|----------|
|                       | all           | Mooncake | sessions | AgentX |       |        |                |          |
| M1, seed 2 (selected) | 0.165         | 0.221    | 0.205    | 0.045  | 1.191 | 13/14  | 0.147 (0.044)  | 5        |
| M1, seed 1            | 0.162         | 0.220    | 0.200    | 0.043  | 1.187 | 13/14  | 0.142 (0.044)  | 4        |
| llm-d-precise-prefix  | 0.135         | 0.124    | 0.203    | 0.064  | 1.153 | 14/14  | 0.135 (0.034)  | 1        |
| ramjet                | 0.135         | 0.154    | 0.194    | 0.038  | 1.152 | 14/14  | 0.122 (0.038)  | 0        |
| lmetric               | 0.133         | 0.116    | 0.208    | 0.060  | 1.150 | 14/14  | 0.132 (0.036)  | 0        |
| M0, seed 2            | 0.129         | 0.129    | 0.194    | 0.047  | 1.145 | 14/14  | 0.124 (0.035)  | 0        |
| M0, seed 1 (selected) | 0.128         | 0.128    | 0.198    | 0.041  | 1.144 | 14/14  | 0.121 (0.037)  | 0        |
| round_robin           | -0.577        | -0.273   | -0.892   | -0.562 | 0.571 | 0/14   | -0.473 (0.066) | 0        |

**Table 49:** Pilot, validation split, head to head (*preliminary (pilot, validation split)*): the selected M1 and M0 against each untuned heuristic, paired on the same replicates  $k = 3, \dots, 10$ . Columns as in Table 48; positive values favor the tuned model. Noise: whether the pair differs beyond noise under the rule defined above. MDE = 0.038 in the units of the segment mean.

|                              | Mean $\Delta$ |          |          |        | Ratio | Better | Seg. mean (SE) | Noise               |
|------------------------------|---------------|----------|----------|--------|-------|--------|----------------|---------------------|
| Against                      | all           | Mooncake | sessions | AgentX |       |        |                |                     |
| <i>M1 (seed 2, selected)</i> |               |          |          |        |       |        |                |                     |
| llm-d-precise-prefix         | 0.030         | 0.097    | 0.002    | −0.019 | 1.032 | 10/14  | 0.009 (0.019)  | no                  |
| ramjet                       | 0.030         | 0.067    | 0.011    | 0.007  | 1.031 | 12/14  | 0.021 (0.011)  | closed <sup>a</sup> |
| lmetric                      | 0.032         | 0.105    | −0.003   | −0.015 | 1.034 | 6/14   | 0.012 (0.020)  | no                  |
| <i>M0 (seed 1, selected)</i> |               |          |          |        |       |        |                |                     |
| llm-d-precise-prefix         | −0.007        | 0.004    | −0.005   | −0.023 | 0.993 | 3/14   | −0.012 (0.006) | no                  |
| ramjet                       | −0.007        | −0.026   | 0.004    | 0.003  | 0.993 | 7/14   | −0.001 (0.005) | no                  |
| lmetric                      | −0.005        | 0.012    | −0.010   | −0.019 | 0.995 | 4/14   | −0.010 (0.005) | lanes <sup>a</sup>  |

<sup>a</sup> Not beyond noise overall; beyond noise in this load mode only.

segment mean on validation, in units of (ratio  $- 1$ ). At the pilot that SE was 0.019, so the MDE is 0.038. M1’s segment mean was only +0.009: the Mooncake window that carried the gain is 5 of the 14 cells but 1 of the 6 segments. Under the campaign’s noise rule M1 did not differ from this heuristic, overall or in any single load mode. The same held against **ramjet** and **lmetric**, with one exception: M1’s lead over **ramjet** in closed loop (+0.038, SE 0.013, over 3 segments) passed the rule.

**Where the margin came from.** M1’s lead over **llm-d-precise-prefix** came from one of the six validation segments. Per segment, the mean of (ratio  $- 1$ ) was +0.102 on the Mooncake window w1 (5 cells), +0.003 and +0.001 on the session seeds s4 and s5, and -0.014, -0.025 and -0.012 on the three AgentX play subsets V1–V3. On validation, then, M1 gained where requests share a few hot prefixes across users, and lost slightly, but on every segment, on long agent contexts. That AgentX deficit shaped phase 2: M1’s second and third restarts started from a behavior clone of **llm-d-precise-prefix** and from a cache-heavy point (Section 6.4), and the 23-feature set v2 became a mandatory arm, the one that won.

**Table 50:** What the pilot M1 learned (*preliminary (pilot, validation split)*).  $s_f = \text{sd}(x_0)/\text{sd}(x_f)$ , with within-set standard deviations measured on candidate tables reconstructed from  $\pi_0$ ’s train replays;  $\theta_f/s_f$  is the utility change per one within-set standard deviation of feature  $f$ , in units of the anchor’s ( $\theta_0 = -1$ ). The search bound was  $|\theta_f| \leq 4s_f$ .

| $f$ | Feature                 | $s_f$ | $\theta_f$ , seed 2 | $\theta_f/s_f$ , seed 2 | $\theta_f/s_f$ , seed 1 |
|-----|-------------------------|-------|---------------------|-------------------------|-------------------------|
| 0   | default_logit_scaled    | 1     | -1 (pinned)         | -1                      | -1                      |
| 1   | overlap_frac            | 25.6  | 13.60               | 0.532                   | 0.383                   |
| 2   | new_prefill_tokens_k    | 17.7  | -8.71               | -0.493                  | -0.143                  |
| 3   | active_prefill_tokens_k | 5.33  | -0.546              | -0.102                  | -0.158                  |
| 4   | kv_load_frac            | 80.8  | -0.434              | -0.005                  | 0.015                   |
| 5   | active_requests_s       | 110   | -49.2               | -0.449                  | -0.407                  |
| 6   | session_affinity        | 21.2  | 0.995               | 0.047                   | 0.018                   |
| 7   | isl_x_prefill_load      | 3.22  | 0.041               | 0.013                   | 0.021                   |

**The lead came with concentration.** Over all validation records, the selected pilot M1 exceeded the share cap on 5 Mooncake cells (27 of 112 cell-replicate records), with a largest share of 0.646 in one record; `llm-d-precise-prefix` exceeded it on one cell. This flag became the phase-2 midpoint’s gaming audit (Section 10).

**What the pilot M1 learned.** Relative to the default cost they start from, both pilot M1 seeds moved in three directions (Table 50): more weight on the overlap fraction, a penalty on new prefill tokens (strongly in seed 2 only), and a penalty on the number of active requests, a quantity the default cost ignores at its shipped weight  $\beta_{\text{req}} = 0$ . Both tuned M0 configurations moved the same way within their own family: more overlap credit ( $\beta_{\text{ov}} = 1.62$  and  $1.65$ , shipped 1), a larger prefill-load scale ( $\beta_{\text{pf}} = 24.8$  and  $16.9$ , shipped 1) and a nonzero active-request weight ( $\beta_{\text{req}} = 854$  and  $1,152$ , shipped 0), so that each active request costs as much as 34 or 68 prefill blocks. Two model classes tuned independently thus both added a request-count load term, the direction the phase-2 M1-v2 took further with its prefill-by-requests interaction (Section 9.8).

**How the search behaved.** M1’s validation score (the CMA-ES mean, on  $k = 0, 1, 2$ ) reached 0.163–0.165 within 160–240 evaluations and then stayed within 0.005 of each run’s maximum; the two seeds ended 0.0004 apart. M0 plateaued within 80–160 evaluations on train, but in one seed the CMA-ES step size then grew from 0.18 to 10.8 internal units and the generation mean fell to  $-0.23$ , a drift along near-flat directions of the default cost’s knobs. Phase 2 therefore capped the per-coordinate step size at 0.3 internal units and set the budget at 400 evaluations per run, the latest validation plateau among the four pilot runs (Sections 6.2 and 6.3). On 16 candidates, the objective on the 12-cell subset ranked candidates like the objective on all 34 train cells (Spearman 0.947); phase 2 nevertheless tuned on all 34.

## C Reproducibility

The campaign’s durable record is the directory `notes/learned-routing/` on the Dynamo branch `rupei/learned-routing`, whose sanitized copy `rupei/learned-routing-public` is public. `README.md` describes the campaign and its state, `REPRODUCE.md` rebuilds it step by step on a clean Linux x86\_64 machine, and `campaign/` mirrors every small file the steps need: facts, cells, audits, run summaries, calibration scripts and the AgentX base manifests. Bulky artifacts (traces, workload replicates, the replay cache) stay in the campaign root and are regenerated from pinned public

sources. Off the original host, git and the public sources are enough for the steps `REPRODUCE.md` covers. This appendix lists the identifiers; the procedure is in `REPRODUCE.md`.

## C.1 Branches

The campaign’s code and record live on three branches of the Dynamo repository.

- `rupei/learned-routing`: the policies (`learned-choice` and `sticky-session` in the builtin catalog, and the `seed` parameter on `dynamo-default-cost-fn`), the harness (`lr-eval`, `lr-train`, `lr-report` and the remote CPU lane), the live lane, and the campaign notes under `notes/learned-routing/`, including this paper. It starts from the head of the router-policy pull-request stack (`#15450` and `#15453`), which already runs catalog policies in offline replay and passes the device-tier overlap to them.
- `rupei/learned-routing-lag`: the router-state lag knob for offline replay, built separately so that the tuning build and its result cache stay untouched.
- `rupei/learned-routing-ais`: feature set `v3` and the research input for a request’s modeled prefill time, also on its own build, for the AIS league.

Every stage’s facts file records the worktree head it ran at. Only the first branch is published, as the sanitized copy `rupei/learned-routing-public`, in which internal commits appear as position labels; the lag and AIS branches are not, so the lag part of the robustness pass and the AIS league can be checked against the mirrored facts but not yet rebuilt from public sources. `REPRODUCE.md` rebuilds the campaign to its calibrated, frozen cells; the later stages are reproduced from the mirrored facts and the analysis scripts named in the final report’s reproduction section. PENDING [a [REPRODUCE.md revision that walks through the pilot, phase 2, the test pass and the robustness pass](#)]

## C.2 Environment

|                    |                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Python, Rust       | CPython 3.12.3; rustc 1.96.1 (pinned by <code>rust-toolchain.toml</code> ); maturin 1.15.0                                                              |
| Simulator          | <code>aisimulate</code> 0.13.0.dev2026093000000000061 (wheel) and the matching <code>aisimulate-core</code> crate                                       |
| Dynamo             | <code>ai-dynamo</code> and <code>ai-dynamo-runtime</code> 1.6.0, editable from the worktree                                                             |
| Bindings           | built with feature <code>ais-forward-pass</code> (required for AIS engine timing); <code>_core.abi3.so</code> SHA-256 1d67c131..., build ID 6955b0ee... |
| Lag-check bindings | separate build, <code>.so</code> 90464b9b..., build ID a6c6b04e...                                                                                      |
| Harness            | package <code>learned_routing</code> , harness version <code>lrh-4</code>                                                                               |
| Python packages    | <code>cma</code> 4.5.0, <code>numpy</code> 2.5.3, <code>scipy</code> 1.18.1, <code>pandas</code> 3.0.6, <code>matplotlib</code> 3.11.2                  |
| Host               | Ubuntu 24.04, glibc 2.39, 24 cores; local replays share a pool of 20 slots                                                                              |

The build ID is a hash of the bindings’ shared object together with the simulator version, and it is part of every result-cache key. A rebuild almost always yields a new ID, because the shared object embeds build-tree paths, so cached results from another build are never reused by accident.

## C.3 The rebuild procedure in brief

Table 51 condenses `REPRODUCE.md`. “Verified” there means the step was re-run while the notes were written, on the campaign host or in a fresh private root.

## C.4 Frozen inputs

- Engine configuration `config/engine.json`, SHA-256 7b5ff164...: the single source of truth for model, hardware, backend, parallelism, context length and engine arguments.

**Table 51:** Sections of REPRODUCE.md and what each establishes.

| §     | Step                  | What it establishes, and how it was checked                                                                                                                                                                       |
|-------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Branch and SHAs       | worktree at the snapshot commit; base is an ancestor                                                                                                                                                              |
| 2–3   | Environment, bindings | pinned versions; the KV-capacity estimate (18,863 blocks) and a one-request replay whose TTFT equals the AIS value prove AIS timing is in effect                                                                  |
| 4     | Harness and tests     | Rust policy tests (127 pass, 2 pre-existing ignored) and harness tests                                                                                                                                            |
| 5     | Campaign root         | <code>config/engine.json</code> copied from git and checked by SHA-256; every <code>DYN_ROUTER_*</code> variable unset, since replay workers inherit them and they change policies without changing the cache key |
| 6     | Traces                | source traces regenerated byte-identically (106 of 106 files, verified)                                                                                                                                           |
| 7     | Cells                 | frozen cells <code>lr-cells-v5</code> installed from git, not regenerated                                                                                                                                         |
| 8     | AgentX lowering       | six base manifests from git; lowering checked field for field by an independent re-implementation for all 82 plays and 6 transforms                                                                               |
| 9     | Derived traces        | every trace the frozen cells reference regenerated byte-identically (34/34 train, 14/14 validation, 60/60 test cells, verified); then a parity smoke against a mirrored reference (4 identical, 0 different)      |
| 10    | Calibration           | mirrored driver scripts; every load level follows from the mirrored load curves (171/171 levels and 108/108 cell loads, verified)                                                                                 |
| 11    | Frozen test set       | SHA-256 of <code>cells/test.jsonl</code> and of every test trace                                                                                                                                                  |
| 12    | Replicates            | the replicate protocols, policy seeds and the noise rule                                                                                                                                                          |
| 13    | Entry points          | <code>lr-eval</code> , <code>lr-train</code> , <code>lr-report</code> ; the remote bundle flow and the cluster CPU lane                                                                                           |
| 14    | Gate and headline     | the pilot gate and the pre-registered headline test                                                                                                                                                               |
| 15    | Later stages          | listed as not yet run when the notes were written (see above)                                                                                                                                                     |
| 16–17 | Costs, mirrored files | measured costs; the small files that used to live only in the campaign root                                                                                                                                       |

- Cells `lr-cells-v5`: test file `cells/test.jsonl`, SHA-256 `7b998b8e...`; split manifest, SHA-256 `0e84edf1....`. The freeze record `facts/test_freeze.json` also hashes the content of every test cell and all 39 test traces (2,130,865,265 bytes).
- Source traces, pinned by SHA-256 in `traces/MANIFEST.json` (`b2616d0d...`).
- The headline test, pre-registered in `facts/HEADLINE_TEST.json` before any training.
- Before the single test pass the test stage verified all 149 freeze hashes, and `facts/finalists.json` was frozen (SHA-256 `e0bae29d...`) before the first test record.

## C.5 Determinism

Replay is deterministic: the same policy on the same cell and replicate reproduces every per-request row. The campaign relies on this in four places.

- **Seeds.** The shipped default router broke ties with an unseeded random number generator, so identical runs of one Mooncake cell gave different goodput (2.8934 against 2.8290 for the catalog policy). The branch adds a `seed` parameter. The harness passes seed  $k + 1$  to the seeded policy types (`dynamo-default-cost-fn`, `learned-choice`, `sticky-session`); every other type takes no seed.
- **Replicates.** Variation comes from workload replicates shared by every policy: replicate  $k$  permutes arrivals logged at the same instant (protocol `crn-order-v1`), re-draws each session’s first arrival inside its logging slot (`crn-spread-v1`, for traces logged on a grid), or permutes recycled AgentX copies.
- **Parity with the default.** `learned-choice` at  $\theta = -e_0$ , with the tie-break that replays the default picker’s random stream, chose the same worker as the seeded `dynamo-default-cost-fn`

on every request of 88 of 88 (cell, replicate) pairs, and again on 22 of 22 after the bindings were rebuilt. With the tie-break used in training (`one_draw`) individual ties resolve differently, but goodput did not differ beyond noise.

- **Re-runs and resumes.** An independent calibration audit re-ran replicates 0 and 1 of 11 cells for  $\pi_0$  and round-robin and matched the cache on 44 of 44 records, every field and every per-request byte; new replicates run twice were identical on 44 of 44. A harness audit killed `lr-train` three times by process ID and the resumed runs were bit-identical, including their continuation.

Results are cached under a SHA-256 of the canonical policy, the cell identifier and content, the replicate and its protocol, the harness version and the bindings build ID. One residual nondeterminism is harmless: replay orders its per-request output by random request identifiers, so summary means can differ in the last bits ( $\sim 10^{-15}$  relative); the harness recomputes every metric from a sorted copy of the rows.

## C.6 Cross-host parity

Replay calls the host’s math library, so bit-exact results require the same OS and C library image. Most tuning ran on a pool of shared CPU nodes, one run per node, from a self-contained bundle holding the exact local shared object, the harness, the traces and the cells. Before any node image was used, a parity smoke compared its records with the workstation’s.

- **Lane validation.** Ten comparisons on AMD EPYC 7702P and EPYC 9654P nodes (Ubuntu 24.04.5, glibc 2.39, CPython 3.12.3) were all identical: 6,312 remote records with no field differences, and 48,901,384 per-request rows byte-identical. They covered seven cells (Mooncake open and closed loop, sessions, lowered AgentX, including one AgentX cell at  $N = 16$ ), six replicas of the calibration reference batch, and two short tuning runs.
- **Tuning numerics.** With default settings, `numpy` and OpenBLAS dispatch AVX-512 kernels on the EPYC 9654P, and the CMA-ES state differed from the workstation’s in the last bits from the first generation. The node scripts pin the dispatch (`OPENBLAS_CORETYPE=Haswell`, `OPENBLAS_NUM_THREADS=1`, `NPY_DISABLE_CPU_FEATURES`); with the pins, a 60-generation check matched the workstation on every generation on both CPU types, and the remote tuning smoke runs wrote byte-identical histories, best points and policy files.
- **Test pass.** Tasks that ran twice, once in a first wave and again after a timeout fix, were bit-identical on different nodes in every field and per-request byte.
- **Pilot and phase 2.** The 384 records of the pilot’s 16 first-generation candidates on the 12 subset cells were identical locally and on the cluster. A node image added during phase 2 passed its smoke (28 of 28 records, 186,548 rows) before it ran any tuning.
- **Lag build.** The separate lag-check build was validated on five node images before the robustness pass (the first: 192 of 192 records, 1,214,112 rows).

## C.7 Entry points and stage facts

Three command-line tools carry every number: `lr-eval` evaluates policies on cells with replicates, through the slot pool and the cache; `lr-train` runs Algorithm 2; and `lr-report` builds paired tables with segment-bootstrap intervals and refuses to pool records from different builds or cell contents. Every long job is resumable and takes `--max-wall-seconds`. A policy is one JSON or YAML object, and the tuned result `best_policy.yaml` is itself a valid policy for `lr-eval`.

Each stage writes one facts file under `facts/` and one line in the append-only `facts/STATE.md`: setup, the three build stages and their fixes, AgentX lowering, noise, calibration and its fix, the test freeze, the pre-registered headline test, the cluster lane, the pilot, the gate, phase-2 preparation, the lag patch, the AIS features (`facts/ais_features.json`), the tier-A and tier-B/C tuning (`facts/tierA.json`, `facts/tierBC.json`) and the phase-2 midpoint fix (`facts/phase2_mid_fix.json`), the SLO-transfer constants (`facts/sla_transfer_spec.json`), the frozen finalists (`facts/finalists.json`), the single test pass (`facts/test_results.json` with its `report_must_state` wording), the robustness pass (`facts/robustness.json`), the secondary metrics (`facts/secondary_metrics_test.json`), the in-class heuristic diagnostic (`facts/fairness_in_class_heuristic.json`), the live lane (`facts/live.json`, `facts/live_smoke.json`, `facts/live_loadgen_parity.json`), the live finalist runs (the registration `facts/live_plan.json` and the analysis `facts/live_results.json`) and this paper's derived numbers (`facts/paper_draft.json`, `facts/paper_final.json`). Every audit report is under `audits/`, and the final report is `report/REPORT.md`; its live section is also `report/LIVE.md`. The live analysis is reproducible from the raw load-generator exports: `runs/live/finalists/scripts/live_results.py` verify re-scores every live run and requires identical scores, and `build` re-runs the registered analysis and writes the live facts, `report/LIVE.md` and the report's table fragment, from which `make tables` copies the live tables of Section 11.

**Footprint.** On 2026-10-03 the campaign root held 95 GB of traces and 247 GB of run outputs, mostly workload replicates (183 GB) and the result cache (37 GB); all of it regenerates from the steps above. Compute per stage is in Table 29.

## C.8 Building this document

The sources are in `notes/learned-routing/paper/`; `make` there runs `latexmk` with `pdflatex` and BibTeX and writes `build/main.pdf`, and `make tables`, given the campaign root, regenerates `generated/` from the final report's table fragments. The figures in `fig/` are copies of the final report's. Every number in the text carries a source comment (`% src:`) naming the campaign file it comes from, and results that do not exist yet are marked with a boxed PENDING tag.